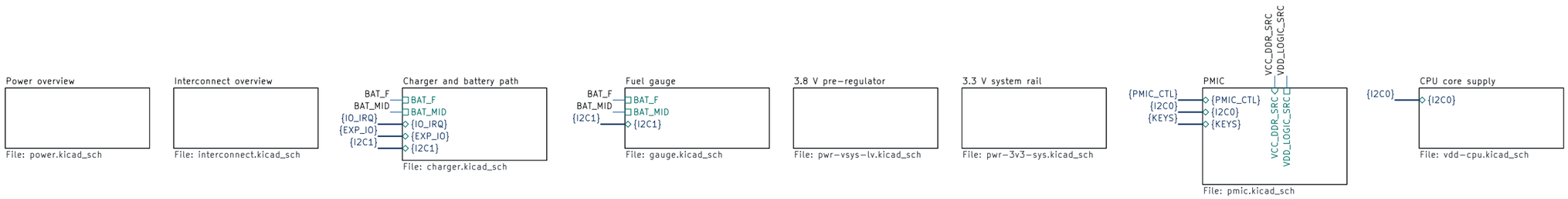
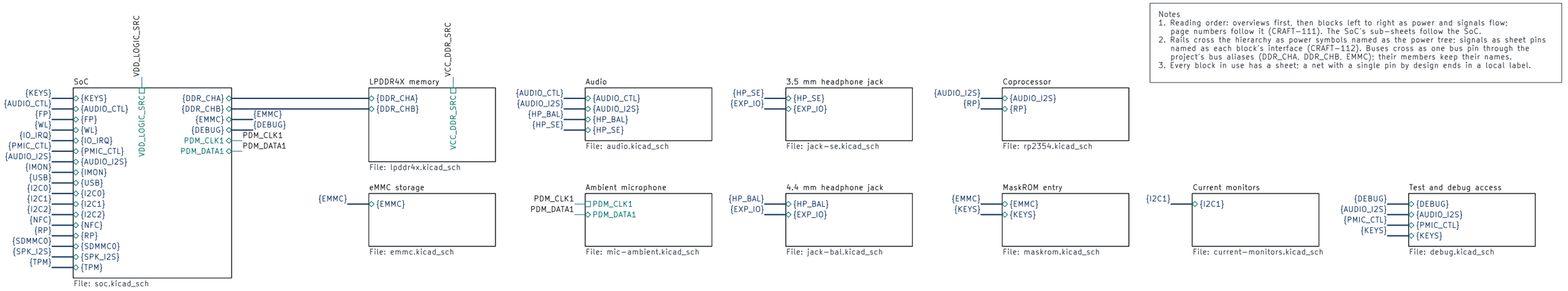
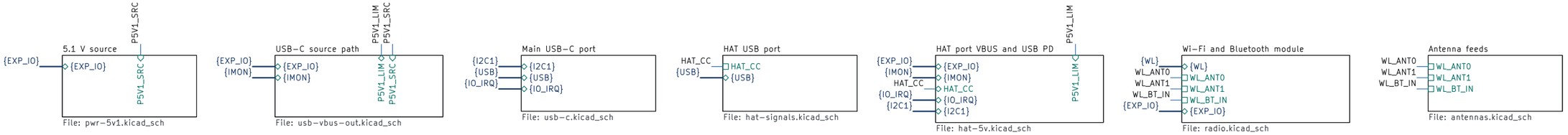
Overviews, then the power blocks in the order power flows



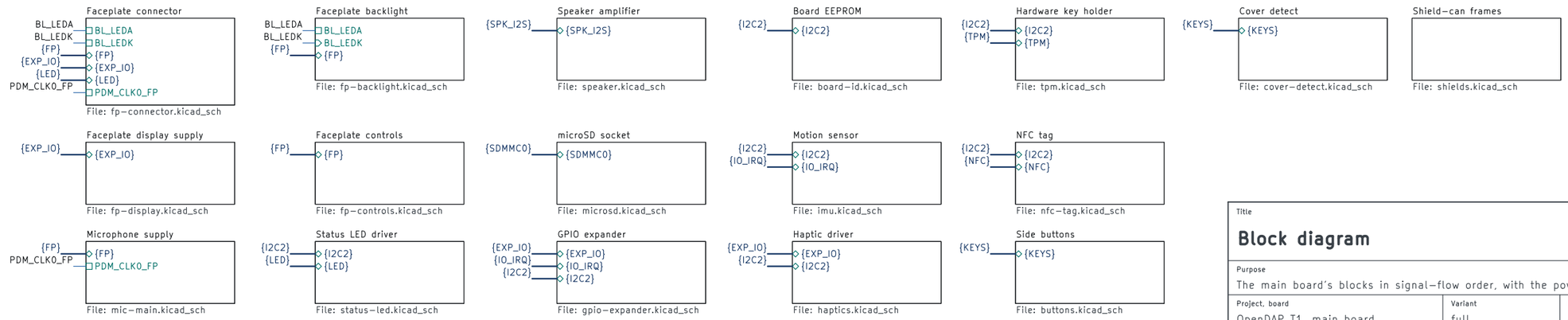
SoC, memory, the coprocessor and MaskROM entry; audio, from the DACs to the jacks; instrumentation and debug



USB ports (the 5.1 V source, USB-C, the HAT port) and the radio



Faceplate interface; control, storage and sensors; the shield frames



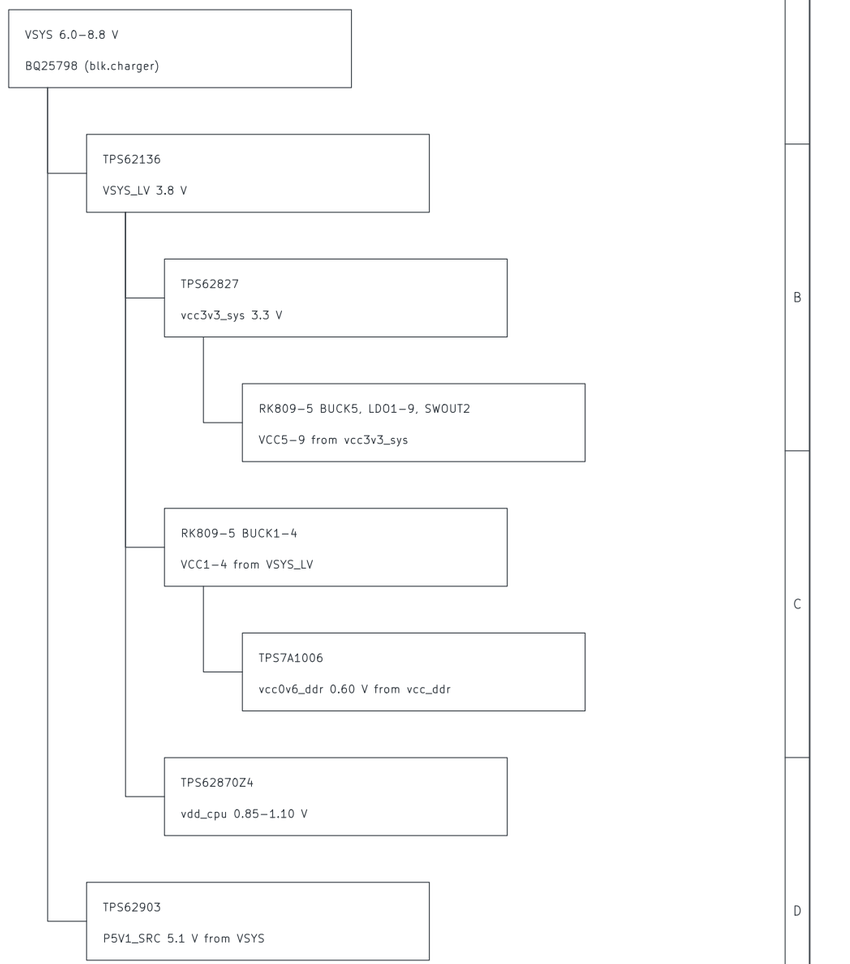
Title	**Block diagram**
Block	root
Purpose	The main board's blocks in signal-flow order, with the power and interconnect overviews
Project, board	OpenDAP T1, main board
Variant	full
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Rail table (power tree v0.11 §3, §4.12)

Rail	Voltage	Source	Switch or enable	Consumers	Net class	Measure
USBC_VBUS	5–12 V	USB-C sink; eFuse #C source (blk.usb-vbus-out)	PD contract; USB_SRC_ON	TPS2121 IN1, FUSB302B	PWR_VBUS	eFuse #C IMON
HAT_5V	5 V	HAT port: eFuse #A source, or a source HAT	HAT5V_EN (TCPM)	TPS2121 IN2, FUSB302B01	PWR_HAT5V	eFuse #A IMON
VBUS_CHG	5–12 V	TPS2121 (blk.charger-input)	input priority	BQ25798 VBUS	PWR_VBUS	–
VBAT	6.0–8.4 V	BQ25798 BAT, 2S pack (blk.charger-core)	ship FET to the pack	BQ25798 BATFET, gauge PACK	PWR_BAT	gauge (blk.gauge)
VSYS	6.0–8.8 V	BQ25798 SYS (VSYS_SRC) through the 10 mΩ shunt	ship FET	TPS62136, TPS62903, LMR33630C	PWR_VSYS	INA236 0x40
VSYS_SRC	6.0–8.8 V	BQ25798 SYS (blk.charger-core)	–	VSYS shunt 10 mΩ (blk.current-monitors)	PWR_VSYS	INA236 0x40 (IN+)
VSYS_AUDIO	6.0–8.8 V	VSYS through 20 mΩ (blk.current-monitors)	–	LMR33630C and its EN shifter (blk.hpa-rails-pre)	PWR_VSYS	INA236 0x48
VSYS_P5V1	6.0–8.8 V	VSYS through 20 mΩ (blk.current-monitors)	–	TPS62903 VIN (blk.pwr-5v1)	PWR_VSYS	INA236 0x43
VSYS_LV	3.8 V	TPS62136 (blk.pwr-vsys-lv)	always on (EN = VSYS)	RK809 VCC1-4, TPS62870Z4, TPS62827, audio, display	–	H5 link
VSYS_LV_AUDIO	3.8 V	VSYS_LV through 20 mΩ (blk.current-monitors)	–	DAC_VP switch, TPS62840, link #27, TPS65131	–	INA236 0x41
VSYS_LV_BL	3.8 V	VSYS_LV through 100 mΩ (blk.current-monitors)	–	TPS61160 VIN (blk.fp-backlight)	–	INA236 0x49
vcc3v3_sys	3.3 V	TPS62827 (blk.pwr-3v3-sys)	always on (EN = VSYS_LV)	RK809 VCC5-9, eMMC VCC, SoC VCCIO1/5/7, USB	–	H5 link
vdd_cpu	0.85–1.10 V	TPS62870Z4 (blk.vdd-cpu)	EN from vcc3v3_pmu; I2C DVS	SoC CPU	–	H5 link
vdd_logic	0.90 V	RK809 BUCK1 (blk.pmic)	off in suspend	SoC logic	–	H5 link (blk.soc)
vdd_gpu	0.90 V	RK809 BUCK2	DT regulator	SoC GPU	–	–
vcc_ddr	1.113 V	RK809 BUCK3 (47k / 120k)	on in suspend	DRAM VDD2, SoC DDR_VDDQ, TPS7A1006	–	H5 link (blk.lpddr4x)
vdd_npu	0.90 V	RK809 BUCK4	off by default	SoC NPU	–	–
vcc_1v8	1.80 V	RK809 BUCK5	off in suspend	SoC VCCIO2/4, eMMC VCCQ, vccio6_1v8	–	–
vccio6_1v8	1.80 V	vcc_1v8 through a 0 Ω link	follows vcc_1v8	SoC VCCIO6	–	–
vcc0v6_ddr	0.60 V	TPS7A1006 (blk.lpddr4x)	EN = vcc_ddr	DRAM VDDQ, SoC DDR_VDDQL	–	H20 link
vcc1v8_ddr	1.80 V	vcca1v8_pmu through a link	always on	DRAM VDD1, TPS7A1006 BIAS	–	M link
vdda0v9_image	0.9 V	RK809 LDO1	unused (RFC 0014)	none: CSI RX has no consumer	–	–
vdda_0v9	0.9 V	RK809 LDO2	on	SoC USB/PCIe/DSI PHY	–	–
vdda0v9_pmu	0.9 V	RK809 LDO3	always on	SoC PMU	–	–
FP_VCI_SRC	2.8 V	RK809 LDO4	RK809 I2C	FP_VCI switch (blk.fp-display)	–	M link
FP_VCI	2.8 V	TPS22945 from FP_VCI_SRC	FP_IO_EN	panel VCI (contact 44)	–	–
vccio_sd	1.8/3.3 V	RK809 LDO5	SD driver	SoC VCCIO3	–	–
vcc3v3_pmu	3.3 V	RK809 LDO6	always on	SoC PMUIO1, FUSB302B, TPS62870Z4 EN	–	–
vcca_1v8	1.8 V	RK809 LDO7	on	SoC PLL and PHY analog	–	–
vcca1v8_pmu	1.8 V	RK809 LDO8	always on	SoC PMUIO0/2, vcc1v8_ddr, pull-ups	–	–
vcca1v8_image	1.8 V	RK809 LDO9	unused (RFC 0014)	none: CSI RX has no consumer	–	–
vcc3v3_sd	3.3 V	RK809 SWOUT2	RK809 I2C	microSD VDD	–	M link
vcc3v3_wl	3.3 V	TPS22916C from vcc3v3_sys (blk.radio-power)	WL_PWR_EN	Type 2EL AVDD33, AVDD18 buck EN	–	H20 link #11
vcc1v8_wl	1.80 V	TPS62826 from VSYS_LV (blk.radio-power)	EN from vcc3v3_wl	Type 2EL AVDD18	–	H20 link #12
FP_3V3	3.3 V	TPS22950 from vcc3v3_sys (blk.fp-connector)	FP_IO_EN	faceplate logic, pull-ups	–	–
FP_1V8	1.80 V	TPS22945 from vcc_1v8 (blk.fp-connector)	FP_IO_EN	faceplate IOVCC	–	–
MIC_1V8	1.80 V	TPS22945 from vcc_1v8 (blk.mic-main)	MIC_EN	both microphone branches	–	–
MIC_1V8_FP	1.80 V	MIC_1V8 through link #34 (blk.mic-main)	MIC_EN	faceplate microphone (contact 40)	–	L10 link #34

Power tree (drawn blocks)



Notes
1. Rail names are the power tree's; every power symbol on every sheet names one (CRAFT-84, -85).
2. Only the rails of the blocks drawn so far are listed; the rest join with their blocks.
3. Measure: the block measurement links of BOM v1 line 137 (shunt in full, 0 Ω in production, H5 kept); the five INA236 channels and their shunts are on blk.current-monitors.
4. PWR_FLAG: KiCad 10.0.7's ERC reports a flag on a net island of power symbols as pin_not_connected, so each flag sits on the sheet that feeds its rail, on a branch of the rail's wire outside the block's groups (link load side, inductor output). Local nets fed through a passive part (TPM_VDD, RP_DVDD) take their flag the same way on their own sheet.
5. Net class '-': the class rules of the product apply through the switch nodes and DDR classes; rail copper widths come with the layout (PCB design constraints).

Title	Block
Power overview	power

Purpose: Rails of the main board drawn so far: source, switch, consumers, net class and measurement

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	2 / 69

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/Power overview/

External connectors

Connector	Block	Pin map
USB-C (USB4105-GF-A)	blk.usb-c	added when the block is drawn
HAT connector	blk.hat-signals	follows the connector decision (RFC 0014; hats standard/ interface v2)
Faceplate (FH35C-51S)	blk.fp-connector	added when the block is drawn (faceplate interface v1)
Battery contacts (6 x 70-9258)	blk.charger	added when the block is drawn (ADR 0020)
Headphone jacks 3.5 / 4.4 mm	blk.jack-se, blk.jack-bal	added when the blocks are drawn
microSD (DM3AT)	blk.microsd	added when the block is drawn

Notes
1. One pin-map table per external connector (pin, net, direction, block, notes), named as on the block sheets (CRAFT-110). No external connector is drawn yet: the blocks drawn so far (power converters, PMIC, vdd_cpu, SoC, LPDDR4X, eMMC) are internal to the board.

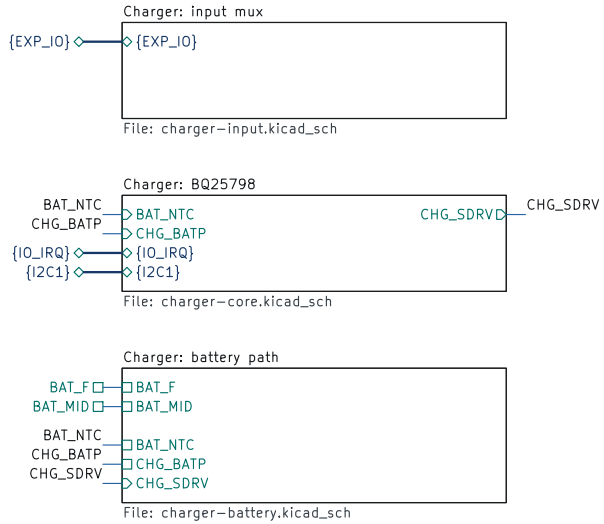
Title	Block
Interconnect overview	interconnect

Purpose: External connections of the main board: one pin map per connector

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	3 / 69

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/Interconnect overview/



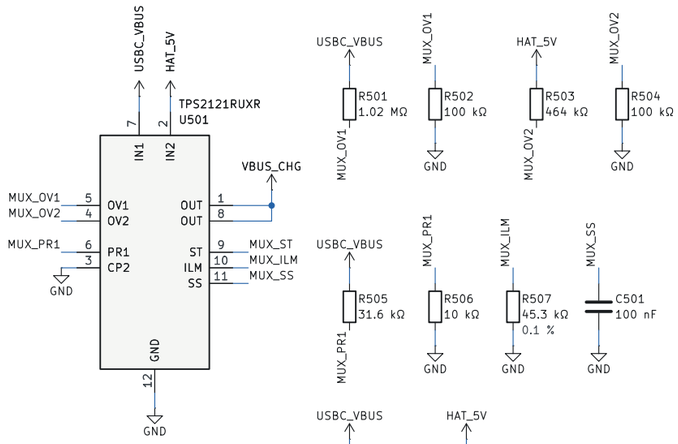
Notes
1. Function: TPS2121 picks USB−C (IN1, priority) or the HAT port (IN2) into VBUS_CHG; the BQ25798 charges the 2S pack and makes VSYS (6.0−8.8 V); the ship FET (SDRV) and the 5 A fuse join the BQ25798 BAT node to the six battery contacts (feature blocks #7; ADR 0014, 0020).
2. Absent state: never omitted. Pack out: with no adapter the BQ25798 is unpowered and the ship FET off, so an inserted pack meets a soft start; with an adapter it reports no battery.
3. Drawn on three sub−sheets (blk.charger−*), sheets of blk.charger as the SoC's sub−sheets.
4. SYS leaves as VSYS_SRC; the VSYS INA236 shunt (10 mΩ, blk.current−monitors) makes VSYS.

Title	Block
Charger and battery path	blk.charger
Purpose: TPS2121 input mux, BQ25798 2S charger, ship FET, fuse and the six battery contacts	

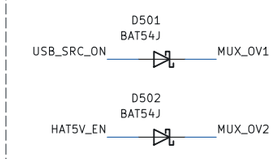
Project, board	Variant	Rev	Date	Sheet
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/Charger and battery path/

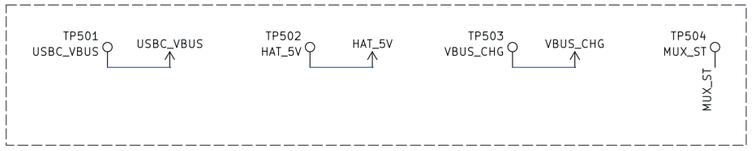
Input mux: TPS2121, USB-C on IN1 with priority (BOM v1 line 45)



OV interlocks (BOM v1 line 45)



Test pads (PCB-T-02)



{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
USBC_VBUS	Supply in	From blk.usb-c: 5–12 V PD sink; TVS2200 there
HAT_5V	Supply in / out	HAT port VBUS (blk.hat-5v): IN2 sinks 5 V only (BOM v1.14)
USB_SRC_ON, HAT5V_EN	Input	Expander P03, P02: high while the T1 sources the port; forces OVx high
VBUS_CHG	Supply out	To the BQ25798 VBUS, VAC1, VAC2 (blk.charger-core)

Rails

Rail	Capacitors	Requirement and source
USBC_VBUS, HAT_5V	1 µF 25 V each at IN1, IN2	SLVSEA3F §11 (value: capture choice)

Notes
1. Function: TPS2121 picks USB-C (IN1, priority above 4.4 V) or the HAT port (IN2) into VBUS_CHG.
2. Absent state: both inputs absent: VBUS_CHG off, the charger unpowered.
3. OV1 1.02 MΩ / 100 kΩ (11.1–12.6 V), OV2 464 kΩ / 100 kΩ (5.60–6.31 V), RILM 45.3 kΩ 0.1 % (1.94–2.95 A); USB_SRC_ON and HAT5V_EN force OV1 and OV2 through a BAT54J each, so the T1's own 5 V never re-enters the charger. PR1 31.6 kΩ / 10 kΩ (4.3–4.6 V), SS 100 nF, CP2 to GND: capture choices (SLVSEA3F Eq. 5, Table 9-1, pin table).
4. ST to a test pad instead of GND (SLVSEA3F §10.2.3): high when IN1 feeds the output, read at bring-up with a pull-up on the probe.

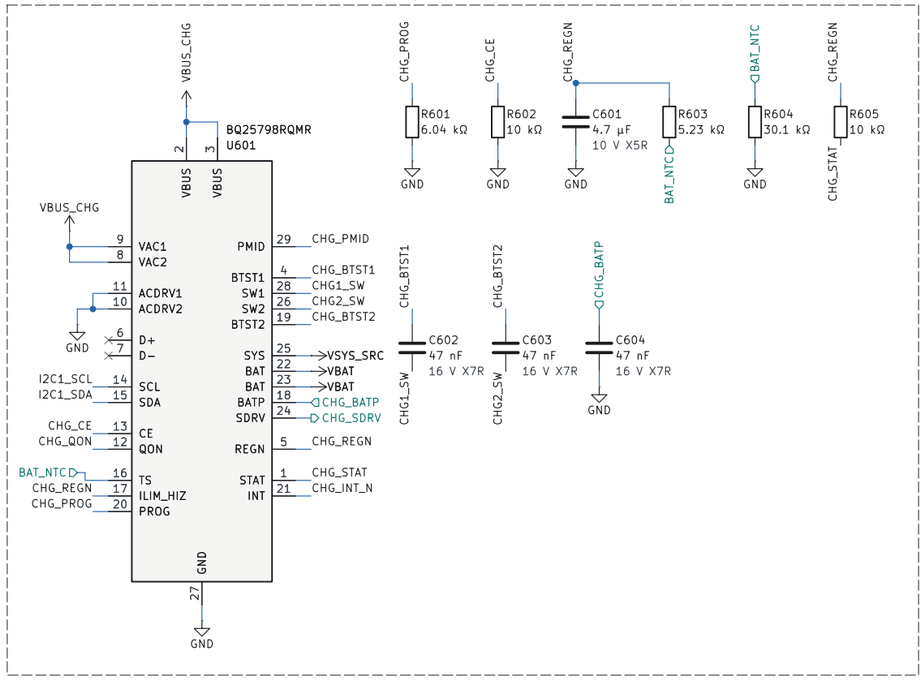
Title	Block
Charger: input mux	blk.charger-input
Purpose	TPS2121 picks USB-C or the HAT port into VBUS_CHG; the OV interlocks

Project, board	Variant	Rev	Date	Sheet
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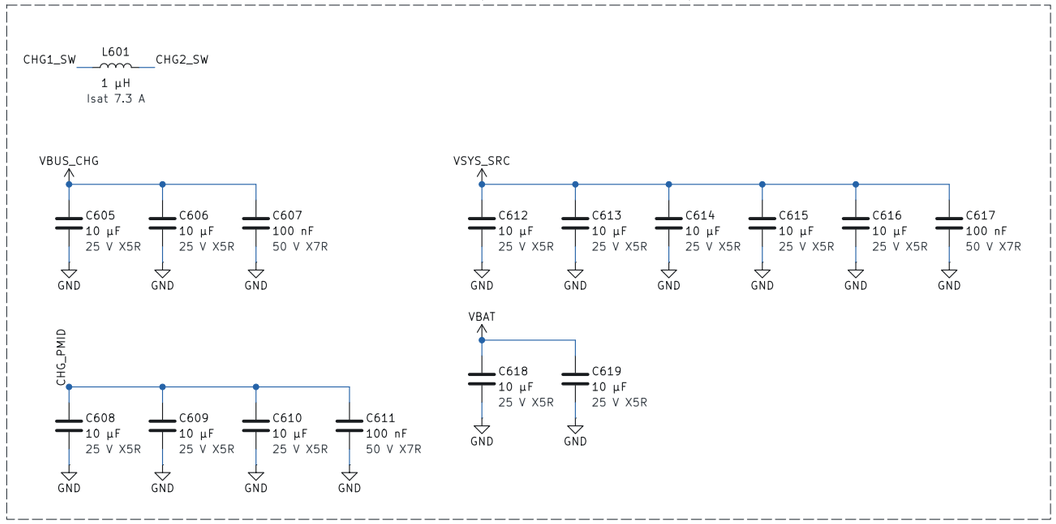
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/Charger and battery path/Charger: input mux/

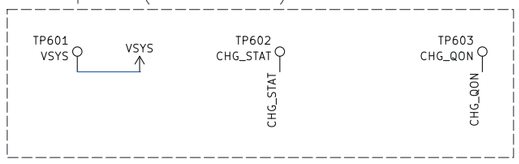
Charger: BQ25798, 1.5 MHz, 2S (BOM v1 lines 39, 115)



Inductor, input and output capacitors (SLUSDV2C §8.2.2)



Test pads (PCB-T-02)



{IO_IRQ}

{I2C1}

Interface

Net	Direction	Absent state (block not fitted)
VBUS_CHG	Supply in	From the TPS2121 (blk.charger-input)
VSYS_SRC	Supply out	SYS output to the VSYS INA236 shunt (blk.current-monitors); never omitted
VBAT	Passive	BAT node to the ship FET (blk.charger-battery)
CHG_SDRV, CHG_BATP, BAT_NTC	Passive	Ship FET gate, pack-side sense, pack NTC (blk.charger-battery)
I2C1_SCL, I2C1_SDA	Bidirectional	BQ25798 at 0x6B; pull-ups on blk.soc-pmuio
CHG_INT_N	Output	Open drain, 256 µs pulses; pull-up on blk.soc-pmuio

Rails

Rail	Capacitors	Requirement and source
VBUS_CHG	2 x 10 µF 25 V + 100 nF	SLUSDV2C Table 5-1, §8.2.2.3
CHG_PMID	3 x 10 µF 25 V + 100 nF	SLUSDV2C Table 5-1 (no backup mode)
VSYS_SRC	5 x 10 µF + 100 nF (25 V parts)	SLUSDV2C §8.2.2.4: 16 V or higher at 2S
VBAT	2 x 10 µF	SLUSDV2C Table 5-1
CHG_REGN	4.7 µF 10 V	SLUSDV2C Table 5-1

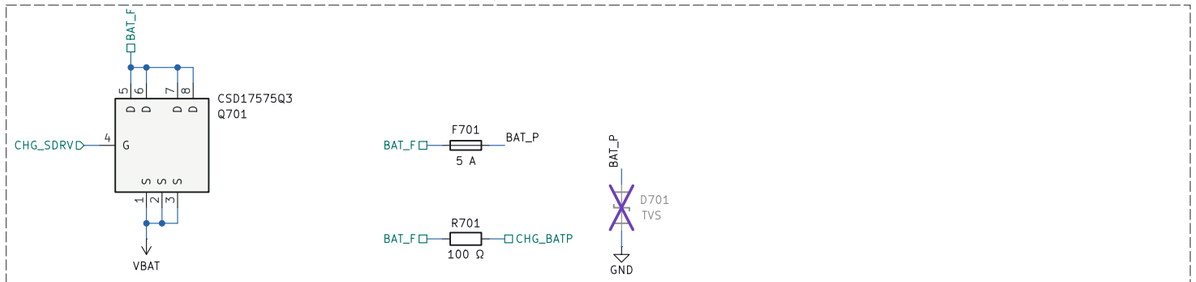
Notes
1. Function: BQ25798 buck-boost charger for the 2S pack; VSYS 6.0-8.8 V.
2. Absent state: never omitted. Unpowered without an adapter; the pack feeds VSYS through BATFET.
3. PROG 6.04 kΩ (1.5 MHz, 2S, Table 7-1): ILIM_HIZ to REGN (input limit by IINDPM, BOM v1.11); TS: 5.23 kΩ / 30.1 kΩ for a 103AT at 0-60 °C (§7.3.9.5.1), pack NTC type UNVERIFIED.
4. VAC1, VAC2 on VBUS, ACDRV to GND (the TPS2121 selects the input); D+/D- open (BC1.2 not used, line 39); CE pulled low (charging by EN_CHG); QON and STAT on test pads; STAT 10 kΩ to REGN.

Title: **Charger: BQ25798**				Block: blk.charger-core
Purpose: 2S buck-boost charger, its inductor and capacitors; VSYS out				
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 6 / 69

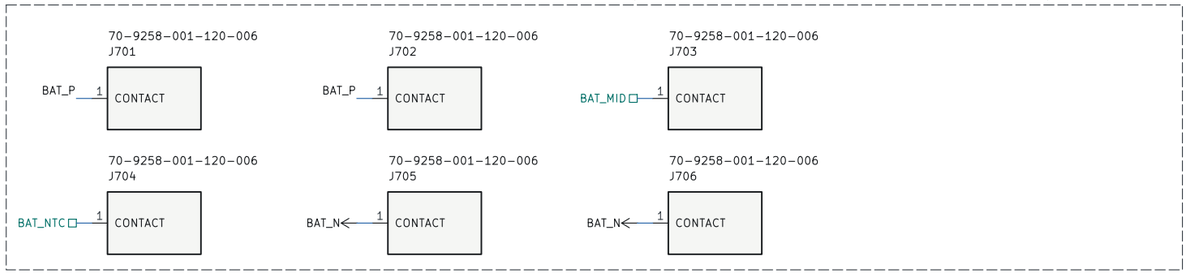
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/Charger and battery path/Charger: BQ25798/

Battery path: ship FET, 5 A fuse, BATP (ADR 0014 item 6, ADR 0017, ADR 0020 item 6)



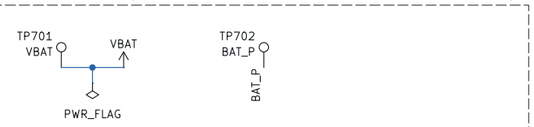
Battery contacts, face compression, 3.5 mm pitch (ADR 0020 item 2; BOM v1 line 47)



Contact ESD, within 3 mm (BOM v1 line 142)



Test pads (PCB-T-02; CB-07, PCB-L-09)



Interface

Net	Direction	Absent state (block not fitted)
VBAT	Passive	BQ25798 BAT node (blk.charger-core)
CHG_SDRV, CHG_BATP	Passive	Ship FET gate drive and BATP sense (blk.charger-core)
BAT_NTC	Passive	Pack NTC contact to the BQ25798 TS divider
BAT_F, BAT_MID, BAT_N	Passive	To blk.gauge: VC2/PBI after the fuse, VC1, cell negative

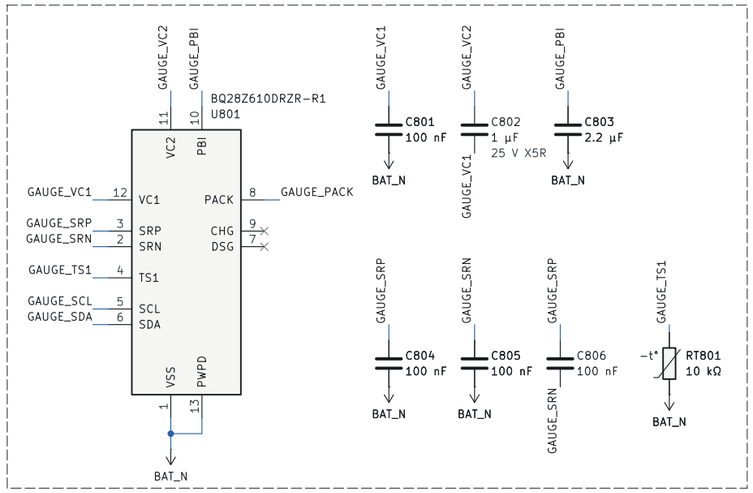
Notes
1. Function: the ship FET (BQ25798 SDRV) and the 5 A fuse join the BAT node to the six battery contacts (ADR 0014 item 6, ADR 0020).
2. Absent state: never omitted. Ship mode: SDRV low, the FET off, the pack disconnected.
3. Pack side of the ship FET: only the gauge (VC2, PBI, VC1, sense), BATP (100 Ω, 47 nF) and the FET (ADR 0017). P+ TVS position DNP until CB-07 and CB-10 ask for it (line 142).
4. Open: the Mill-Max 0965 pogo fallback land (line 47); bring-up terminal board (ADR 0020 item 9).

Title	**Charger: battery path**
Block	blk.charger-battery
Purpose	Ship FET, 5 A fuse, BATP and the six battery contacts with their ESD
Project, board	OpenDAP T1, main board
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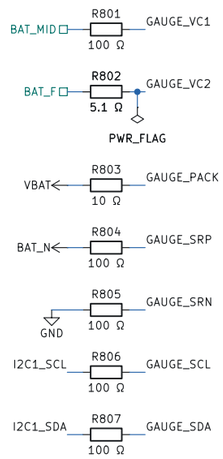
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/Charger and battery path/Charger: battery path/

Gauge: BQ28Z610-R1, gauge only (ADR 0014, ADR 0017)



Inputs (SLUSE07B Figure 9-1)



Sense resistor, low side, Kelvin (ADR 0017 item 4)



{I2C1}

Interface

Net	Direction	Absent state (block not fitted)
BAT_F	Passive	Pack side of the ship FET, after the 5 A fuse (blk.charger)
BAT_MID	Passive	Mid-cell tap contact (blk.charger)
BAT_N	Passive	P− contacts: cell negative, the gauge's VSS
VBAT	Passive	BQ25798 BAT node: PACK senses the switched side (ADR 0017 item 2)
I2C1_SCL, I2C1_SDA	Bidirectional	0x55; pull-ups on blk.soc-pmuio

Rails

Rail	Capacitors	Requirement and source
GAUGE_VC2	1 µF to VC1 (5 Ω in)	SLUSE07B Figure 9-1
GAUGE_PBI	2.2 µF	SLUSE07B §7.3, Figure 9-1

Notes
1. Function: BQ28Z610-R1 reads both cells (VC2 after the fuse, VC1 on the tap) and counts charge on a 3 mΩ low-side sense resistor; the pack PCM protects and balances (ADR 0014, ADR 0017).
2. Absent state: never omitted. Storage: Blackbird OS sends MAC 0x0010 twice, then the BQ25798 enters shutdown mode; the ship FET opens, PACK falls and the gauge shuts down (ADR 0017 item 3).
3. VSS, PWPD and every gauge capacitor return to BAT_N, the cell side of the sense resistor; SRP to the cell side, SRN to GND, each through 100 Ω, 100 nF at the pins (SLUSE07B Figure 9-1).
4. CHG and DSG open (no FETs: SLUSE07B §8.3.14; TI E2E 903283, forum guidance).
5. TS1: board NTC NCP15XH103F03RC against the pocket floor (BOM v1 line 48).
6. Golden image, CC Gain calibration per board: ADR 0017 items 4–5.

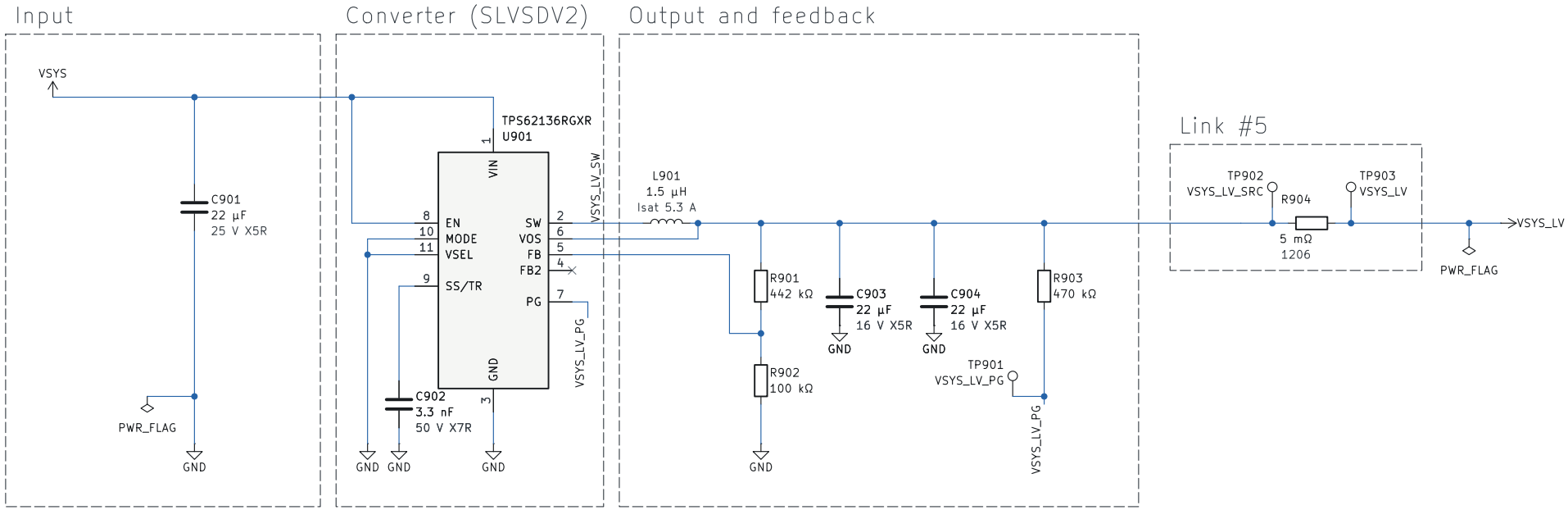
Title	Block
Fuel gauge	blk.gauge

Purpose: BQ28Z610-R1 on the pack side, its 3 mΩ sense resistor and board NTC (ADR 0017)

Project, board	Variant	Rev	Date	Sheet
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/Fuel gauge/



Interface

Net	Direction	Absent state (block not fitted)
VSYS	Supply in	Not applicable: the block is never omitted (BRD−049)
VSYS_LV	Supply out	Not applicable: always on; feeds the PMIC, vdd_cpu and vcc3v3_sys

Rails

Rail	Capacitors	Requirement and source
VSYS	22 µF 25 V 0805	CIN 10 µF nominal (SLVSDV2 §10.1.4.2); 3 µF effective minimum (§7.3): 3.9 µF at 8.8 V (BOM v1.18 item 1)
VSYS_LV	2 x 22 µF 16 V 0805	COUT 47 µF nominal with 1.5 µH (SLVSDV2 Table 2)

Notes
1. Function: TI TPS62136RGXR, VSYS (6.0−8.8 V, 2S) to VSYS_LV 3.8 V, 4 A. Every part on VSYS_LV is rated 5.5 V or less (ADR 0014; BOM v1 line 113).
2. Absent state: not applicable. EN tied to VSYS: always on (feature blocks #10).
3. Output: R1 442 kΩ / R2 100 kΩ with VFB 0.7 V: 3.794 V (SLVSDV2 Eq. 8). L: XGL4020−152MEC 1.5 µH (BOM v1 line 114). CSS 3.3 nF, PG 470 kΩ (Table 1).
4. MODE low (auto PFM/PWM), VSEL low (FB2 unused): capture choices, never floating.
5. Link H5 5 mΩ in every variant (BOM line 137); feedback on its source side.
6. PWR_FLAG: VSYS and GND here until blk.charger is drawn; VSYS_LV at the link.
7. Constraints: PCB−P−02, −07, −08. Unless noted: resistors 1 % 0402.

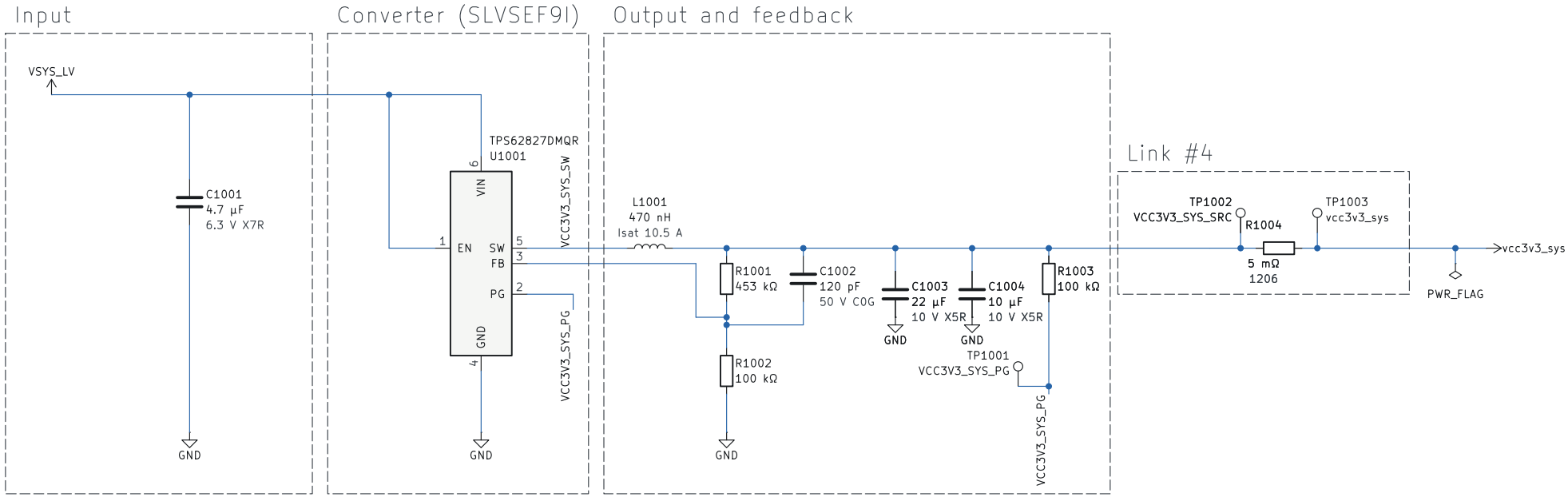
Title
3.8 V pre−regulator

Block
blk.pwr−vsys−lv

Purpose
VSYS (2S battery or charger, 6.0−8.8 V) to the always−on VSYS_LV 3.8 V rail, 4 A

Project, board	Variant	Rev	Date	Sheet
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/3.8 V pre−regulator/



Interface

Net	Direction	Absent state (block not fitted)
VSYS_LV	Supply in	Not applicable: never omitted
vcc3v3_sys	Supply out	Not applicable: always on (EN = VSYS_LV)

Rails

Rail	Capacitors	Requirement and source
VSYS_LV	4.7 µF 6.3 V X7R 0603	C1 4.7 µF (SLVSEF9I Fig. 8–2); >= 3 µF effective (p. 14)
vcc3v3_sys	22 µF + 10 µF 0603	>= 20 µF effective for the TPS62827 (p. 14; BOM D26); DC bias UNVERIFIED

Notes
1. Function: TI TPS62827DMQR, VSYS_LV to vcc3v3_sys 3.3 V, 4 A, always on (BOM v1 line 6): RK809 VCC5–9, eMMC, SoC IO banks, radio switch (power tree §3).
2. Absent state: not applicable. EN tied to VSYS_LV (RK809 EXT_EN unused).
3. Feedback: 453 kΩ / 100 kΩ with VFB 0.6 V: 3.318 V (SLVSEF9I Eq. 4; the 450 kΩ of Eq. 4 is not an E96 value). C3 120 pF = 12 µs / R2 (Eq. 5).
4. L: Coilcraft XGL4015-471MEC 0.47 µH, 1.5 mm (compaction study, 2026-10-27; was Würth 744373240047 of BOM v1 line 106, 1.8 mm, kept as the other-land alternate).
5. Link H5 5 mΩ in every variant (power tree §4.12 #4); feedback on its source side.
6. Unless noted: resistors 1 % 0402; capacitors X5R/X7R, at least twice the rail.

Title
3.3 V system rail

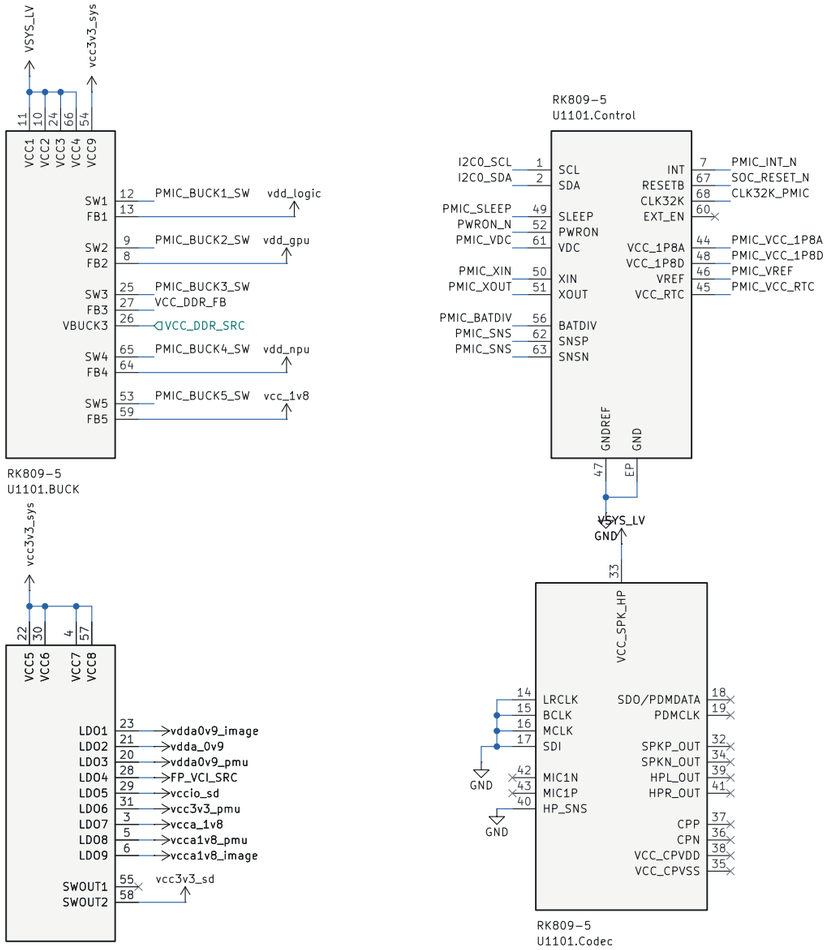
Block
blk.pwr-3v3-sys

Purpose
VSYS_LV to the always-on vcc3v3_sys 3.3 V rail, 4 A

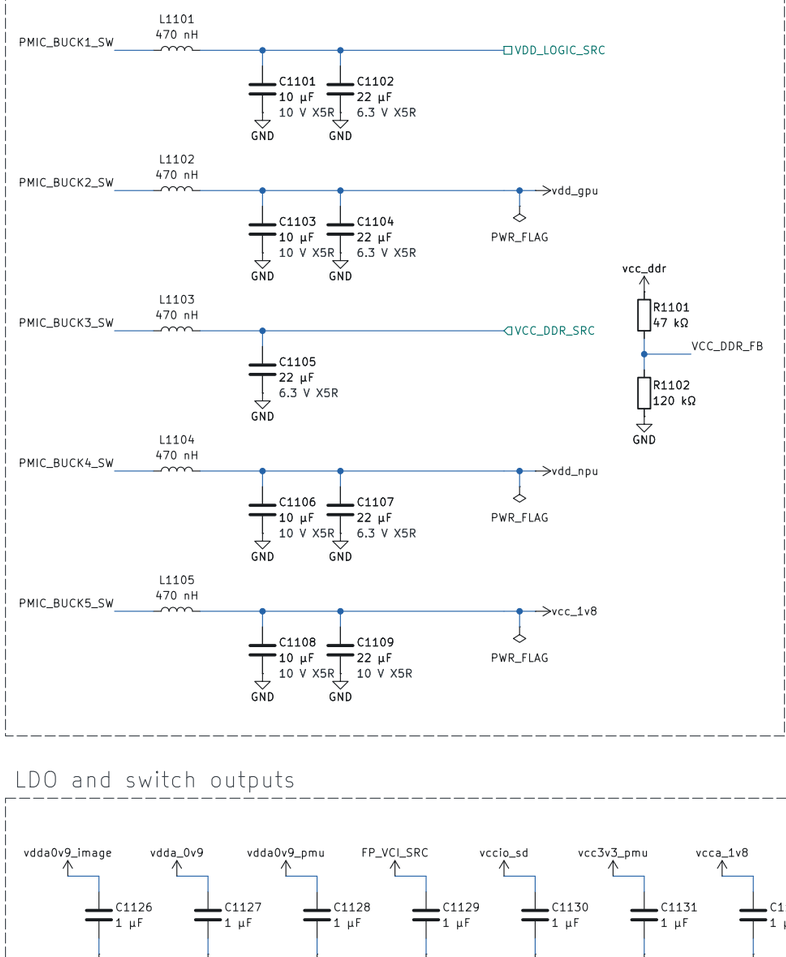
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	10 / 69

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/3.3 V system rail/

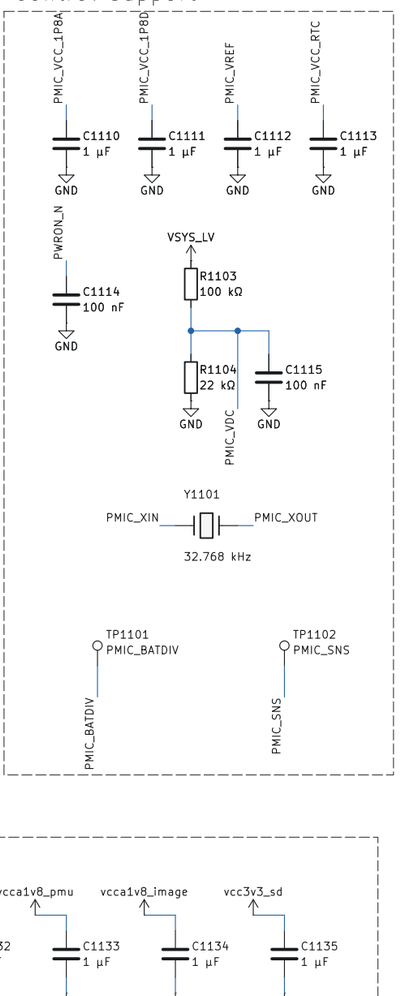
RK809-5



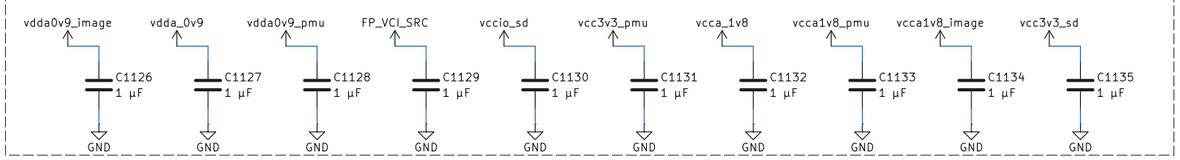
Buck output stages



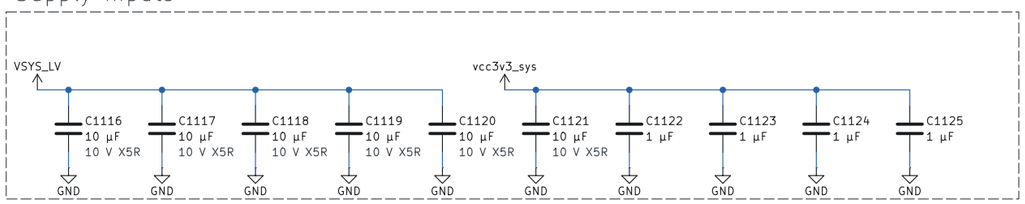
Control support



LDO and switch outputs



Supply inputs



{I2C0} {KEYS} {PMIC_CTL}

Interface

Net	Direction	Absent state (block not fitted)
I2C0_SCL, I2C0_SDA	Bidirectional	Not applicable (never omitted); pull-ups on blk.soc-pmuio
PMIC_INT_N, CLK32K_PMIC, SOC_RESET_N	Output	Open drain; pull-ups on blk.soc-pmuio
PMIC_SLEEP	Input	SoC ball pull-down at reset (pinout v1)
VDD_LOGIC_SRC, VCC_DDR_SRC	Supply out	BUCK1 and BUCK3 outputs, to their links

Rails

Rail	Capacitors	Requirement and source
VSYS_LV	5 x 10 µF	VCC1-VCC4 10 µF each, VCC_SPK 10 µF (RK809 DS Fig. 1-1)
vcc3v3_sys	10 µF + 4 x 1 µF	VCC9 10 µF; VCC5-VCC8 1 µF each (Fig. 1-1)
Buck outputs	10 µF + 22 µF (BUCK3 22 µF)	HDG 1.1 near the source; Fig. 1-1 33 µF / 22 µF
LDO outputs	1 µF each	Fig. 1-1

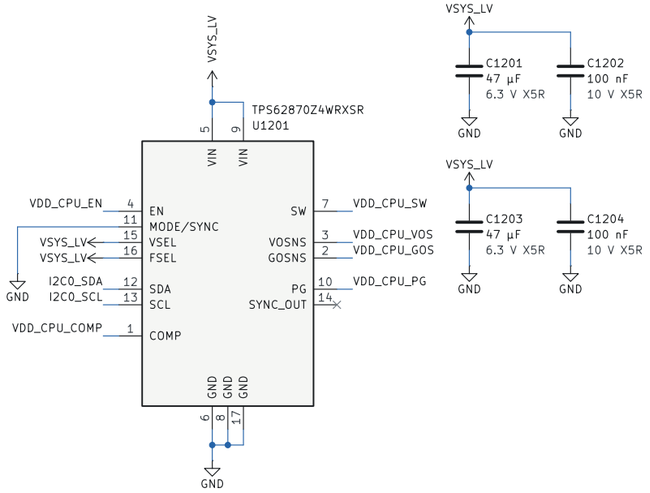
Notes
1. Function: Rockchip RK809-5 (BOM v1 line 2): BUCK1 vdd_logic, BUCK2 vdd_gpu, BUCK3 vcc_ddr, BUCK4 vdd_npu, BUCK5 vcc_1v8; LDO1-9 and SWOUT2 per power tree §3. VCC1-4 from VSYS_LV, VCC5-9 from vcc3v3_sys (ROCK 3C v1.4). SWOUT1 and EXT_EN unused. I2C0 at 0x20.
2. Absent state: not applicable (never omitted). OTP defaults of the -5 part UNVERIFIED.
3. FB1 senses vdd_logic and FB3 the 47k / 120k divider on vcc_ddr, both on the load side of their links (blk.soc-power, blk.lpddr4x); BUCK2, 4, 5 sense their outputs.
4. L: Murata DFE252012F-R47M 0.47 µH (BOM v1 line 5). Output capacitors 10 + 22 µF (HDG near the source); BUCK3 22 µF (Fig. 1-1). Fig. 1-1's 33 µF is met as 10 + 22 µF.
5. VDC: 100k / 22k from VSYS_LV with 100 nF as Fig. 1-1 from VCC_SYS: 0.69 V > 0.55 V, so the PMIC starts when VSYS_LV first rises (bring-up confirms).
6. Codec unused: MCLK, BCLK, LRCLK, SDI, HP_SNS to GND; outputs and MIC1 open (BOM line 2, HDG p. 73). PDMCLK is an output (DS p. 14): left open, not grounded as line 2 lists it. VCC_SPK_HP from VSYS_LV with 10 µF (Fig. 1-1).
7. Open: BATDIV and SNSP/SNSN (joined, Fig. 1-1) on test pads until the ROCK 3C sheet 28 wiring is read (UNVERIFIED); 32.768 kHz load capacitors (22 pF optional in Fig. 1-1) not fitted with the 7 pF SC-32S (UNVERIFIED); PWRON_N to blk.buttons and blk.maskrom.
8. Groups follow RK809 DS Fig. 1-1 and power tree §3 (BOM v1 line 5): LDO outputs 1 µF each. Constraints: PCB-P-07, PCB-P-08 (PMIC_BUCKn_SW). Unless noted: resistors 1 % 0402.

Title	PMIC
Block	blk.pmic
Purpose	RK809-5: five bucks, nine LDOs and a switch for the RK3566 domains; RTC, power key, reset
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026-10-25
Sheet	11 / 69

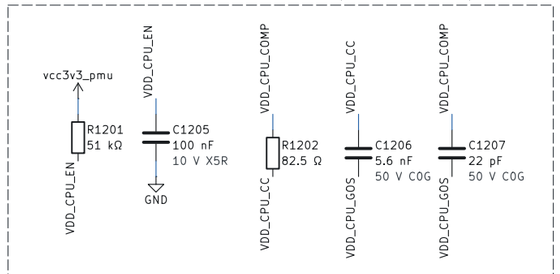
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/PMIC/

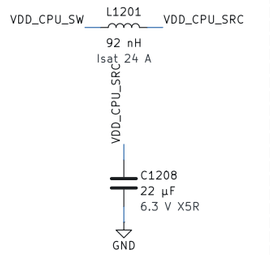
TPS62870Z4 at I2C0 0x42 (ADR 0026 item 1)



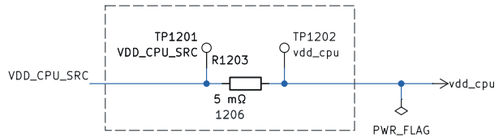
Enable and compensation (items 1, 4)



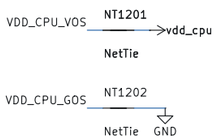
Output (items 2, 4)



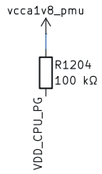
Link #1



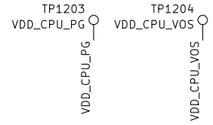
Remote sense at the SoC capacitor (item 5)



Power good pull-up (item 1)



Test pads



{I2C0}

Interface

Net	Direction	Absent state (block not fitted)
I2C0_SDA, I2C0_SCL	Bidirectional	Not applicable (never omitted); pull-ups on blk.soc-pmuio
VSYS_LV	Supply in	Not applicable
vdd_cpu	Supply out	Not applicable

Rails

Rail	Capacitors	Requirement and source
VSYS_LV	47 µF X5R 0603 + 100 nF at each VIN pin	ADR 0026 item 3: >= 5 µF effective per pin (SLVSGC5E p. 9): 8.65 µF
VDD_CPU_SRC	22 µF 0603	ADR 0026 item 4: 3 x 22 µF + 2 x 4.7 µF at the SoC (load >= 2 x device)

Notes
1. Function: TI TPS62870Z4WRXSR, 6 A, I2C0 0x42, +-1 % over TJ -40..125 C, differential remote sense, from VSYS_LV (ADR 0026). VSEL, FSEL to VIN: 0.875 V start-up, 3 MHz.
2. Absent state: not applicable (never omitted). EN from vcc3v3_pmu through 51 kΩ (>= 15 kΩ, SLVSGC5E p. 6) and 100 nF: vdd_cpu follows VCC3V3_PMU (HDG p. 34): on in suspend.
3. MODE/SYNC to GND: power save by pin; forced PWM through FPWMEN (driver mode FAST). SYNC_OUT open (single device). PG open drain: monitor only, never a reset.
4. Compensation COMP to GOSNS (never PGND): RZ 82.5 Ω, CC 5.6 nF C0G, CC2 22 pF (item 4). Output 45.7 µF effective at 1.1 V with the SoC set: 4.8 A release peaks at 1.149 V.
5. Sense (item 5): VOSNS and GOSNS as a differential pair from the 22 µF at the SoC, load side of link #1 (net ties), away from VDD_CPU_SW (SLVSGC5E pp. 53-54).
6. BRD-049 safety limit: devicetree regulator-max-microvolt = <1100000> (no 1800 MHz / 1150 mV point) until bring-up shows DC top + load-step peak <= 1.170 V at the SoC. The converter accepts up to 3.35 V over I2C: the kernel and U-Boot limits are the ceiling.
7. L XEL4012-920NEC 92 nH on the 4020-family land (UNVERIFIED). Link #1 H5 inside the sense loop. Constraints: PCB-P-07, -08. Unless noted: resistors 1 % 0402.

Title
CPU core supply

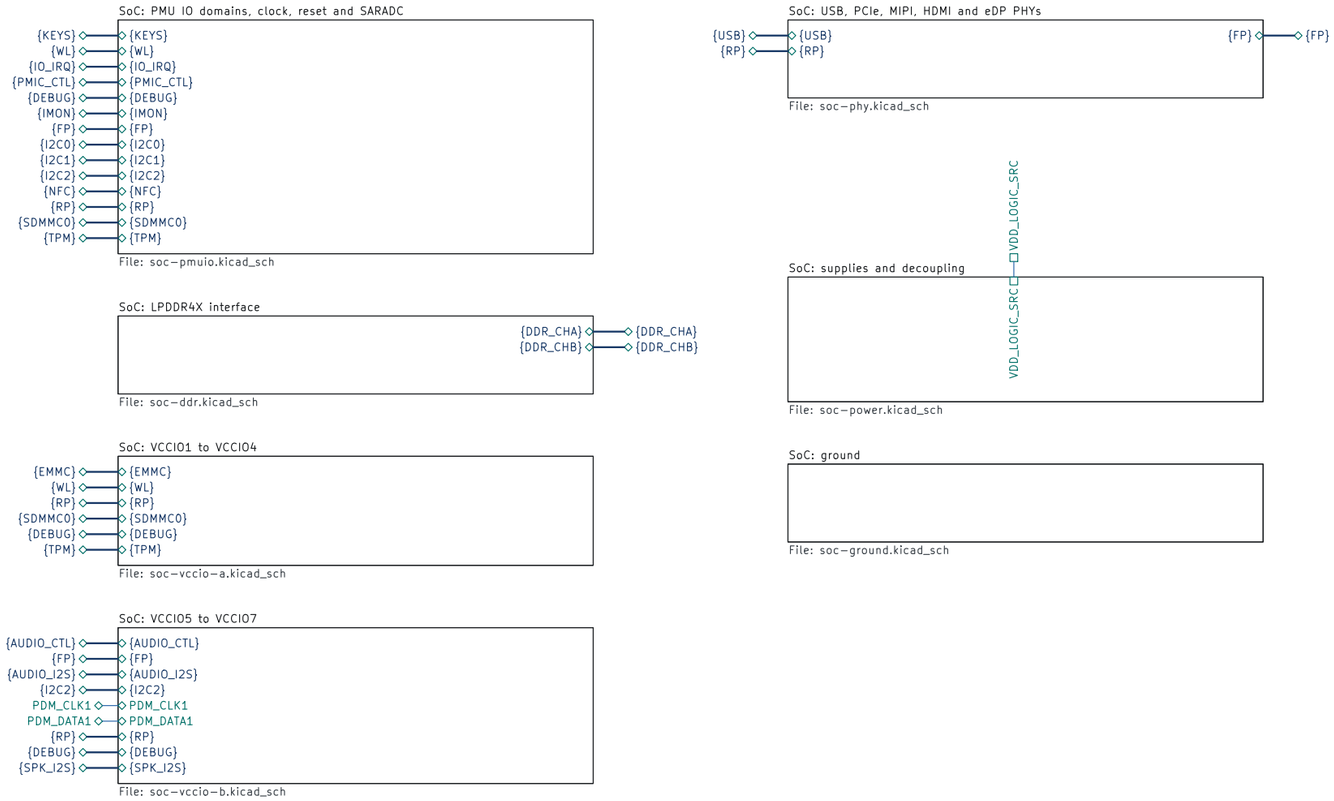
Block
blk.vdd-cpu

Purpose
vdd_cpu 0.85-1.10 V (cap) for the RK3566 CPU cluster, I2C DVS on I2C0 at 0x42, 6 A

Project, board	Variant	Rev	Date	Sheet
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/CPU core supply/



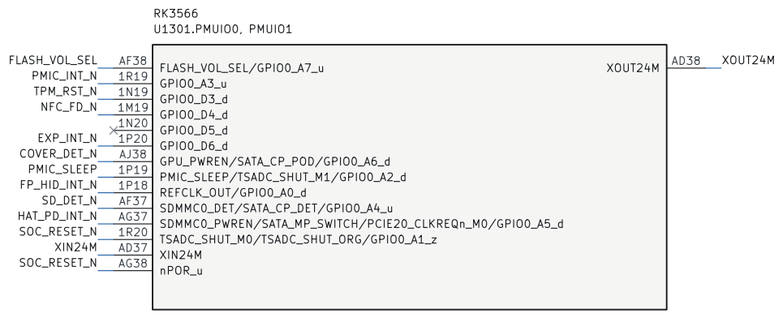
Notes
1. Function: the RK3566 SoC (board ADR 0003), drawn as one sub−sheet per IO domain or function (CRAFT−126): each sub−sheet holds the RK3566 units of that domain and their support parts.
2. Absent state: not applicable. No variant omits the SoC; another SoC is another board (RFC 0011 §3).
3. Sub−sheet IDs (blk.soc−*) are sheets of blk.soc: their parts carry Block = blk.soc−* so the craft check can resolve them; the catalog block is blk.soc (feature−blocks spec #1).
4. Constraints: PCB−I−05 (LPDDR4X), PCB−I−06 (eMMC); decoupling per RK3566 HDG 1.1 pp. 35−41.
5. Decisions: ADR 0003 (RK3566), ADR 0026 (vdd_cpu), ADR 0022 (single−rank LPDDR4X).

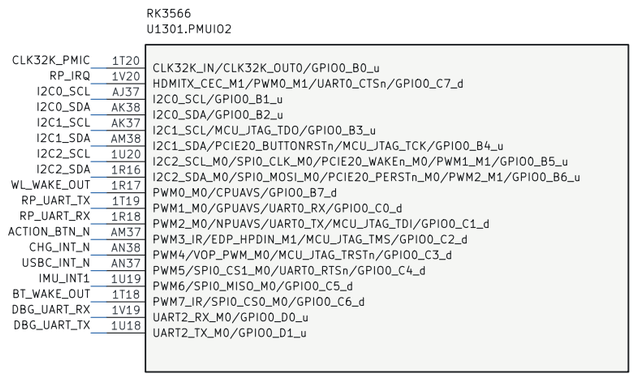
Title	Block
SoC	blk.soc

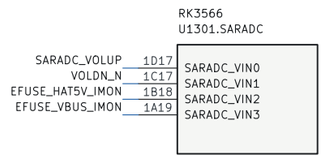
Purpose: The RK3566 application processor, one sub−sheet per IO domain or function

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026−10−25	13 / 69

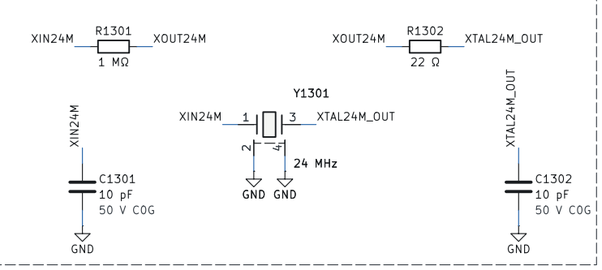
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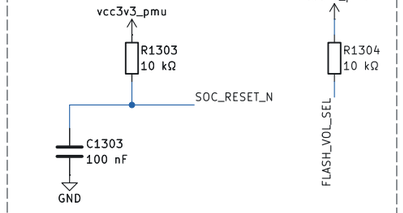




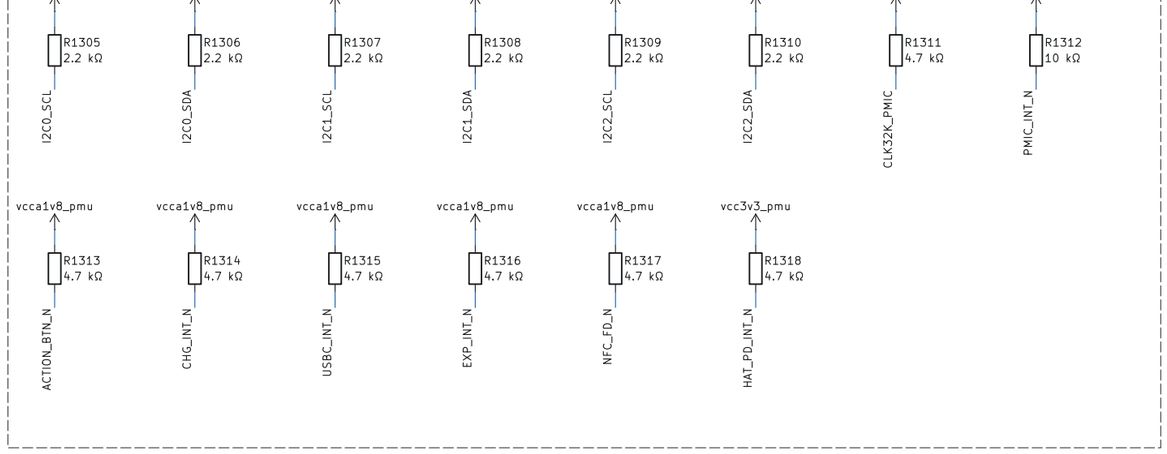
24 MHz crystal (HDG 1.1 pp. 16–17)



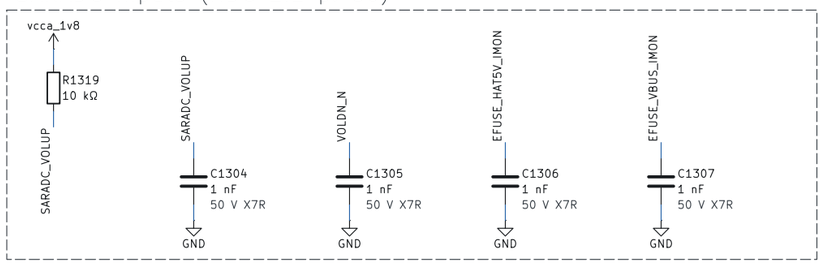
Reset and boot strap



Bus and interrupt pull-ups (host side)



SARADC inputs (HDG 1.1 p. 60)



{NFC} {PMIC_CTL} {RP} {SDMMC0} {TPM} {WL}

{DEBUG} {FP} {I2C0} {I2C1} {I2C2} {IMON} {IO_IRQ} {KEYS}

Rails

Rail	Capacitors	Requirement and source
vcca_1v8	4 x 1 nF on VIN0–VIN3	1 nF per input; 10 kΩ on VIN0 (HDG 1.1 p. 60)

Notes
1. Function: RK3566 PMUIO0 (crystal), PMUIO1 (3.3 V fixed: reset, PMIC and HAT port interrupts), PMUIO2 (1.8 V: I2C0–2, wake inputs, debug UART) and the SARADC (pinout v1 §1, §3).
2. Absent state: not applicable (the SoC is never omitted). The pull-ups here hold every open-drain interrupt and bus of the blocks they serve, so those blocks may be unfitted.
3. Crystal NX3225GA-24MHZ-STD-CRG-2, CL 8 pF (NDK 2008E): C = 2 (CL − 3 pF stray) = 10 pF C0G; 1 MΩ and 22 Ω per HDG 1.1 pp. 16–17. Bring-up trims C against the 24 MHz error.
4. Pull-ups: I2C 2.2 kΩ (BOM v1 §5); interrupts 4.7 kΩ (pinout v1 B9); PMIC_INT_N 10 kΩ (RK809 DS Fig. 1–1); SOC_RESET_N and FLASH_VOL_SEL 10 kΩ: capture choice, HDG gives none.
5. A net with one pin by design ends in a local label (ERC warning isolated_pin_label).
6. Unless noted: resistors 1 % 0402; capacitors 0402 X7R (100 nF on rails 0201 X5R 10 V), C0G where marked.

Title	**SoC: PMU IO domains, clock, reset and SARADC**
Block	blk.soc-pmuio
Purpose	RK3566 units on PMUIO0–2 and the SARADC, with the 24 MHz crystal, reset, straps and bus pull-ups
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026-10-25
Sheet	14 / 69

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/SoC/SoC: PMU IO domains, clock, reset and SARADC/

RK3566
U1301.DDR channel A

DDR_DM0_A	1E2	DDR_DM0_A/DDR4_DML_A/LPDDR4_DM0_A/DDR3_DM0/LPDDR3_DM1	
DDR_DM1_A	1G1	DDR_DM1_A/DDR4_DMU_A/LPDDR4_DM1_A/DDR3_DM1/LPDDR3_DM3	
DDR_DQ0_A	G2	DDR_DQ0_A/DDR4_DQL0_A/LPDDR4_DQ0_A/DDR3_DQ0/LPDDR3_DQ15	
DDR_DQ1_A	F2	DDR_DQ1_A/DDR4_DQL2_A/LPDDR4_DQ1_A/DDR3_DQ1/LPDDR3_DQ14	
DDR_DQ2_A	F1	DDR_DQ2_A/DDR4_DQL4_A/LPDDR4_DQ2_A/DDR3_DQ2/LPDDR3_DQ10	
DDR_DQ3_A	E1	DDR_DQ3_A/DDR4_DQL6_A/LPDDR4_DQ3_A/DDR3_DQ3/LPDDR3_DQ9	
DDR_DQ4_A	M2	DDR_DQ4_A/DDR4_DQL7_A/LPDDR4_DQ4_A/DDR3_DQ4/LPDDR3_DQ13	
DDR_DQ5_A	K1	DDR_DQ5_A/DDR4_DQL5_A/LPDDR4_DQ5_A/DDR3_DQ5/LPDDR3_DQ12	
DDR_DQ6_A	K2	DDR_DQ6_A/DDR4_DQL3_A/LPDDR4_DQ6_A/DDR3_DQ6/LPDDR3_DQ8	
DDR_DQ7_A	1E1	DDR_DQ7_A/DDR4_DQL1_A/LPDDR4_DQ7_A/DDR3_DQ7/LPDDR3_DQ11	
DDR_DQ8_A	P2	DDR_DQ8_A/DDR4_DQU3_A/LPDDR4_DQ8_A/DDR3_DQ8/LPDDR3_DQ25	
DDR_DQ9_A	R1	DDR_DQ9_A/DDR4_DQU1_A/LPDDR4_DQ9_A/DDR3_DQ9/LPDDR3_DQ24	
DDR_DQ10_A	R2	DDR_DQ10_A/DDR4_DQU7_A/LPDDR4_DQ10_A/DDR3_DQ10/LPDDR3_DQ28	
DDR_DQ11_A	T2	DDR_DQ11_A/DDR4_DQU5_A/LPDDR4_DQ11_A/DDR3_DQ11/LPDDR3_DQ29	
DDR_DQ12_A	M1	DDR_DQ12_A/DDR4_DQU2_A/LPDDR4_DQ12_A/DDR3_DQ12/LPDDR3_DQ26	
DDR_DQ13_A	1F2	DDR_DQ13_A/DDR4_DQU4_A/LPDDR4_DQ13_A/DDR3_DQ13/LPDDR3_DQ31	
DDR_DQ14_A	1F1	DDR_DQ14_A/DDR4_DQU6_A/LPDDR4_DQ14_A/DDR3_DQ14/LPDDR3_DQ30	
DDR_DQ15_A	1G2	DDR_DQ15_A/DDR4_DQU0_A/LPDDR4_DQ15_A/DDR3_DQ15/LPDDR3_DQ27	
DDR_DQS0N_A	H1	DDR_DQS0N_A/DDR4_DQSL_N_A/LPDDR4_DQS0N_A/DDR3_DQS0N/LPDDR3_DQS1N	
DDR_DQS0P_A	H2	DDR_DQS0P_A/DDR4_DQSL_P_A/LPDDR4_DQS0P_A/DDR3_DQS0P/LPDDR3_DQS1P	
DDR_DQS1N_A	N2	DDR_DQS1N_A/DDR4_DQSU_N_A/LPDDR4_DQS1N_A/DDR3_DQS1N/LPDDR3_DQS3N	
DDR_DQS1P_A	N1	DDR_DQS1P_A/DDR4_DQSU_P_A/LPDDR4_DQS1P_A/DDR3_DQS1P/LPDDR3_DQS3P	
		DDR4_A2/LPDDR4_A1_A/DDR3_A4/LPDDR3_A6/AC2	C1 DDR_A1_A
		DDR4_A3/LPDDR4_CKE1_A/DDR3_A3/-/AC3	1C3
		DDR4_A8/LPDDR4_ODT0_CA_A/DDR3_A6/LPDDR3_A9/AC8	E2 DDR_ODT0_CA_A
		DDR4_A11/LPDDR4_A0_A/DDR3_A7/LPDDR3_A8/AC11	C2 DDR_A0_A
		DDR4_A12/LPDDR4_A3_A/DDR3_BA2/-/AC12	1B3 DDR_A3_A
		DDR4_A14_WEn/LPDDR4_A4_A/DDR3_A15/LPDDR3_A5/AC14	1B1 DDR_A4_A
		DDR4_A15_CASn/LPDDR4_A2_A/DDR3_A0/-/AC15	1C1 DDR_A2_A
		DDR4_A16_RASn/LPDDR4_A5_A/DDR3_RASn/LPDDR3_A7/AC16	1A1 DDR_A5_A
		DDR4_BG1/LPDDR4_ODT1_CA_A/DDR3_BA0/-/AC21	1C2
		DDR4_CKE/LPDDR4_CKE0_A/DDR3_CKE/LPDDR3_CKE/AC22	B4 DDR_CKE0_A
		DDR4_CLKN/LPDDR4_CLKN_A/DDR3_CLKN/LPDDR3_CLKN/AC24	B5 DDR_CLKN_A
		DDR4_CLKP/LPDDR4_CLKP_A/DDR3_CLKP/LPDDR3_CLKP/AC23	A5 DDR_CLKP_A
		DDR4_CS0n/LPDDR4_CS0n_A/DDR3_ODT1/LPDDR3_ODT0/AC25	B3 DDR_CS0_A_N
		DDR4_CS1n/LPDDR4_CS1n_A/DDR3_CS1n/LPDDR3_ODT1/AC26	A3

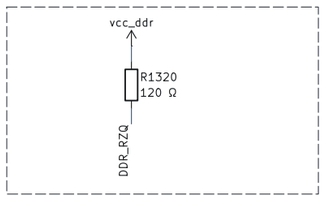
RK3566
U1301.DDR channel B

DDR_DM0_B	1B8	DDR_DM0_B/DDR4_DMU_B/LPDDR4_DM0_B/DDR3_DM2/LPDDR3_DM0	
DDR_DM1_B	1A11	DDR_DM1_B/DDR4_DML_B/LPDDR4_DM1_B/DDR3_DM3/LPDDR3_DM2	
DDR_DQ0_B	B12	DDR_DQ0_B/DDR4_DQU7_B/LPDDR4_DQ0_B/DDR3_DQ16/LPDDR3_DQ1	
DDR_DQ1_B	B10	DDR_DQ1_B/DDR4_DQU5_B/LPDDR4_DQ1_B/DDR3_DQ17/LPDDR3_DQ5	
DDR_DQ2_B	1A8	DDR_DQ2_B/DDR4_DQU3_B/LPDDR4_DQ2_B/DDR3_DQ18/LPDDR3_DQ6	
DDR_DQ3_B	1A7	DDR_DQ3_B/DDR4_DQU1_B/LPDDR4_DQ3_B/DDR3_DQ19/LPDDR3_DQ4	
DDR_DQ4_B	B13	DDR_DQ4_B/DDR4_DQU0_B/LPDDR4_DQ4_B/DDR3_DQ20/LPDDR3_DQ2	
DDR_DQ5_B	1A10	DDR_DQ5_B/DDR4_DQU6_B/LPDDR4_DQ5_B/DDR3_DQ21/LPDDR3_DQ3	
DDR_DQ6_B	1B9	DDR_DQ6_B/DDR4_DQU4_B/LPDDR4_DQ6_B/DDR3_DQ22/LPDDR3_DQ7	
DDR_DQ7_B	1A9	DDR_DQ7_B/DDR4_DQU2_B/LPDDR4_DQ7_B/DDR3_DQ23/LPDDR3_DQ0	
DDR_DQ8_B	B19	DDR_DQ8_B/DDR4_DQL0_B/LPDDR4_DQ8_B/DDR3_DQ24/LPDDR3_DQ18	
DDR_DQ9_B	A19	DDR_DQ9_B/DDR4_DQL2_B/LPDDR4_DQ9_B/DDR3_DQ25/LPDDR3_DQ19	
DDR_DQ10_B	A20	DDR_DQ10_B/DDR4_DQL4_B/LPDDR4_DQ10_B/DDR3_DQ26/LPDDR3_DQ22	
DDR_DQ11_B	B20	DDR_DQ11_B/DDR4_DQL6_B/LPDDR4_DQ11_B/DDR3_DQ27/LPDDR3_DQ23	
DDR_DQ12_B	A15	DDR_DQ12_B/DDR4_DQL7_B/LPDDR4_DQ12_B/DDR3_DQ28/LPDDR3_DQ16	
DDR_DQ13_B	B15	DDR_DQ13_B/DDR4_DQL5_B/LPDDR4_DQ13_B/DDR3_DQ29/LPDDR3_DQ17	
DDR_DQ14_B	1C10	DDR_DQ14_B/DDR4_DQL1_B/LPDDR4_DQ14_B/DDR3_DQ30/LPDDR3_DQ20	
DDR_DQ15_B	B16	DDR_DQ15_B/DDR4_DQL3_B/LPDDR4_DQ15_B/DDR3_DQ31/LPDDR3_DQ21	
DDR_DQS0N_B	A12	DDR_DQS0N_B/DDR4_DQSU_N_B/LPDDR4_DQS0N_B/DDR3_DQS2N/LPDDR3_DQS0N	
DDR_DQS0P_B	A13	DDR_DQS0P_B/DDR4_DQSU_P_B/LPDDR4_DQS0P_B/DDR3_DQS2P/LPDDR3_DQS0P	
DDR_DQS1N_B	B17	DDR_DQS1N_B/DDR4_DQSL_N_B/LPDDR4_DQS1N_B/DDR3_DQS3N/LPDDR3_DQS2N	
DDR_DQS1P_B	A17	DDR_DQS1P_B/DDR4_DQSL_P_B/LPDDR4_DQS1P_B/DDR3_DQS3P/LPDDR3_DQS2P	
		DDR4_A0/LPDDR4_CLKP_B/DDR3_A9/-/AC0	A7 DDR_CLKP_B
		DDR4_A4/LPDDR4_A3_B/DDR3_BA1/LPDDR3_A3/AC4	1B6 DDR_A3_B
		DDR4_A5/LPDDR4_A5_B/DDR3_A11/LPDDR3_A2/AC5	B8 DDR_A5_B
		DDR4_A6/LPDDR4_A1_B/DDR3_A13/LPDDR3_A1/AC6	B9 DDR_A1_B
		DDR4_A7/LPDDR4_ODT0_CA_B/DDR3_A8/-/AC7	A10 DDR_ODT0_CA_B
		DDR4_A9/LPDDR4_CLKN_B/DDR3_A5/-/AC9	B7 DDR_CLKN_B
		DDR4_A10/LPDDR4_CKE0_B/DDR3_A10/-/AC10	1A3 DDR_CKE0_B
		DDR4_A13/LPDDR4_A0_B/DDR3_A14/LPDDR3_A0/AC13	A9 DDR_A0_B
		DDR4_ACTn/LPDDR4_CKE1_B/DDR3_CASn/-/AC17	1A2
		DDR4_BA0/LPDDR4_A2_B/DDR3_A1/-/AC18	1B7 DDR_A2_B
		DDR4_BA1/LPDDR4_A4_B/DDR3_A12/LPDDR3_A4/AC19	1A6 DDR_A4_B
		DDR4_BG0/LPDDR4_ODT1_CA_B/DDR3_WEn/-/AC20	1A4
		DDR4_ODT0/LPDDR4_CS1n_B/DDR3_ODT0/LPDDR3_CS1n/AC27	1A5
		DDR4_ODT1/LPDDR4_CS0n_B/DDR3_CS0n/LPDDR3_CS0n/AC28	1B5 DDR_CS0_B_N

RK3566
U1301.DDR common

	DDR4_A1/-/DDR3_A2/-/AC1	1D2
	DDR4_RESETn/LPDDR4_RESETn/DDR3_RESETn/AC29	1C7 DDR_RESET_N
	DDR_RZQ	1F3 DDR_RZQ
	DDR_VREFOUT	1H3

DDR calibration



{DDR_CHB}

{DDR_CHA}

Rails

Rail	Capacitors	Requirement and source
vcc_ddr	R (ZQ) only	Decoupling of DDR_VDDQ is on blk.soc-power

Notes
1. Function: RK3566 DDR PHY, channels A and B and the common unit (DS 1.1 Table 2-1).
2. Absent state: not applicable (blk.lpddr4x is never omitted).
3. Every LPDDR4X ball goes to the same-named DRAM ball in the Rockchip template order, which may not be swapped (HDG 1.1 p. 22; pinout v1 CSV). DDR_RZQ: 120 Ω 1 % to vcc_ddr (HDG p. 25).
4. Rank-1 outputs (CS1, CKE1, ODT1_CA of both channels) take a no-connect flag: the DRAM is single rank (board ADR 0022 item 4). DDR_VREFOUT is not used with LPDDR4X (HDG p. 25).
5. Constraints: PCB-I-05 (net classes DDR_50SE, DDR_100D).
6. Net names as pinout v1: active low ends in _N, upper case (CRAFT-84).

Title
SoC: LPDDR4X interface

Block
blk.soc-ddr

Purpose
RK3566 DDR PHY units: channels A and B, reset and ZQ calibration

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	15 / 69

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/SoC/SoC: LPDDR4X interface/

RK3566
U1301.VCCIO1

SPARE_1A1	B22	I2C3_SCL_M0/UART3_TX_M0/AUDIOPWM_LOUT_N/GPIO1_A1_u
SPARE_1A0	A22	I2C3_SDA_M0/UART3_RX_M0/AUDIOPWM_LOUT_P/GPIO1_A0_u
SPARE_1A6	1A14	I2S1_LRCK_RX_M0/UART4_TX_M0/PDM_CLK0_M0/AUDIOPWM_ROUT_P/GPIO1_A6_d
SPARE_1A5	B24	I2S1_LRCK_TX_M0/UART4_RTSn_M0/SCR_RST/GPIO1_A5_d
RP_PWR_EN	A23	I2S1_MCLK_M0/UART3_RTSn_M0/SCR_CLK/GPIO1_A2_d
SPARE_1A4	1A13	I2S1_SCLK_RX_M0/UART4_RX_M0/PDM_CLK1_M0/SPDIF_TX_M0/GPIO1_A4_d
SPARE_1A3	B23	I2S1_SCLK_TX_M0/UART3_CTSn_M0/SCR_IO/GPIO1_A3_d
SPARE_1B3	1B14	I2S1_SDI0_M0/PDM_SDI0_M0/GPIO1_B3_d
SPARE_1A7	B25	I2S1_SDO0_M0/UART4_CTSn_M0/SCR_DET/AUDIOPWM_ROUT_N/GPIO1_A7_d
	1B13	I2S1_SDO1_M0/I2S1_SDI3_M0/PDM_SDI3_M0/PCIE20_CLKREQn_M2/GPIO1_B0_d
	A26	I2S1_SDO2_M0/I2S1_SDI2_M0/PDM_SDI2_M0/PCIE20_WAKEn_M2/GPIO1_B1_d
	B26	I2S1_SDO3_M0/I2S1_SDI1_M0/PDM_SDI1_M0/PCIE20_PERSTn_M2/GPIO1_B2_d

RK3566
U1301.VCCIO2

EMMC_CLK	A29	EMMC_CLKOUT/FLASH_DQS/GPIO1_C5_d
EMMC_CMD	A27	EMMC_CMD/FLASH_WRn/GPIO1_C4_u
EMMC_D0	A32	EMMC_D0/FLASH_D0/GPIO1_B4_u
EMMC_D1	B27	EMMC_D1/FLASH_D1/GPIO1_B5_u
EMMC_D2	B32	EMMC_D2/FLASH_D2/GPIO1_B6_u
EMMC_D3	B29	EMMC_D3/FLASH_D3/GPIO1_B7_u
EMMC_D4	B33	EMMC_D4/FLASH_D4/GPIO1_C0_u
EMMC_D5	A30	EMMC_D5/FLASH_D5/GPIO1_C1_u
EMMC_D6	B30	EMMC_D6/FLASH_D6/GPIO1_C2_u
EMMC_D7	A33	EMMC_D7/FLASH_D7/GPIO1_C3_u
EMMC_DS	1A16	EMMC_DATA_STROBE/FSPI_CS1n/FLASH_CLE/GPIO1_C6_d
EMMC_RST_N	1B16	EMMC_RSTn/FSPI_D2/FLASH_WPn/GPIO1_C7_d
SPARE_1D0	1A15	FSPI_CLK/FLASH_ALE/GPIO1_D0_d
SPARE_1D3	1B17	FSPI_CS0n/FLASH_CS0n/GPIO1_D3_u
SPARE_1D1	1A17	FSPI_D0/FLASH_RDY/GPIO1_D1_u
TPM_PIRQ_N	1A18	FSPI_D1/FLASH_RDn/GPIO1_D2_u
	1C15	FSPI_D3/FLASH_CS1n/GPIO1_D4_u

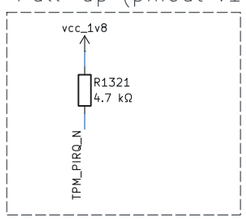
RK3566
U1301.VCCIO3

SD_CLK	G38	SDMMC0_CLK/TEST_CLKOUT/UART5_TX_M0/GPIO2_A2_d
SD_CMD	1E19	SDMMC0_CMD/PWM10_M1/UART5_RX_M0/GPIO2_A1_u
SD_D0	1E20	SDMMC0_D0/UART2_TX_M1/UART6_TX_M1/PWM8_M1/GPIO1_D5_u
SD_D1	1F19	SDMMC0_D1/UART2_RX_M1/UART6_RX_M1/PWM9_M1/GPIO1_D6_u
SD_D2	1D20	SDMMC0_D2/ARM_JTAG_TCK/UART5_CTSn_M0/GPIO1_D7_u
SD_D3	1F18	SDMMC0_D3/ARM_JTAG_TMS/UART5_RTSn_M0/GPIO2_A0_u

RK3566
U1301.VCCIO4

	B38	CLK32K_OUT1/UART8_RX_M0/SPI1_CS1_M0/GPIO2_C6_d
	A37	I2S2_LRCK_RX_M0/UART6_CTSn_M0/SPI1_CS0_M0/GPIO2_C0_d
	1D19	I2S2_LRCK_TX_M0/UART9_RTSn_M0/SPI2_MOSI_M0/GPIO2_C3_d
	B36	I2S2_MCLK_M0/ETH0_REFCLKO_25M/UART7_RTSn_M0/SPI2_CLK_M0/GPIO2_C1_d
SPARE_2B7	D38	I2S2_SCLK_RX_M0/UART6_RTSn_M0/SPI1_MOSI_M0/GPIO2_B7_d
	C37	I2S2_SCLK_TX_M0/UART7_CTSn_M0/SPI2_MISO_M0/GPIO2_C2_d
BT_WAKE_IN	D37	I2S2_SDI_M0/UART8_TX_M0/SPI2_CS1_M0/GPIO2_C5_d
	1B20	I2S2_SDO_M0/UART9_CTSn_M0/SPI2_CS0_M0/GPIO2_C4_d
WL_SD_CLK	B37	SDMMC1_CLK/UART9_TX_M0/GPIO2_B0_d
WL_SD_CMD	B35	SDMMC1_CMD/UART9_RX_M0/GPIO2_A7_u
WL_SD_D0	1C19	SDMMC1_D0/UART6_RX_M0/GPIO2_A3_u
WL_SD_D1	1E18	SDMMC1_D1/UART6_TX_M0/GPIO2_A4_u
WL_SD_D2	1D18	SDMMC1_D2/UART7_RX_M0/GPIO2_A5_u
WL_SD_D3	1A20	SDMMC1_D3/UART7_TX_M0/GPIO2_A6_u
SPARE_2B2	B34	SDMMC1_DET/I2C4_SCL_M1/UART8_CTSn_M0/GPIO2_B2_u
SPARE_2B1	A35	SDMMC1_PWREN/I2C4_SDA_M1/UART8_RTSn_M0/GPIO2_B1_d
BT_UART_RTS	G37	UART1_CTSn_M0/SPI1_MISO_M0/GPIO2_B6_u
BT_UART_CTS	F37	UART1_RTSn_M0/SPI1_CLK_M0/GPIO2_B5_u
BT_UART_TXD	1C20	UART1_RX_M0/GPIO2_B3_u
BT_UART_RXD	F38	UART1_TX_M0/GPIO2_B4_u

Pull-up (pinout v1 B9)



{WL}
{TPM}
{SDMMC0}
{RP}
{EMMC}
{DEBUG}

Notes
1. Function: RK3566 IO banks VCCIO1 (3.3 V), VCCIO2 (1.8 V, eMMC), VCCIO3 (vccio_sd, microSD) and VCCIO4 (1.8 V, radio SDIO and UART) (pinout v1 §1).
2. Absent state: not applicable (the SoC is never omitted). Balls with no function take a no-connect flag; unused-pin handling per RK3566_PinOut is open (pinout v1 O2).
3. Series parts of HDG 1.1 (22 Ω on SDMMC0, eMMC CLK) sit on the sheets of their blocks.
4. A net with one pin by design ends in a local label (ERC warning isolated_pin_label).

Title: **SoC: VCCIO1 to VCCIO4**				Block: blk.soc-vccio-a
Purpose: RK3566 IO banks VCCIO1-4: eMMC, microSD, the radio SDIO and UART, and spare GPIO				
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 16 / 69

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/SoC/SoC: VCCIO1 to VCCIO4/

RK3566
U1301.VCCIO5

SPARE_3B7	AP5	PWM12_M0/GMAC1_TXEN_M0/UART3_TX_M1/PDM_SDI2_M2/GPIO3_B7_d
SPARE_3C0	AR4	PWM13_M0/GMAC1_MCLKINOUT_M0/UART3_RX_M1/PDM_SDI3_M2/GPIO3_C0_d
SPARE_3C4	1P3	PWM14_M0/VOP_PWM_M1/GMAC1_MDC_M0/UART7_TX_M1/PDM_CLK1_M2/GPIO3_C4_d
SPARE_3C5	1P4	PWM15_IR_M0/SPDIF_TX_M1/GMAC1_MDIO_M0/UART7_RX_M1/I2S1_LRCK_RX_M2/GPIO3_C5_d
SPK_SD_MODE	1U4	VOP_BT1120_CLK/GMAC1_TXCLK_M0/I2S3_SDI_M0/SDMMC2_CLK_M1/GPIO3_A6_d
FP_SPI_CS_N	1U5	VOP_BT1120_D0/SPI1_CS0_M1/SDMMC2_D0_M1/GPIO3_A1_d
FP_DBI_DC	AR10	VOP_BT1120_D1/GMAC1_TXD2_M0/I2S3_MCLK_M0/SDMMC2_D1_M1/GPIO3_A2_d
SPK_BCLK	AP10	VOP_BT1120_D2/GMAC1_TXD3_M0/I2S3_SCLK_M0/SDMMC2_D2_M1/GPIO3_A3_d
SPK_LRCK	AR9	VOP_BT1120_D3/GMAC1_RXD2_M0/I2S3_LRCK_M0/SDMMC2_D3_M1/GPIO3_A4_d
SPK_SDIN	AP9	VOP_BT1120_D4/GMAC1_RXD3_M0/I2S3_SDO_M0/SDMMC2_CMD_M1/GPIO3_A5_d
FP_LCD_RST_N	1V3	VOP_BT1120_D5/GMAC1_RXCLK_M0/SDMMC2_DET_M1/GPIO3_A7_d
FP_LCD_TE	AR7	VOP_BT1120_D6/ETH1_REFCLKO_25M_M0/SDMMC2_PWREN_M1/GPIO3_B0_d
BL_PWM	AP7	VOP_BT1120_D7/GMAC1_RXD0_M0/UART4_RX_M1/PWM8_M0/GPIO3_B1_d
FP_ALS_INT_N	AR6	VOP_BT1120_D8/GMAC1_RXD1_M0/UART4_TX_M1/PWM9_M0/GPIO3_B2_d
FP_TP_INT	AP6	VOP_BT1120_D9/GMAC1_RXDV_CRS_M0/I2C5_SCL_M0/PDM_SDI0_M2/GPIO3_B3_d
FP_TP_RST_N	1U3	VOP_BT1120_D10/GMAC1_RXER_M0/I2C5_SDA_M0/PDM_SDI1_M2/GPIO3_B4_d
FP_I2C_SCL	1T4	VOP_BT1120_D11/GMAC1_TXD0_M0/I2C3_SCL_M1/PWM10_M0/GPIO3_B5_d
FP_I2C_SDA	1V2	VOP_BT1120_D12/GMAC1_TXD1_M0/I2C3_SDA_M1/PWM11_IR_M0/GPIO3_B6_d
FP_SPI_MOSI	AP4	VOP_BT1120_D13/SPI1_MOSI_M1/PCIE20_PERSTn_M1/I2S1_SDO2_M2/GPIO3_C1_d
FP_SPI_MISO	AP3	VOP_BT1120_D14/SPI1_MISO_M1/UART5_TX_M1/I2S1_SDO3_M2/GPIO3_C2_d
FP_SPI_SCLK	AR2	VOP_BT1120_D15/SPI1_CLK_M1/UART5_RX_M1/I2S1_SCLK_RX_M2/GPIO3_C3_d

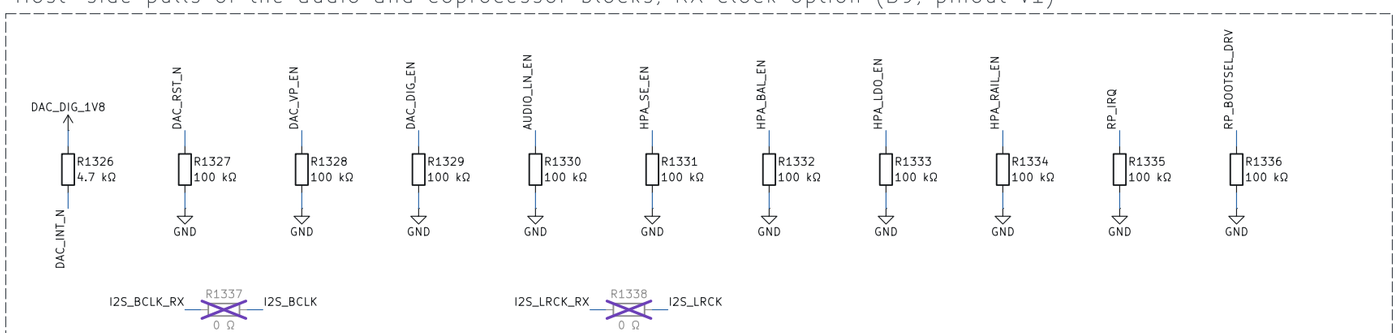
RK3566
U1301.VCCIO6

I2S_LRCK_RX	1N1	CAM_CLKOUT0/EBC_SDCE1/GMAC1_RXD0_M1/SPI3_CS1_M0/I2S1_LRCK_RX_M1/GPIO4_A7_d
DSD_L	1L2	CAM_CLKOUT1/EBC_SDCE2/GMAC1_RXD1_M1/SPI3_MISO_M0/I2S1_SDO1_M1/GPIO4_B0_d
RP_RUN	AB1	CIF_CLKIN/EBC_SDCLK/GMAC1_MCLKINOUT_M1/UART1_CTSn_M1/I2S2_SCLK_RX_M1/GPIO4_C1_d
MIC_EN	AB2	CIF_CLKOUT/EBC_GDCLK/PWM11_IR_M1/GPIO4_C0_d
RP_BOOTSEL_DRV	1T2	CIF_D0/EBC_SDDO0/SDMMC2_D0_M0/I2S1_MCLK_M1/VOP_BT656_D0_M1/GPIO3_C6_d
I2S_BCLK	1U1	CIF_D1/EBC_SDDO1/SDMMC2_D1_M0/I2S1_SCLK_TX_M1/VOP_BT656_D1_M1/GPIO3_C7_d
I2S_LRCK	AP1	CIF_D2/EBC_SDDO2/SDMMC2_D2_M0/I2S1_LRCK_TX_M1/VOP_BT656_D2_M1/GPIO3_D0_d
I2S_SDO0	AN2	CIF_D3/EBC_SDDO3/SDMMC2_D3_M0/I2S1_SDO0_M1/VOP_BT656_D3_M1/GPIO3_D1_d
	AM2	CIF_D4/EBC_SDDO4/SDMMC2_CMD_M0/I2S1_SDI0_M1/VOP_BT656_D4_M1/GPIO3_D2_d
	AM1	CIF_D5/EBC_SDDO5/SDMMC2_CLK_M0/I2S1_SDI1_M1/VOP_BT656_D5_M1/GPIO3_D3_d
	1R2	CIF_D6/EBC_SDDO6/SDMMC2_DET_M0/I2S1_SDI2_M1/VOP_BT656_D6_M1/GPIO3_D4_d
	1T1	CIF_D7/EBC_SDDO7/SDMMC2_PWREN_M0/I2S1_SDI3_M1/VOP_BT656_D7_M1/GPIO3_D5_d
PDM_CLK0	AL2	CIF_D8/EBC_SDDO8/GMAC1_TXD2_M1/UART1_TX_M1/PDM_CLK0_M1/GPIO3_D6_d
PDM_DATA0	AL1	CIF_D9/EBC_SDDO9/GMAC1_TXD3_M1/UART1_RX_M1/PDM_SDI0_M1/GPIO3_D7_d
PDM_CLK1	AK2	CIF_D10/EBC_SDDO10/GMAC1_TXCLK_M1/PDM_CLK1_M1/GPIO4_A0_d
PDM_DATA1	1P2	CIF_D11/EBC_SDDO11/GMAC1_RXD2_M1/PDM_SDI1_M1/GPIO4_A1_d
DAC_RST_N	1R1	CIF_D12/EBC_SDDO12/GMAC1_RXD3_M1/UART7_TX_M2/PDM_SDI2_M1/GPIO4_A2_d
DAC_INT_N	AJ2	CIF_D13/EBC_SDDO13/GMAC1_RXCLK_M1/UART7_RX_M2/PDM_SDI3_M1/GPIO4_A3_d
RP_SWCLK	1N2	CIF_D14/EBC_SDDO14/GMAC1_TXD0_M1/UART9_TX_M2/I2S2_LRCK_TX_M1/GPIO4_A4_d
RP_SWDIO	AJ1	CIF_D15/EBC_SDDO15/GMAC1_TXD1_M1/UART9_RX_M2/I2S2_LRCK_RX_M1/GPIO4_A5_d
HPA_SE_EN	1L1	CIF_HREF/EBC_SDLE/GMAC1_MDC_M1/UART1_RTSn_M1/I2S2_MCLK_M1/GPIO4_B6_d
HPA_BAL_EN	AC2	CIF_VSYNC/EBC_SDOE/GMAC1_MDIO_M1/I2S2_SCLK_TX_M1/GPIO4_B7_d
	AD1	I2C2_SCL_M1/EBC_SDSHR/I2S1_SDO3_M1/GPIO4_B5_d
AUDIO_LN_EN	AF1	I2C2_SDA_M1/EBC_GDSP/ISP_FLASH_TRIGIN/VOP_BT656_CLK_M1/GPIO4_B4_d
DAC_VP_EN	AE2	I2C4_SCL_M0/EBC_GDOE/ETH1_REFCLKO_25M_M1/SPI3_CLK_M0/I2S2_SDO_M1/GPIO4_B3_d
DAC_DIG_EN	AF2	I2C4_SDA_M0/EBC_VCOM/GMAC1_RXER_M1/SPI3_MOSI_M0/I2S2_SDI_M1/GPIO4_B2_d
I2S_BCLK_RX	AH2	ISP_FLASHTRIGOUT/EBC_SDCE0/GMAC1_TXEN_M1/SPI3_CS0_M0/I2S1_SCLK_RX_M1/GPIO4_A6_d
DSD_R	AG2	ISP_PRELIGHT_TRIG/EBC_SDCE3/GMAC1_RXDV_CRS_M1/I2S1_SDO2_M1/GPIO4_B1_d

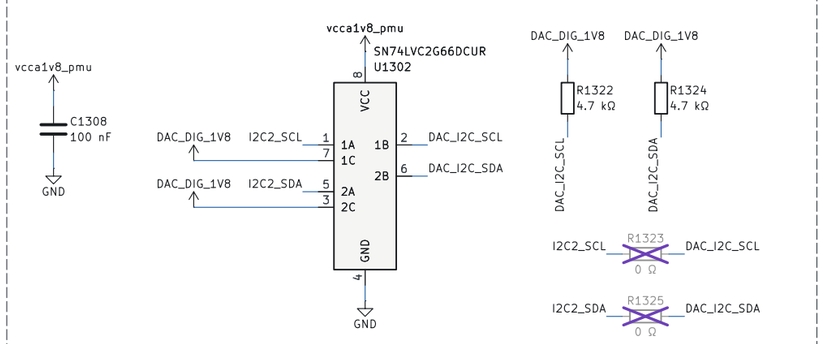
RK3566
U1301.VCCIO7

MCLK_DIV_RUN	1V6	EDP_HPDIN_M0/SPDIF_TX_M2/SATA2_ACT_LED/I2S3_LRCK_M1/GPIO4_C4_d
SPARE_4D1	1V5	HDMITX_CEC_M0/SPI3_CS1_M1/GPIO4_D1_u
SPARE_4C7	AR12	HDMITX_SCL/I2C5_SCL_M1/GPIO4_C7_u
SPARE_4D0	AP11	HDMITX_SDA/I2C5_SDA_M1/GPIO4_D0_u
HPA_RAIL_EN	1U6	PWM12_M1/SPI3_MISO_M1/SATA1_ACT_LED/UART9_TX_M1/I2S3_SDO_M1/GPIO4_C5_d
HPA_LDO_EN	1T6	PWM13_M1/SPI3_CS0_M1/SATA0_ACT_LED/UART9_RX_M1/I2S3_SDI_M1/GPIO4_C6_d
MCLK441_EN	1U7	PWM14_M1/SPI3_CLK_M1/I2S3_MCLK_M1/GPIO4_C2_d
MCLK48_EN	AP12	PWM15_IR_M1/SPI3_MOSI_M1/I2S3_SCLK_M1/GPIO4_C3_d

Host-side pulls of the audio and coprocessor blocks, RX clock option (B9; pinout v1)



DAC I2C isolation (BOM v1 line 133, G2)



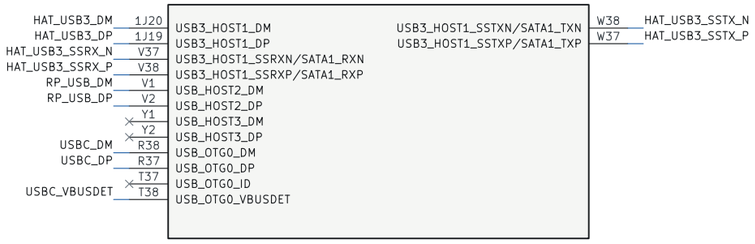
{SPK_I2S}
{RP}
{I2C2}
{FP}
{DEBUG}
{AUDIO_I2S}
{AUDIO_CTL}

Notes
1. Function: RK3566 IO banks VCCIO5 (3.3 V, faceplate), VCCIO6 (vccio6_1v8, audio) and VCCIO7 (3.3 V, audio clock and amplifier enables) (pinout v1 §1). HATs are USB only (RFC 0014): no IO bank carries HAT lines; the freed balls are spare GPIO.
2. Absent state: not applicable (the SoC is never omitted).
3. Audio host side (feature blocks §1): the SN74LVC2G66 isolates I2C2 from both DACs while DAC_DIG_1V8 is off (BOM v1 line 133, G2; 0 Ω bypasses DNP). DAC_INT_N pulled up on DAC_DIG_1V8, every audio enable and DAC_RST_N pulled down (B9, PCB-A-13; 100 kΩ capture choice); DNP 0 Ω RX-clock fallbacks (pinout v1). I2S series parts sit at the DACs. The RP2354A's RP_IRQ and BOOTSEL gate pull-downs sit here too, so an unfitted coprocessor leaves both low (feature blocks #46).
4. A net with one pin by design ends in a local label (ERC warning isolated_pin_label).

Title: SoC: VCCIO5 to VCCIO7					Block: blk.soc-vccio-b
Purpose: RK3566 IO banks VCCIO5-7: faceplate and audio lines					
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 17 / 69	

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/SoC/SoC: VCCIO5 to VCCIO7/

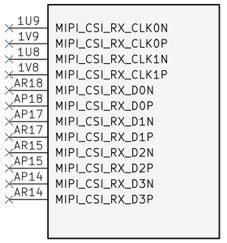
RK3566
U1301.USB



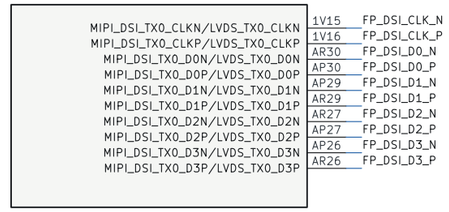
RK3566
U1301.Multi-PHY (PCIe, USB 3, SATA)



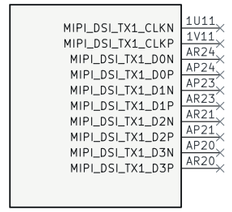
RK3566
U1301.MIPI CSI RX



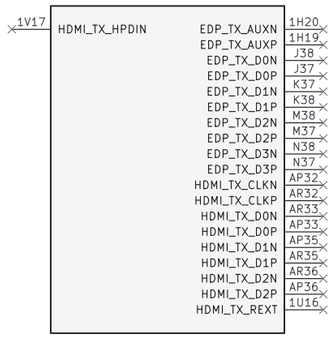
RK3566
U1301.MIPI DSI TX0, LVDS



RK3566
U1301.MIPI DSI TX1



RK3566
U1301.HDMI, eDP



{USB}
{RP}
{FP}

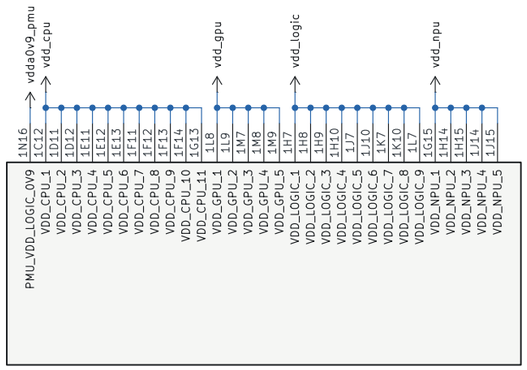
Notes
1. Function: RK3566 PHY units: MIPI DSI TX0 (faceplate), DSI TX1, HDMI and eDP; DSI1, HDMI and eDP are unused in the T1 (pinout v1 O2). Units 13 (USB) and 14 (multi-PHY: USB3 HOST1) are placed with blk.usb-c and blk.hat-signals, the blocks that drive their inputs. PCIe and unit 15 (CSI RX) have no consumer (RFC 0014: HATs are USB only); their balls are NC.
2. Absent state: not applicable (the SoC is never omitted).
3. USB 2 series resistors (2.2 Ω, HDG 1.1 p. 59; the RP2354A end keeps its own 27 Ω, DS Table 1678) and the USB 3 SSTX AC capacitors sit on the sheets of blk.usb-c, blk.hat-signals.
4. A net with one pin by design ends in a local label (ERC warning isolated_pin_label).

Title	Block
SoC: USB, PCIe, MIPI, HDMI and eDP PHYs	blk.soc-phy

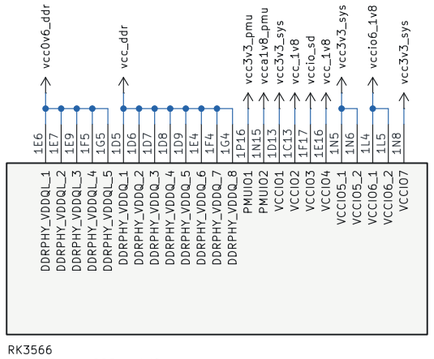
Purpose
RK3566 PHY units: USB 2.0 and USB 3, multi-PHY, MIPI CSI RX, MIPI DSI TX0 and TX1, HDMI and eDP

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	18 / 69

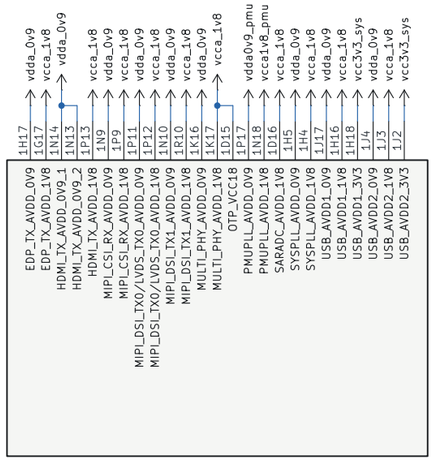
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/SoC/SoC: USB, PCIe, MIPI, HDMI and eDP PHYs/



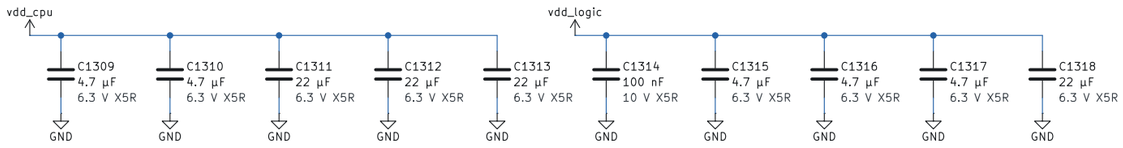
RK3566
U1301.Power: CPU, GPU, NPU, logic

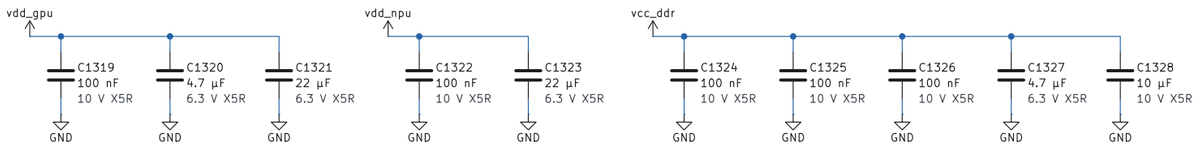


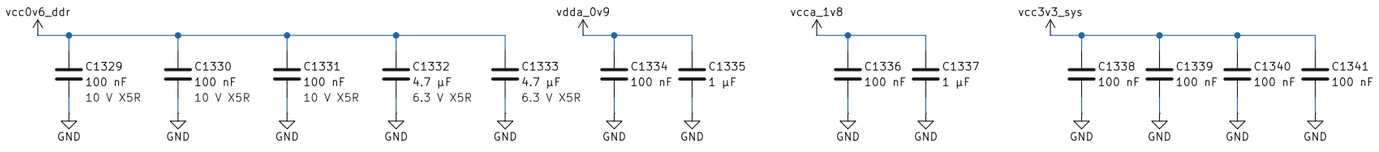
RK3566
U1301.Power: DDR and IO

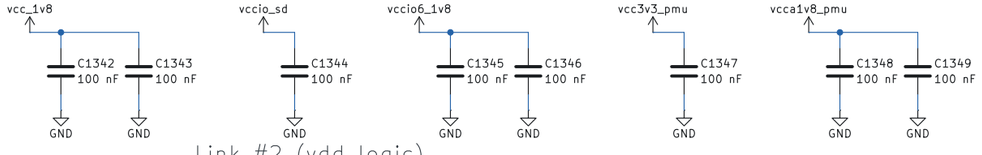


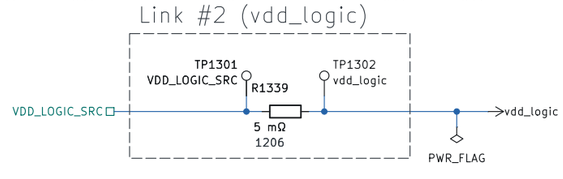
RK3566
U1301.Power: analog

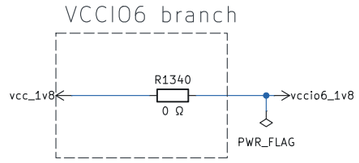












Rails

Rail	Capacitors	Requirement and source
vdd_cpu	C1309 4.7 µF, C1310 4.7 µF, C1311 22 µF, C1312 22 µF, C1313 22 µF	2 x 4.7 µF 0402 + 3 x 22 µF 0603 (HDG Fig. 2–17; ADR 0026 item 4: 35.5 µF effective at 1.1 V, at least twice the converter's, SLVSGC5E p. 54); VOSNS and GOSNS sense at one 22 µF capacitor
vdd_logic	C1314 100 nF, C1315 4.7 µF, C1316 4.7 µF, C1317 4.7 µF, C1318 22 µF	100 nF 0201, 3 x 4.7 µF 0402, 22 µF 0603 (HDG Fig. 2–18)
vdd_gpu	C1319 100 nF, C1320 4.7 µF, C1321 22 µF	100 nF 0201, 4.7 µF 0402, 22 µF 0603 (HDG Fig. 2–19)
vdd_npu	C1322 100 nF, C1323 22 µF	100 nF 0201, 22 µF 0603 (HDG Fig. 2–20)
vcc_ddr	C1324 100 nF, C1325 100 nF, C1326 100 nF, C1327 4.7 µF, C1328 10 µF	DDR_VDDQ: 3 x 100 nF 0201, 4.7 µF 0402, 10 µF 0603, under the SoC (HDG Fig. 2–26)
vcc0v6_ddr	C1329 100 nF, C1330 100 nF, C1331 100 nF, C1332 4.7 µF, C1333 4.7 µF	DDR_VDDQL: 3 x 100 nF 0201, 2 x 4.7 µF 0402, under the SoC (HDG Fig. 2–26)
vdda_0v9	C1334 100 nF, C1335 1 µF	SYSPLL_AVDD_0V9: 1 µF + 100 nF (HDG p. 35)
vcca_1v8	C1336 100 nF, C1337 1 µF	SYSPLL_AVDD_1V8: 1 µF + 100 nF (HDG p. 35)
vcc3v3_sys	C1338 100 nF, C1339 100 nF, C1340 100 nF, C1341 100 nF	VCCIO1, VCCIO5 (2 balls), VCCIO7: 100 nF per supply ball (HDG p. 32)
vcc_1v8	C1342 100 nF, C1343 100 nF	VCCIO2, VCCIO4: 100 nF per supply ball (HDG p. 32)
vccio_sd	C1344 100 nF	VCCIO3: 100 nF (HDG p. 32)
vccio6_1v8	C1345 100 nF, C1346 100 nF	VCCIO6 balls 1L4, 1L5: 100 nF each (HDG p. 32; BOM v1 line 141)
vcc3v3_pmu	C1347 100 nF	PMUIO1: 100 nF at the ball (HDG p. 32)
vcca1v8_pmu	C1348 100 nF, C1349 100 nF	PMUIO2 and PMUPLL_AVDD_1V8 (PMUIO0): 100 nF each (HDG p. 32)

Notes
1. Function: RK3566 supply units and their decoupling, grouped by rail in placement order (smallest value nearest the balls). Values: RK3566 HDG 1.1 pp. 32–41, as in the rail table.
2. Absent state: not applicable (the SoC is never omitted).
3. The capacitors of vdd_cpu, vdd_logic, vdd_gpu, vdd_npu and the DDR rails go on the back, under the SoC, ground pads solid (HDG pp. 36–40, 92–93). None may be removed (HDG).
4. vdd_logic link R: H5 5 mΩ in every variant (BOM v1 line 137); FB1 of the RK809 senses vdd_logic on its load side, so the drop is regulated out (power tree §4.12).
5. PHY analog supplies (vdda_0v9 PHY balls, USB 3.3 V) have no value in the HDG transcription: their decoupling is open (UNVERIFIED) until RK3566_PinOut and HDG Fig. 2–22 are read.
6. Unless noted: capacitors X5R/X7R, rated at least twice the rail.

Title
SoC: supplies and decoupling

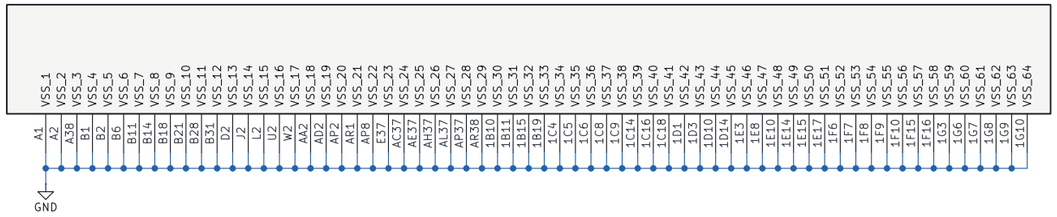
Block
blk.soc-power

Purpose
RK3566 supply units with the decoupling of the hardware design guide, grouped by rail

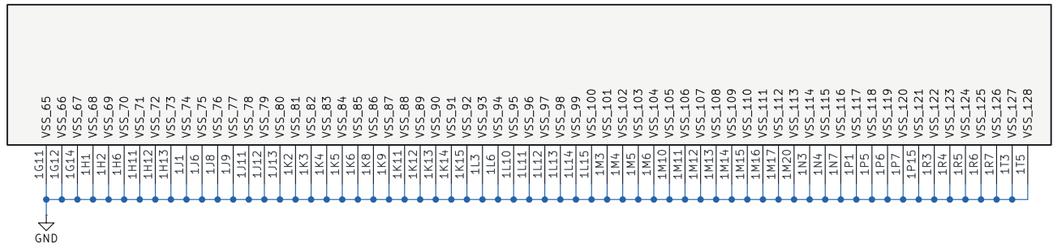
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	19 / 69

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/SoC/SoC: supplies and decoupling/

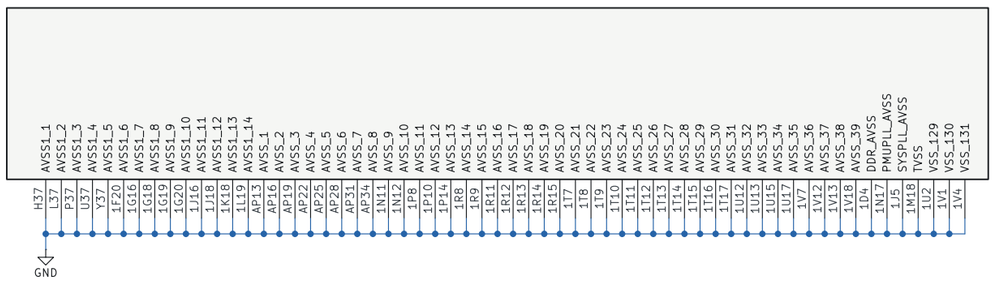
RK3566
U1301.Ground 1



RK3566
U1301.Ground 2



RK3566
U1301.Ground 3



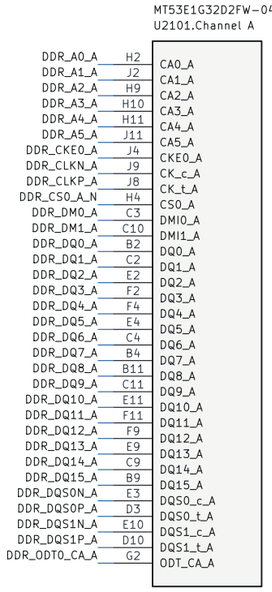
Notes
1. Function: RK3566 ground balls, all on GND.
2. Absent state: not applicable (the SoC is never omitted).
3. Every GND ball has its own via to the adjacent ground plane; balls joined by a # pattern of 0.254 mm traces on the top layer (HDG 1.1 pp. 92–93).

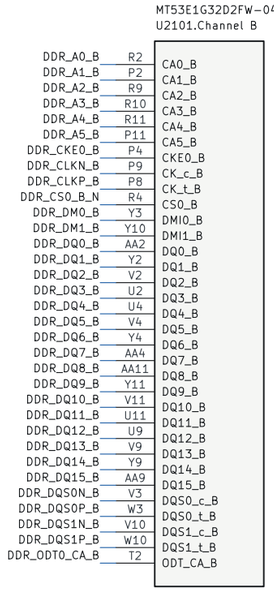
Title	Block
SoC: ground	blk.soc-ground

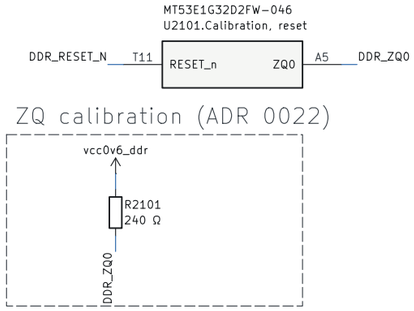
Purpose				
RK3566 ground balls				
Project, board	**Variant**	**Rev**	**Date**	**Sheet**
OpenDAP T1, main board	full	A	2026-10-25	20 / 69

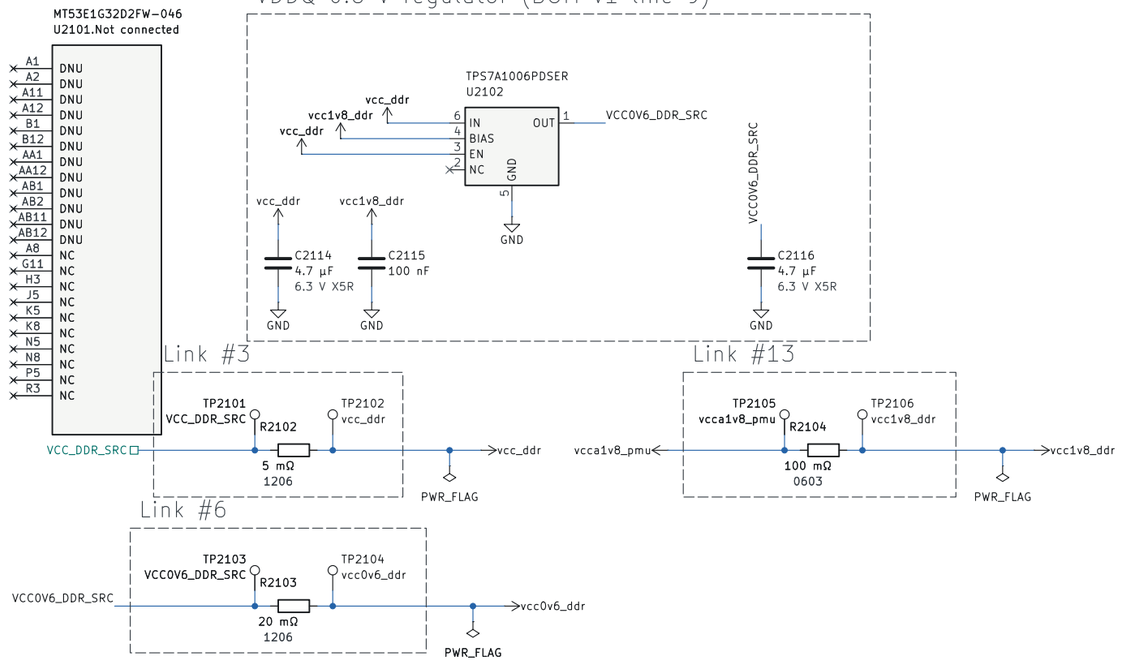
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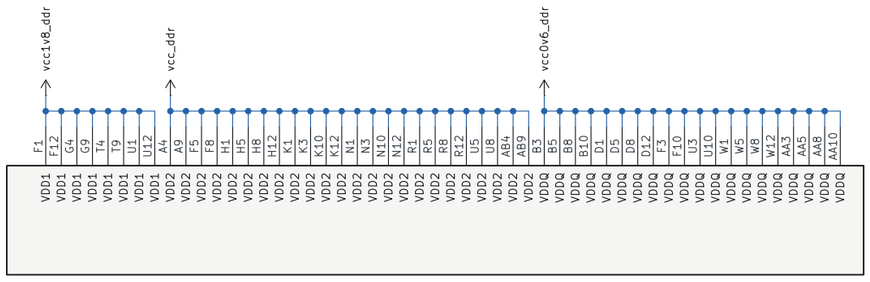
/SoC/SoC: ground/



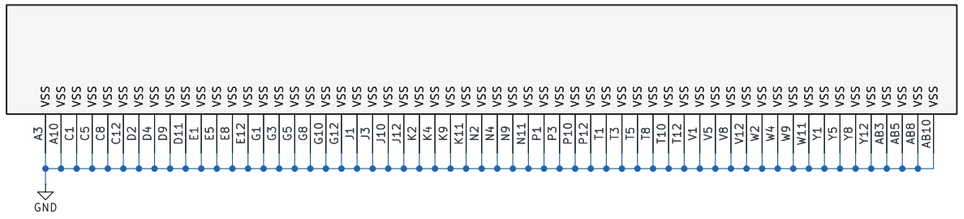




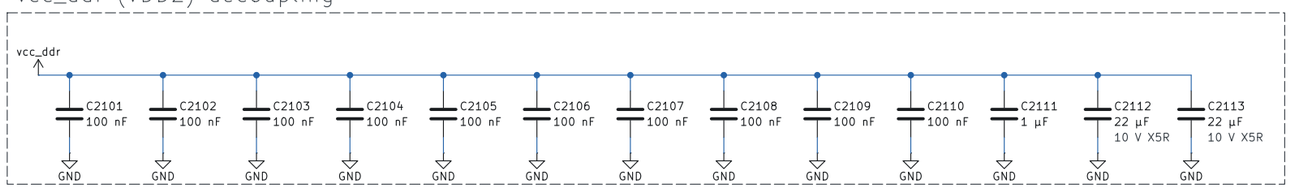




MT53E1G32D2FW-046
U2101.Power



vcc_ddr (VDD2) decoupling



{DDR_CHB}

{DDR_CHA}

Interface

Net	Direction	Absent state (block not fitted)
DDR_*_A, DDR_*_B	SoC to DRAM	Not applicable: blk.lpddr4x is never omitted
DDR_RESET_N	Input	Not applicable
VCC_DDR_SRC	Supply in	RK809 BUCK3 output (blk.pmic), before the link

Rails

Rail	Capacitors	Requirement and source
vcc_ddr	C2101–C2113: 10 x 100 nF, 1 µF, 2 x 22 µF	Radxa ZERO 3W v1.12 reference set (BOM v1 §5)
vcc0v6_ddr	4.7 µF at IN, 100 nF at BIAS, 4.7 µF at OUT	CIN >= 2.2 µF, CBIAS 0.1 µF, COUT 2.2–22 µF; 4.7 µF per the worked example (SBVS314B §8.2.2)
vcc1v8_ddr	none drawn	VDD1 decoupling not given by the DRAM data sheet (UNVERIFIED)

Notes
1. Function: 4 GB LPDDR4X, Micron MT53E1G32D2FW-046 IT:C (WT:C accepted), two 16 Gb dies, one per channel, single rank (board ADR 0022 item 1).
2. Open part, one land: the part choice stays open between Micron and the CXMT CXDB5CBAM-MA-B second source (ADR 0022, DRAM sourcing matrix); it plugs in only through this symbol and the libs kicad/ land, sized for both bodies (10 x 15 mm envelope, 1.1 mm maximum height).
3. Single rank: the RK3566 CS1, CKE1 and ODT1_CA outputs carry no-connect flags (blk.soc-ddr) and ball A8 (ZQ1 on dual-rank parts) has no resistor. A dual-rank part needs a new decision and a symbol change (ADR 0022 item 4).
4. Absent state: not applicable (never omitted; BRD-037). Supplies: VDD1 vcc1v8_ddr (always on), VDD2 vcc_ddr (on in suspend), VDDQ vcc0v6_ddr from the TPS7A1006 (power tree §4.8).
5. Every ball goes to the same-named SoC ball, order not swapped (HDG 1.1 p. 22). ZQ0: 240 Ω 1 % to VDDQ (Micron z42n p. 33). Constraints: PCB-I-05.
6. Links: shunt in full, 0 Ω in production except H5 (BOM v1 line 137). FB3 senses vcc_ddr.
7. Open: VDD1 and VDDQ decoupling at the DRAM (data sheet silent, UNVERIFIED); the DNP TPS7A2018 vcc1v8_ddr option of BOM v1 line 132 is not drawn yet.
8. Unless noted: capacitors 0402 (100 nF 0201) X5R/X7R rated at least twice the rail; resistors 1 %.

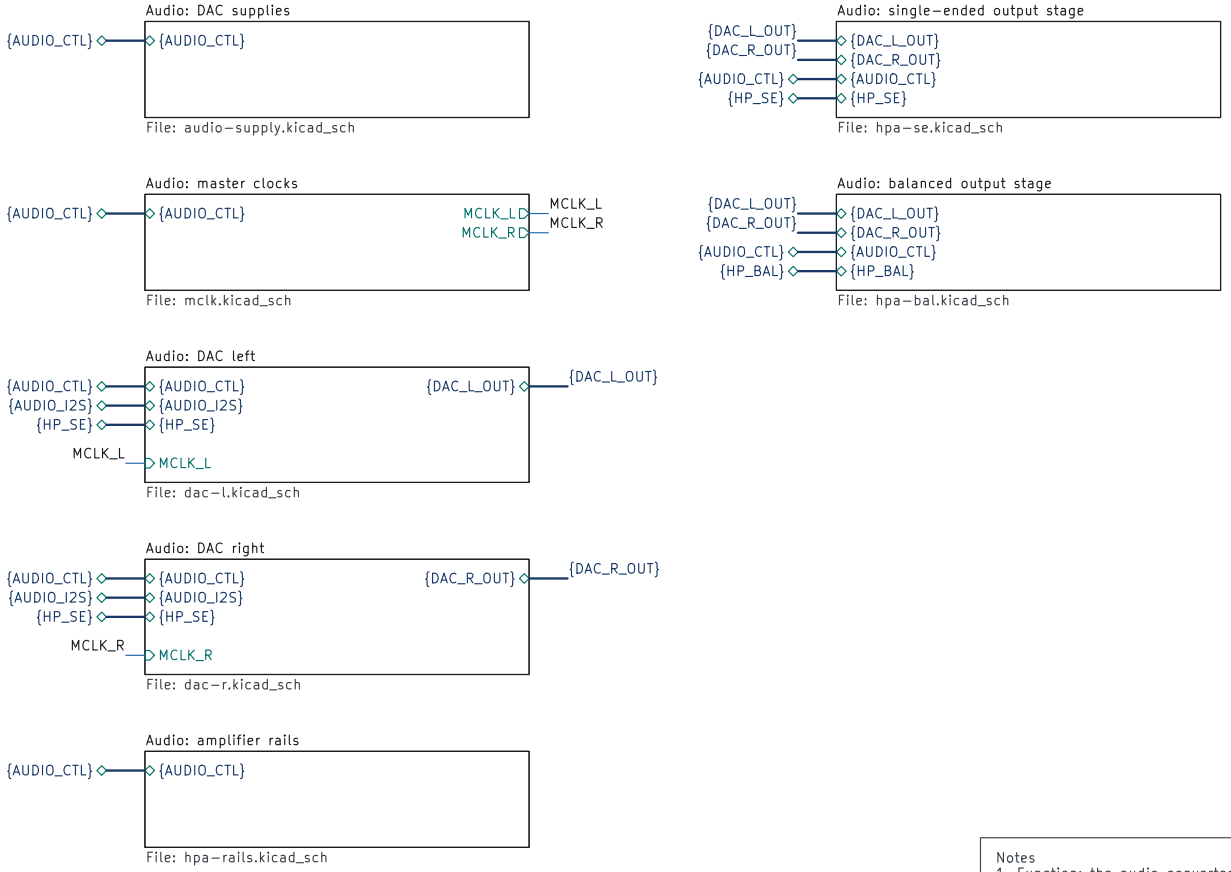
Title	Block
LPDDR4X memory	blk.lpddr4x

Purpose: 4 GB LPDDR4X on both RK3566 channels, its 0.6 V VDDQ regulator and supply links

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	21 / 69

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/LPDDR4X memory/



Notes
1. Function: the audio converter boundary (feature blocks #27): DAC supplies, master clocks, two CS43198 in dual mono, the amplifier rails and both OPA1622 output stages. Another variant may replace any sub-sheet behind this interface (RFC 0011 rule 6).
2. Absent state: never omitted (T1-040). Unpowered: every enable pulled off on the host side (blk.soc-vccio-b, PCB-A-13); I2C2 reaches the DACs only through the SN74LVC2G66 there (G2).
3. Interface buses: AUDIO_I2S (I2S1 and DSD), AUDIO_CTL (enables, reset, interrupt, the isolated DAC I2C), HP_SE and HP_BAL (outputs and references to the jacks).
4. Decisions: ADR 0001, 0010, 0012, 0015, 0019. Constraints: PCB-A-01 to PCB-A-19.

Title	Block
Audio	blk.audio

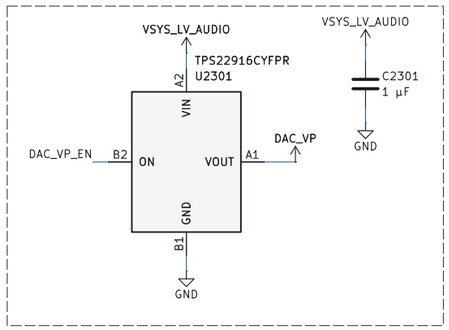
Purpose: The audio converter boundary: DAC supplies, master clocks, two CS43198, the amplifier rails and both output stages

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	22 / 69

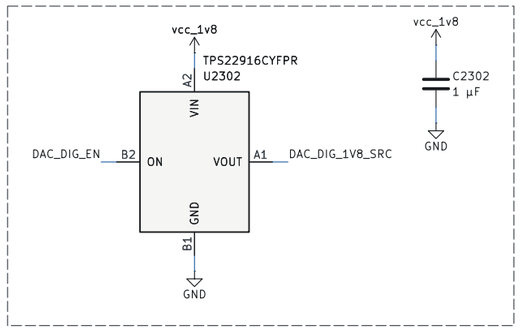
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/Audio/

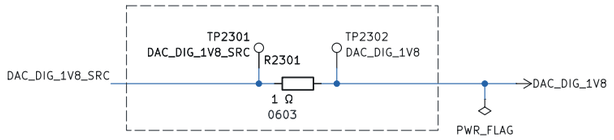
DAC_VP switch



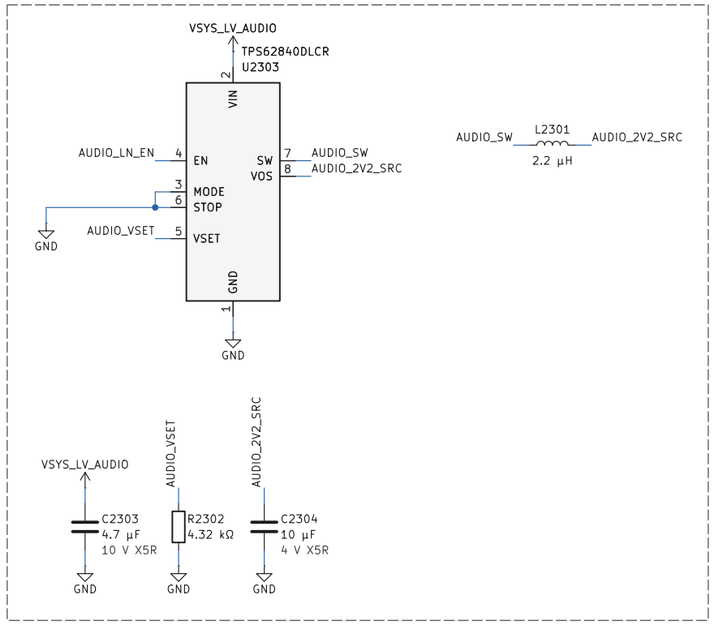
DAC_DIG_1V8 switch



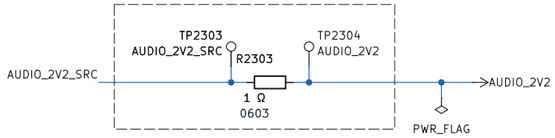
Link #25



AUDIO_2V2 pre-buck (TPS62840)



Link #26



{AUDIO_CTL}

Interface

Net	Direction	Absent state (block not fitted)
DAC_VP_EN, DAC_DIG_EN	Input	Pulled down on the host side (blk.soc-vccio-b): switches off
AUDIO_LN_EN	Input	Pulled down on the host side: TPS62840 off, output discharged

Rails

Rail	Capacitors	Requirement and source
VSYS_LV_AUDIO	1 µF; 4.7 µF 10 V (TPS62840 VIN)	TPS22916C CIN 1 µF (BOM v1 line 50); TPS62840 CI 4.7 µF (SLVSEC6D Table 2)
vcc_1v8	1 µF	TPS22916C CIN 1 µF (BOM v1 line 50)
AUDIO_2V2	10 µF 4 V	TPS62840 CO 10 µF (SLVSEC6D Table 2); each LDO adds 2.2 µF

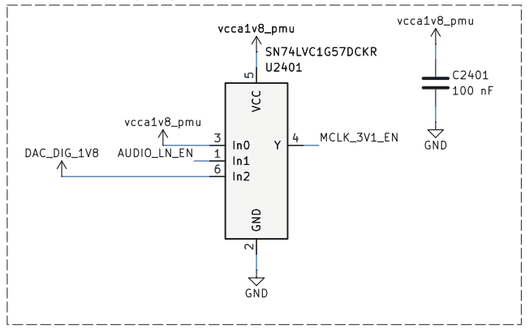
Notes
1. Function: DAC_VP (TPS22916C from VSYS_LV: VP of both CS43198), DAC_DIG_1V8 (TPS22916C from vcc_1v8: VD, VL, VCP, the I2C2 switch controls, the MCLK interlock) and AUDIO_2V2 (TPS62840 at 2.2 V, VSET 4.32 kΩ, SLVSEC6D Table 1) for the VA and MCLK_1V8 LDOs (power tree §2, §3).
2. Absent state: enables pulled down on the host side; QOD tied to VOUT empties DAC_VP and DAC_DIG_1V8 at switch-off (SLVSEN5B §8.3.3); the TPS62840 discharges its output via VOS.
3. Sequence (power tree §6.2): DAC_VP_EN first, then DAC_DIG_EN and AUDIO_LN_EN; VP off last.
4. TPS62840 MODE and STOP low: power save (SLVSEC6D p. 5); MODE high is the fallback if bursts are audible (UNVERIFIED, BOM v1 line 27). Links L1 #25, #26 (0 Ω in production).
5. VSYS_LV_AUDIO: the audio branch of VSYS_LV behind the 0x41 INA236 shunt (blk.current-monitors).
6. Unless noted: resistors 1 % 0402; capacitors X5R/X7R rated at least twice the rail.

Title	Audio: DAC supplies
Block	blk.audio-supply
Purpose	DAC_VP and DAC_DIG_1V8 load switches and the AUDIO_2V2 pre-buck, with their links
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026-10-25
Sheet	23 / 69

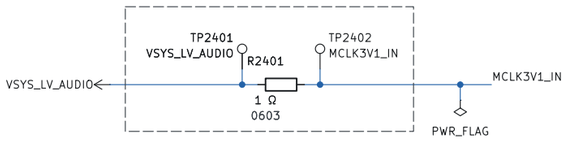
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/Audio/Audio: DAC supplies/

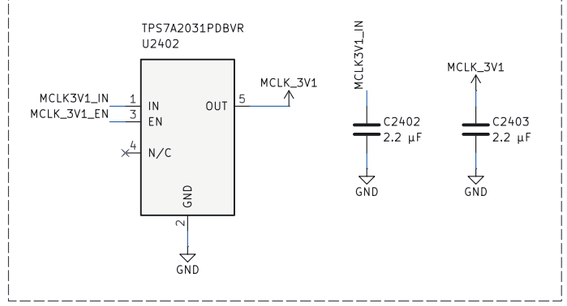
Interlock (G3)



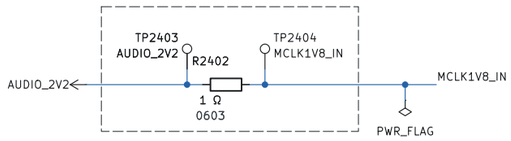
Link #27



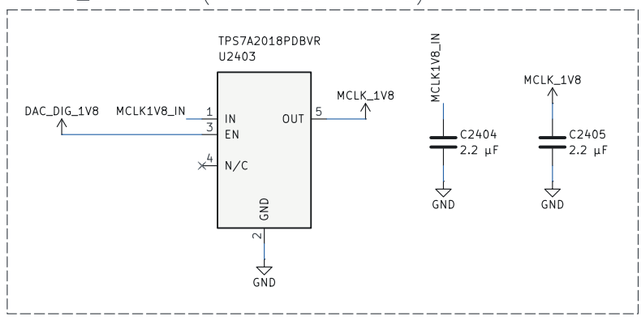
MCLK_3V1 LDO



Link #36



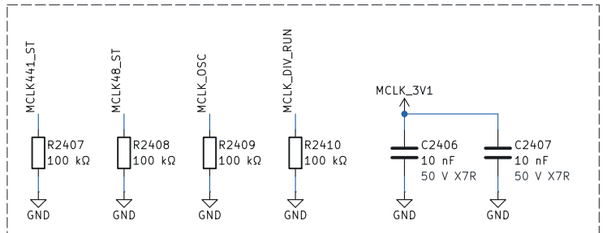
MCLK_1V8 LDO (ADR 0019 item 3)



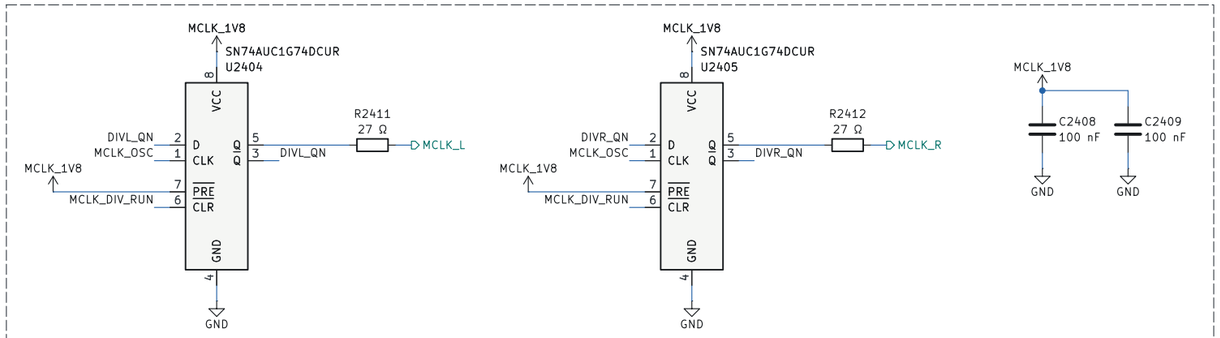
Oscillators (ADR 0019 item 1)



Pulls and bypass



Dividers by two (ADR 0019 item 2)



{AUDIO_CTL}

Interface

Net	Direction	Absent state (block not fitted)
MCLK441_EN, MCLK48_EN	Input	100 kΩ pull-down after the 0 Ω: both oscillators tri-stated
MCLK_DIV_RUN	Input	100 kΩ pull-down: dividers held clear
MCLK_L, MCLK_R	Output	Low while the dividers are held or unpowered (G3)

Rails

Rail	Capacitors	Requirement and source
vcca1v8_pmu	100 nF	Gate VCC bypass (capture choice; SCES414P gives none)
MCLK_3V1	2.2 µF + 2 x 10 nF	TPS7A20 COUT 0.47 µF effective minimum (SBVS338H §5.3; BOM v1.18 item 2); 0.01 µF per oscillator (NZ2520SDA p. 1)
MCLK_1V8	2.2 µF + 2 x 100 nF	TPS7A20 COUT 0.47 µF effective minimum; 100 nF per divider (BOM v1 line 140)
LDO inputs	2.2 µF each	TPS7A20 CIN 0.47 µF effective minimum (SBVS338H §5.3; BOM v1.18 item 2)

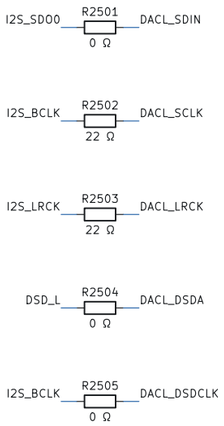
Notes
1. Function: NDK NSC5083D at 45.1584 / 49.152 MHz on MCLK_3V1 (TPS7A2031), one at a time, 33 Ω each into MCLK_OSC (100 kΩ pull-down); one SN74AUC1G74 per DAC divides by two (Q-bar to D, PRE-bar high), Q through 27 Ω to its XTI (ADR 0019: BOM v1 lines 20, 21, 29, 139, 140).
2. Absent state: ST pulled down, dividers held by MCLK_DIV_RUN. MCLK_3V1 EN = AUDIO_LN_EN AND DAC_DIG_1V8 (SN74LVC1G57 with In0 high: Y = In1 AND In2, SCES414P Table 1) and MCLK_1V8 EN = DAC_DIG_1V8, so no clock reaches an unpowered XTI (G3).
3. Start (ADR 0019 item 6): ST and CLR-bar low before the rails; CLR-bar released while MCLK_OSC is low; then one ST high. Family change: DACs to the RCO, CLR-bar low, swap ST.
4. Layout (item 5): one cluster; oscillator to divider <= 10 mm, divider to XTI <= 15 mm.
5. Open: ST rating at 3.3 V drive into a 3.1 V or unpowered oscillator (pinout 011); the 45.1584 MHz value on an NDK document (UNVERIFIED). Links L1 #27, #36.
6. Unless noted: resistors 1 % 0402; capacitors X7R.

Title: **Audio: master clocks**				Block: blk.mclk
Purpose: Two oscillators at twice the MCLK, their LDOs and interlock, one divider per DAC (ADR 0019)				
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 24 / 69

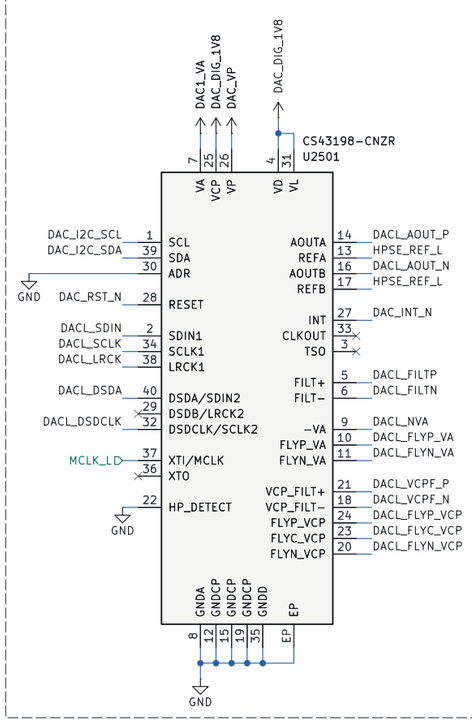
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/Audio/Audio: master clocks/

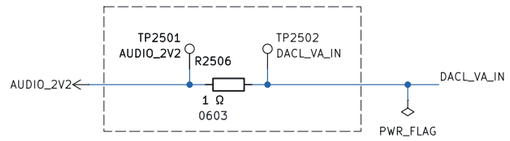
Serial port (BOM v1 lines 134, 141)



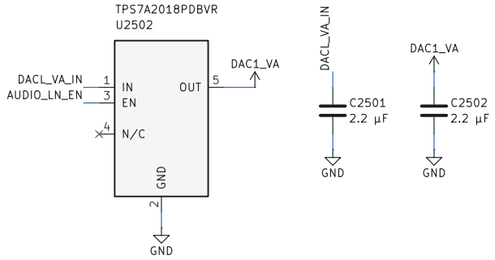
CS43198 in mono mode (DS1156F2 Fig. 2–2)



Link #28



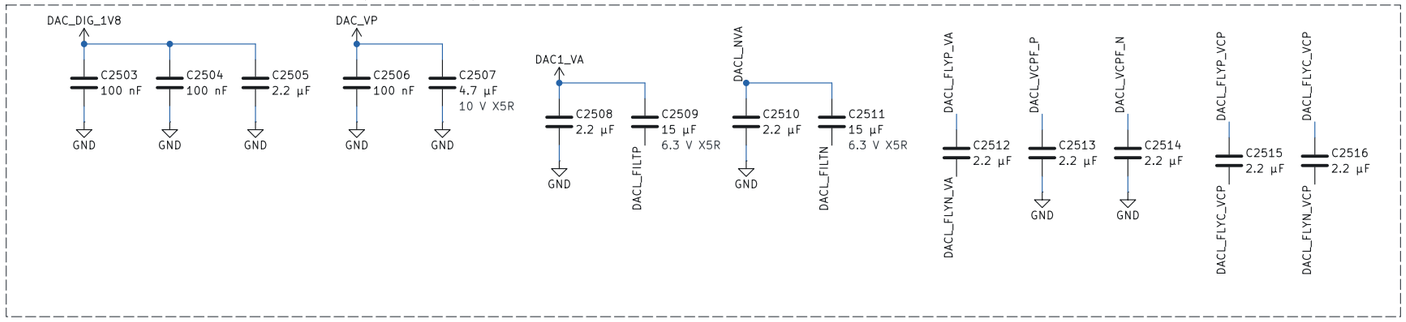
VA LDO (TPS7A2018)



MCLK test pad (BOM v1 line 88)



Supplies and charge pumps (Fig. 2–1)



{AUDIO_CTL} {AUDIO_I2S} {DAC_L_OUT} {HP_SE}

Interface

Net	Direction	Absent state (block not fitted)
I2S_SDO0, I2S_BCLK, I2S_LRCK, DSD_x	Bidirectional	Series parts here; the SoC parks the lines low (G2, G21)
DAC_I2C_SCL, DAC_I2C_SDA	Bidirectional	Isolated on the host side while DAC_DIG_1V8 is off (G2)
DAC_RST_N	Input	Pulled down on the host side: converter in reset (PCB–A–13)
DAC_INT_N	Output	Open drain; pull–up on DAC_DIG_1V8 on the host side
MCLK_L	Input	Low while the dividers are held or off
DACL_AOUT_P, DACL_AOUT_N	Output	Into both output stages
HPSE_REF_L	Passive	REFA, REFB to the 3.5 mm sleeve through its net tie (PCB–A–03, –16)

Rails

Rail	Capacitors	Requirement and source
DAC_DIG_1V8	VL, VD 100 nF each; VCP 2.2 µF	DS1156F2 Fig. 2–1
DAC_VP	100 nF + 4.7 µF 10 V	DS1156F2 Fig. 2–1
DAC1_VA	2.2 µF; FILT+ 15 µF to VA; LDO 2.2 µF	DS1156F2 Fig. 2–1; SBVS338H Fig. 7–4
−VA, VCP_FILT±	2.2 µF each; flying 2.2 µF x 3	DS1156F2 Fig. 2–1, internal VCP_FILT (EXT_VCPFILT = 0)

Notes
1. Function: Cirrus Logic CS43198 for the left channel in mono mode (dual mono, BOM v1 line 18): AOUTA and AOUTB a differential pair into both output stages; I2C2 address 0x30 (ADR to ground).
2. Absent state: never omitted (T1–040). RESET held low by the host–side pull–down; INT pulled up only on DAC_DIG_1V8: I2C isolated while DAC_DIG_1V8 is off (G2).
3. Clock master: SCLK1 and LRCK1 drive I2S_BCLK and I2S_LRCK through 22 Ω (HDG p. 70). XTI/MCLK from its own divider (blk.mclk): XTO, CLKOUT, TSO and DSDB/LRCK2 unused. DSD: DSDA from DSD_x, DSDCLK from I2S_BCLK (native DSD reserved, blackbird–os ADR 0006).
4. HP_DETECT tied low: plug detection is on the jacks (expander P10, P11).
5. VA from its own TPS7A2018 (EN = AUDIO_LN_EN) behind link L1; VL, VD, VCP on DAC_DIG_1V8; VP on DAC_VP (PCB–A–05). GNDA, GNDCP, GNDD and the pad in one ground (PCB–A–01, –02).
6. Open: series values other than 0 Ω (bring–up, G2); CS43198 pins fail–safe unpowered (UNVERIFIED).

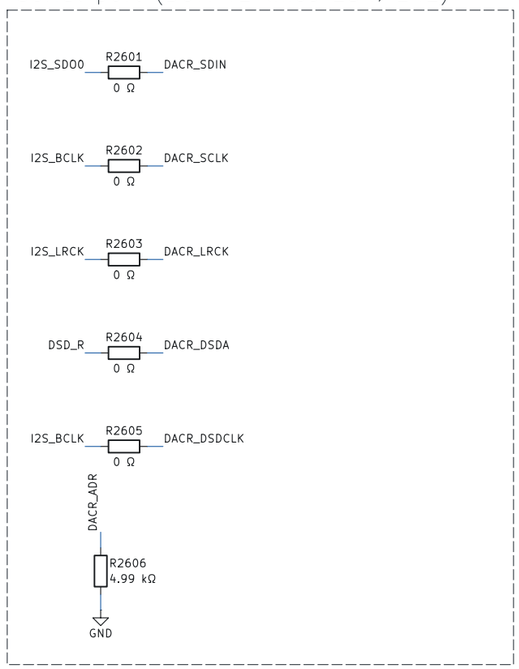
Title	Block
Audio: DAC left	blk.dac–l
Purpose	CS43198 for the left channel in mono mode, its VA LDO and serial–port series parts

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026–10–25	25 / 69

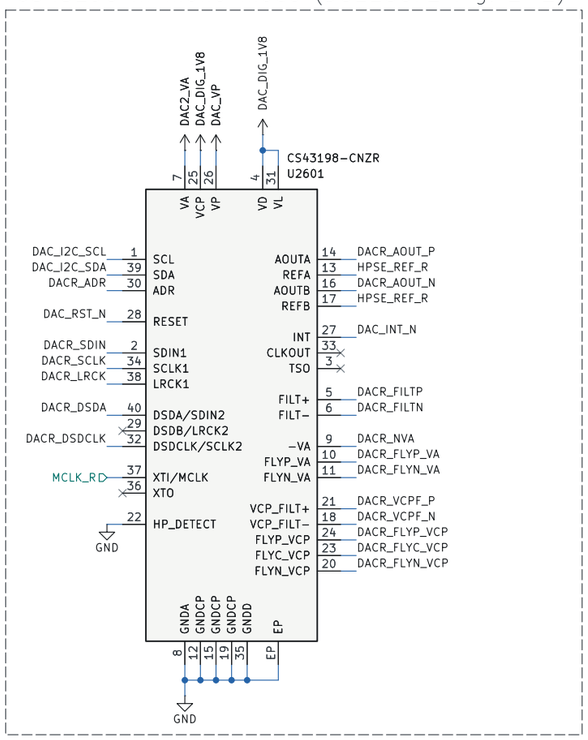
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/Audio/Audio: DAC left/

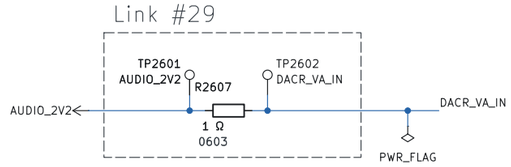
Serial port (BOM v1 lines 134, 141)



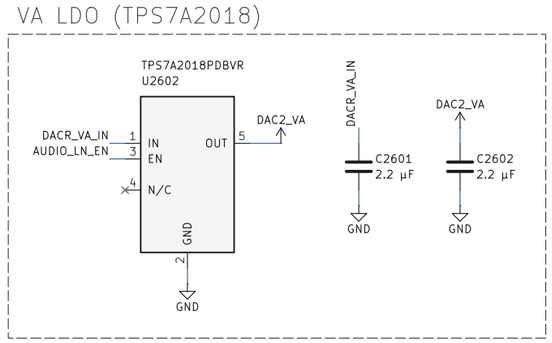
CS43198 in mono mode (DS1156F2 Fig. 2–2)



Link #29



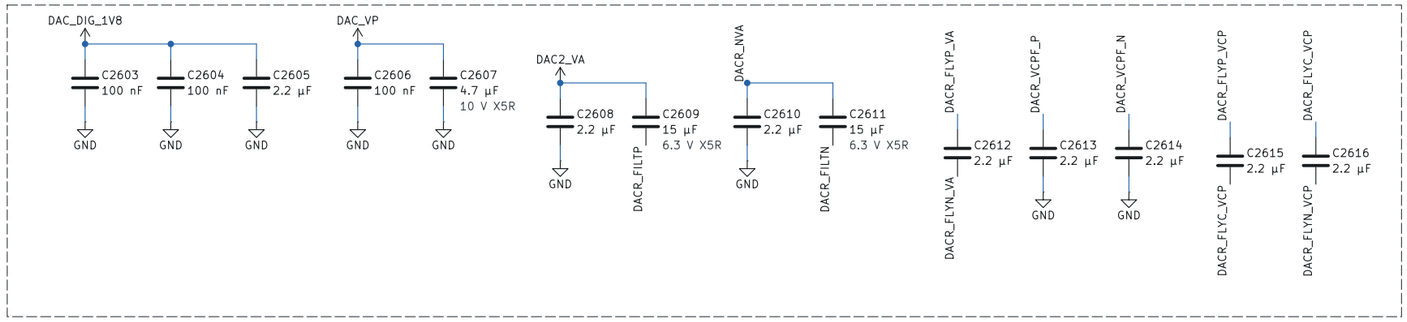
VA LDO (TPS7A2018)



MCLK test pad (BOM v1 line 88)



Supplies and charge pumps (Fig. 2–1)



{AUDIO_CTL} {AUDIO_I2S} {DAC_R_OUT} {HP_SE}

Interface

Net	Direction	Absent state (block not fitted)
I2S_SDO0, I2S_BCLK, I2S_LRCK, DSD_x	Bidirectional	Series parts here; the SoC parks the lines low (G2, G21)
DAC_I2C_SCL, DAC_I2C_SDA	Bidirectional	Isolated on the host side while DAC_DIG_1V8 is off (G2)
DAC_RST_N	Input	Pulled down on the host side: converter in reset (PCB–A–13)
DAC_INT_N	Output	Open drain; pull–up on DAC_DIG_1V8 on the host side
MCLK_R	Input	Low while the dividers are held or off
DACR_AOUT_P, DACR_AOUT_N	Output	Into both output stages
HPSE_REF_R	Passive	REFA, REFB to the 3.5 mm sleeve through its net tie (PCB–A–03, –16)

Rails

Rail	Capacitors	Requirement and source
DAC_DIG_1V8	VL, VD 100 nF each; VCP 2.2 µF	DS1156F2 Fig. 2–1
DAC_VP	100 nF + 4.7 µF 10 V	DS1156F2 Fig. 2–1
DAC2_VA	2.2 µF; FILT+ 15 µF to VA; LDO 2.2 µF	DS1156F2 Fig. 2–1; SBVS338H Fig. 7–4
−VA, VCP_FILT±	2.2 µF each; flying 2.2 µF x 3	DS1156F2 Fig. 2–1, internal VCP_FILT (EXT_VCPFILT = 0)

Notes
1. Function: Cirrus Logic CS43198 for the right channel in mono mode (dual mono, BOM v1 line 18): AOUTA and AOUTB a differential pair into both output stages; I2C2 address 0x31 (ADR 4.99 kΩ to ground).
2. Absent state: never omitted (T1–040). RESET held low by the host–side pull–down; INT pulled up only on DAC_DIG_1V8; I2C isolated while DAC_DIG_1V8 is off (G2).
3. Clock slave: SCLK1 and LRCK1 from the left DAC through 0 Ω. XTI/MCLK from its own divider (blk.mclk); XTO, CLKOUT, TSO and DSDB/LRCK2 unused. DSD: DSDA from DSD_x, DSDCLK from I2S_BCLK (native DSD reserved, blackbird–os ADR 0006).
4. HP_DETECT tied low: plug detection is on the jacks (expander P10, P11).
5. VA from its own TPS7A2018 (EN = AUDIO_LN_EN) behind link L1; VL, VD, VCP on DAC_DIG_1V8; VP on DAC_VP (PCB–A–05). GNDA, GNDCP, GNDD and the pad in one ground (PCB–A–01, –02).
6. Open: series values other than 0 Ω (bring–up, G2); CS43198 pins fail–safe unpowered (UNVERIFIED).

Title
Audio: DAC right

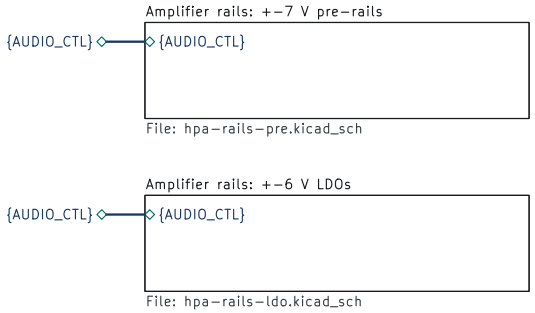
Block
blk.dac–r

Purpose
CS43198 for the right channel in mono mode, its VA LDO and serial–port series parts

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026–10–25	26 / 69

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/Audio/Audio: DAC right/



Notes
1. Function: the amplifier rails (feature blocks #32), drawn on two sub−sheets: the +−7 V pre−rails with the −7 V enable shifter, and the paired +−6 V LDOs (ADR 0015).
2. Absent state: HPA_RAIL_EN and HPA_LDO_EN pulled down: every rail off.
3. Sub−sheet IDs (blk.hpa−rails−*) are sheets of blk.hpa−rails, as the SoC's sub−sheets.

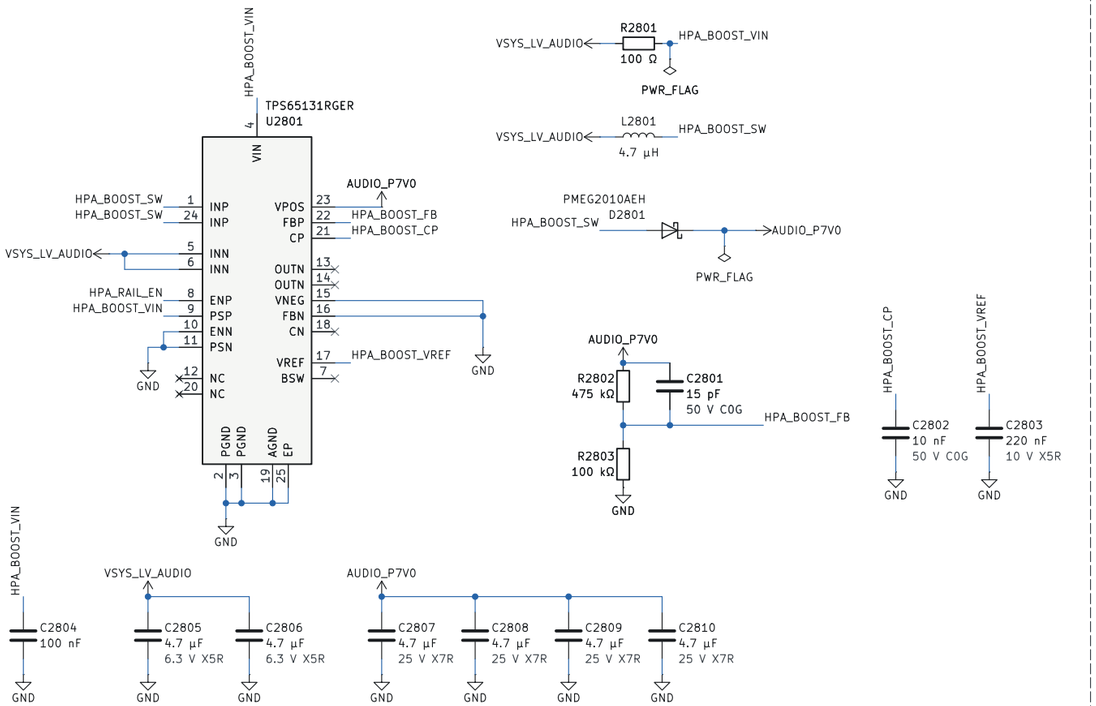
Title	Block
Audio: amplifier rails	blk.hpa−rails

Purpose: +−7 V pre−rails and the paired +−6 V LDOs of the output stages (ADR 0015)

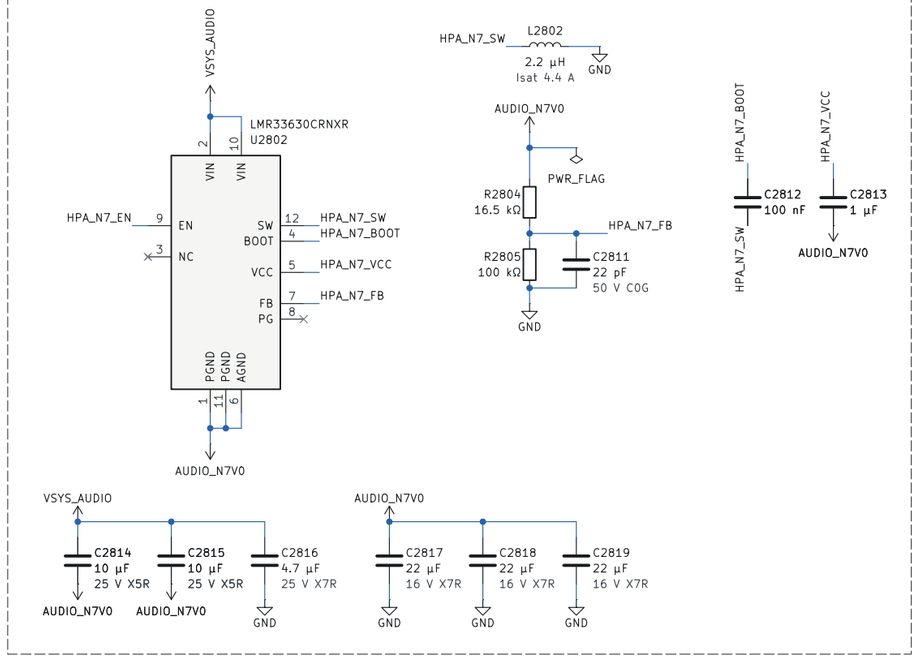
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026−10−25	27 / 69

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/Audio/Audio: amplifier rails/

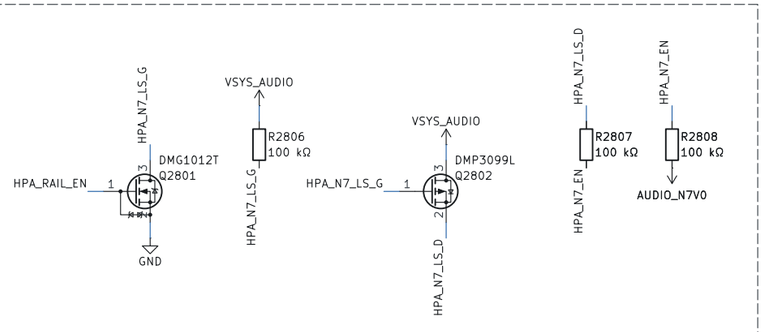
+7 V pre-rail: TPS65131 boost (ADR 0015 item 2)



−7 V pre-rail: LMR33630C inverting buck-boost (ADR 0015 item 1)



−7 V enable level shifter (SNVA856B Fig. 7-1)



{AUDIO_CTL}

Interface

Net	Direction	Absent state (block not fitted)
HPA_RAIL_EN	Input	100 kΩ pull-down on the host side (blk.soc-vccio-b): both pre-rails off
AUDIO_P7V0, AUDIO_N7V0	Supply out	Off; the LDOs behind them off first (power tree §6.3)

Rails

Rail	Capacitors	Requirement and source
VSYS_LV_AUDIO	2 x 4.7 µF (INP, INN); 100 nF at VIN behind 100 Ω	SLVS493E Fig. 8-1, Table 8-1
AUDIO_P7V0	4 x 4.7 µF 25 V X7R; 15 pF feed-forward	BOM v1 line 111 (SLVS493E Eq. 7, 11)
VSYS_AUDIO (−7 V IC)	2 x 10 µF 25 V to −7 V; 4.7 µF 25 V to GND	BOM v1 line 118 (SNVA856B Fig. 3-1)
AUDIO_N7V0	3 x 22 µF 16 V X7R 1206; 22 pF feed-forward	BOM v1 line 118

Notes
1. Function: AUDIO_P7V0 (TPS65131 boost from VSYS_LV_AUDIO, 475 kΩ / 100 kΩ: 6.97 V), AUDIO_N7V0 (LMR33630C inverting buck-boost from VSYS_AUDIO, IC ground = −7 V, 100 kΩ / 16.5 kΩ: −7.06 V), then HPA_P6V0 / HPA_N6V0 from two LT3045 and two LT3094 in parallel (ADR 0015; BOM lines 22–26, 116–119).
2. Absent state: HPA_RAIL_EN and HPA_LDO_EN pulled down on the host side; both pre-rails and the LDOs off.
3. TPS65131 inverter half unused: ENN, PSN, VNEG, FBN to ground, OUTN and CN open, INN on VSYS_LV_AUDIO with 4.7 µF (BOM line 22; no single-output case in SLVS493E, UNVERIFIED).
4. LMR33630C: EN through the two-FET shifter of SNVA856B Fig. 7-1; PG and NC unused.
5. LDOs: SET tied, 2 x 60.4 kΩ (6.04 V), 2 x 0.47 µF (≈ 65 ms); RILIM 523 Ω / 13.0 kΩ; PGFB to IN; PG open; OUT and OUTS joined, then a 20 mΩ trace ballast (net tie) to the shared rail.
6. Sequence: HPA_LDO_EN at least 10 ms after HPA_RAIL_EN (power tree §6.2).
7. Constraints: PCB-A-06, −14, −15. Unless noted: resistors 1 % 0402; X7R capacitors.

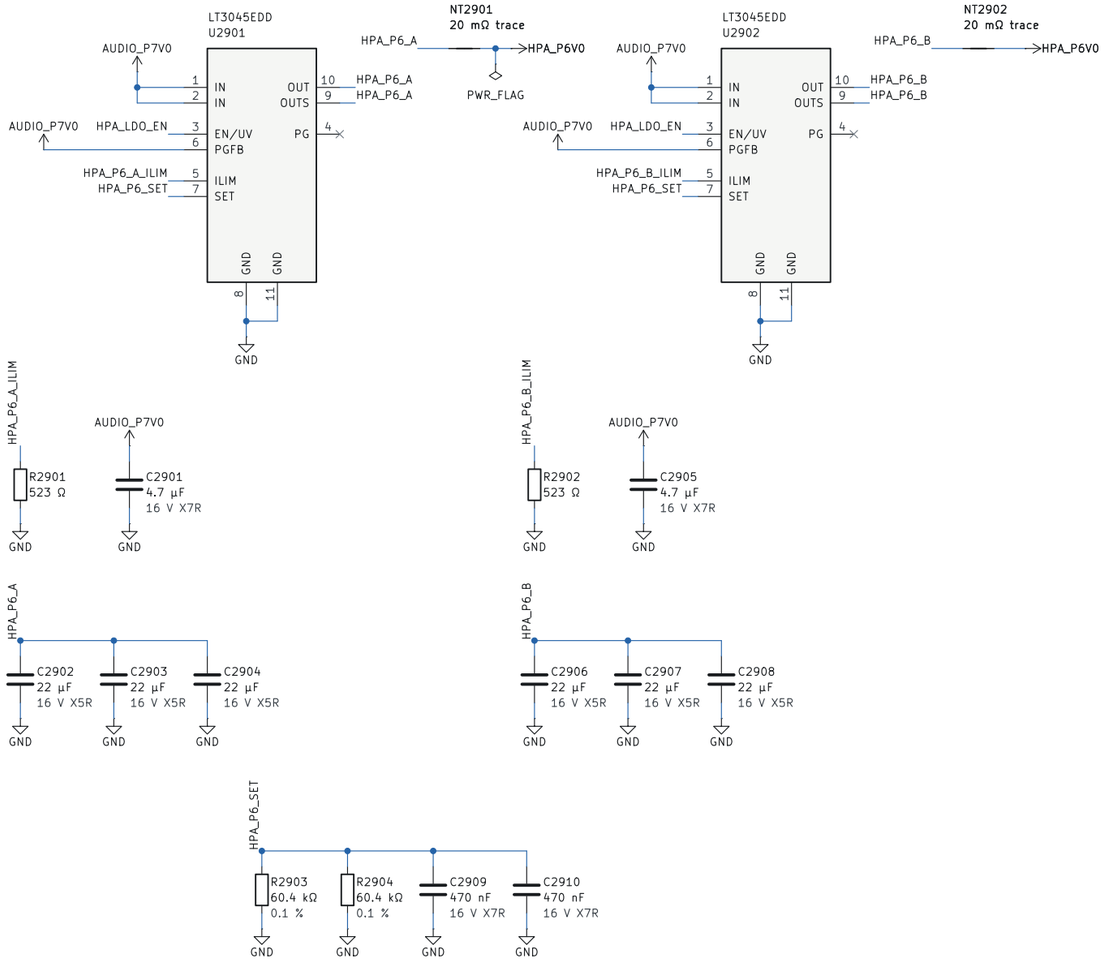
Title	Block
Amplifier rails: +−7 V pre-rails	blk.hpa-rails-pre
Purpose	+7 V TPS65131 boost, −7 V LMR33630C inverting buck-boost and its enable shifter

Project, board	Variant	Rev	Date	Sheet
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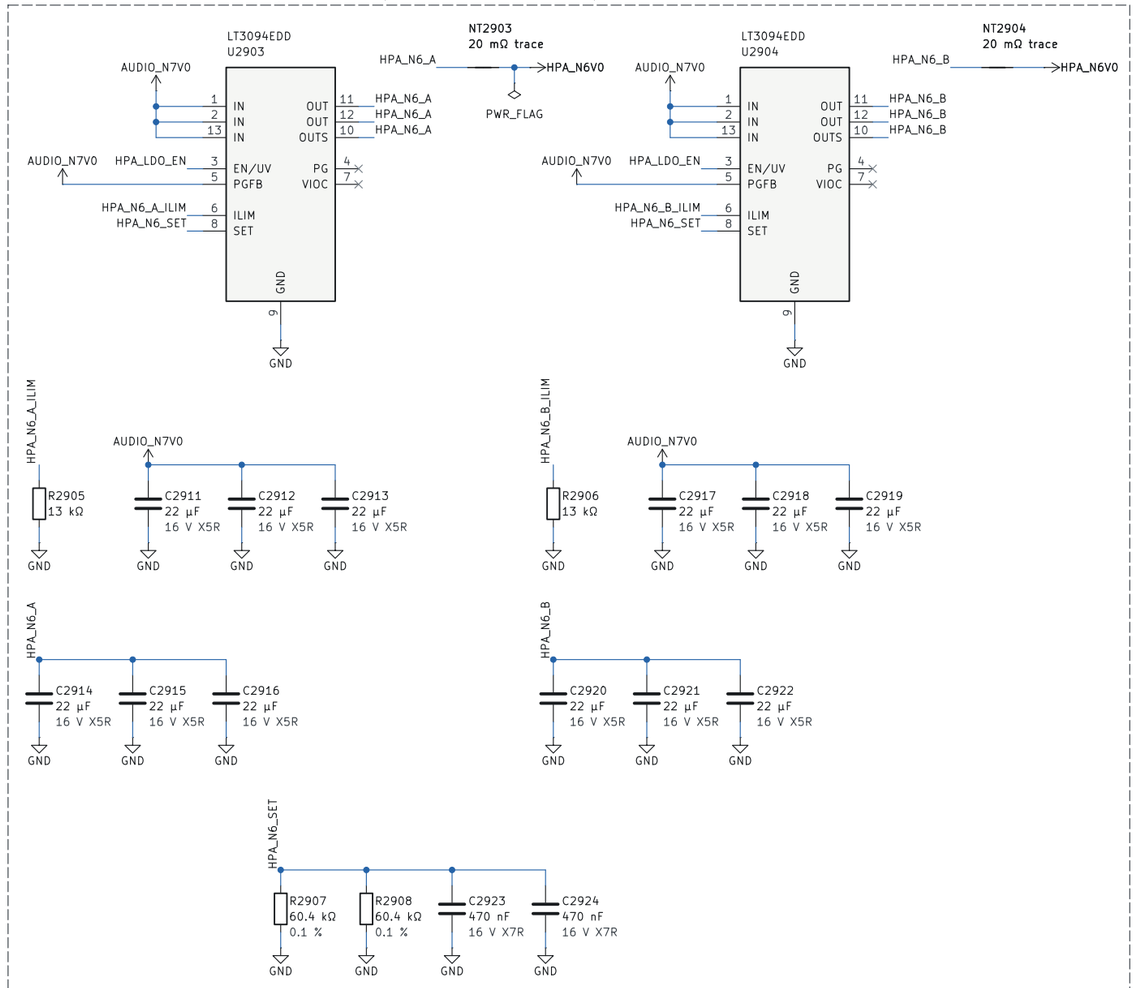
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/Audio/Audio: amplifier rails/Amplifier rails: +-7 V pre-rails/

+6 V: two LT3045 in parallel (ADR 0015 item 3)



−6 V: two LT3094 in parallel (ADR 0015 item 3)



{AUDIO_CTL}

Interface

Net	Direction	Absent state (block not fitted)
HPA_LDO_EN	Input	Pulled down on the host side: the four LDOs off
AUDIO_P7V0, AUDIO_N7V0	Supply in	From blk.hpa−rails−pre
HPA_P6V0, HPA_N6V0	Supply out	Off with HPA_LDO_EN low; OPA1622 EN low first (power tree §6.3)

Rails

Rail	Capacitors	Requirement and source
AUDIO_P7V0	4.7 µF per LT3045	BOM v1 line 110 (LT3045 Rev. D pp. 16–21)
AUDIO_N7V0	3 x 22 µF 16 V 0805 (1.0 mm) per LT3094	10 µF minimum at IN (LT3094 Rev. B p. 20): 11.3 µF effective at 7.06 V (BOM v1.18 item 5)
HPA_P6V0, HPA_N6V0	3 x 22 µF 16 V 0805 (1.0 mm) per device at OUT, before the ballast	10 µF minimum (LT3045 Rev. D p. 16): 13.4 µF effective at 6.04 V (BOM v1.18 item 5); ESR < 20 / 30 mΩ

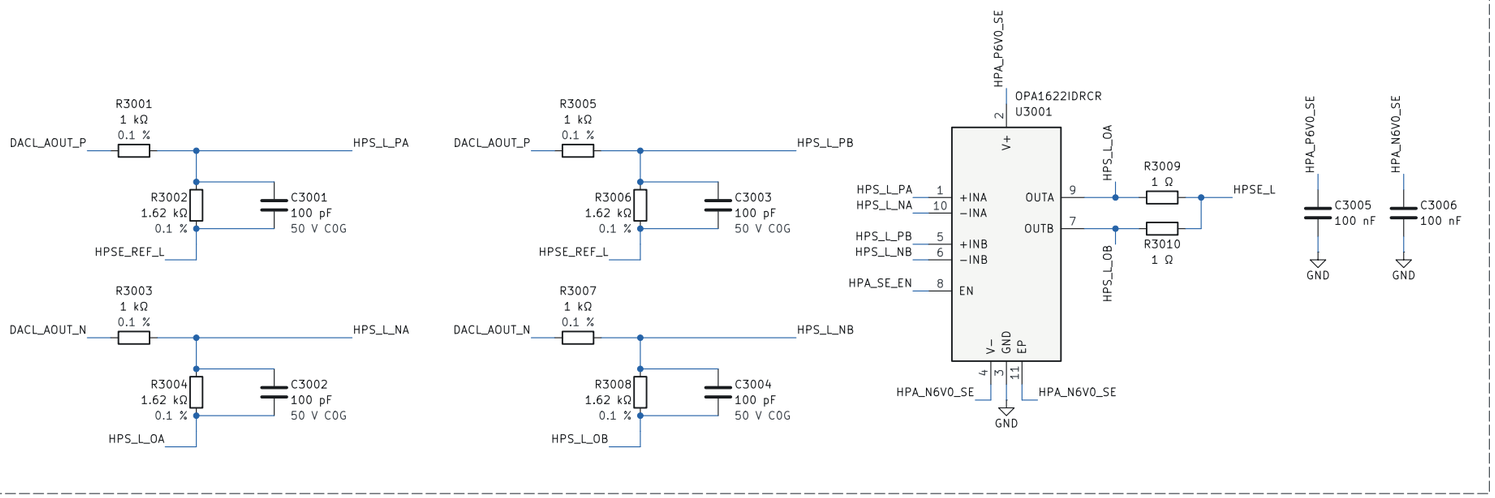
Notes
1. Function: HPA_P6V0 from two LT3045 and HPA_N6V0 from two LT3094 in parallel, as both ADI datasheets describe (LT3045 p. 21 Fig. 7, LT3094 p. 25 Fig. 15; ADR 0015 item 3).
2. Absent state: HPA_LDO_EN pulled down: the four LDOs off; enabled >= 10 ms after HPA_RAIL_EN.
3. SET tied per rail. RSET 2 x 60.4 kΩ (200 µA x 30.2 kΩ = 6.04 V). CSET 2 x 0.47 µF (≈ 65 ms); RILIM 523 Ω (LT3045) / 13.0 kΩ (LT3094) per device: pair limits 516–631 / 519–646 mA.
4. PGFB tied to IN (fast start−up off). PG open; OUT and OUTS joined at each device, then a 20 mΩ PCB trace ballast (net tie) to the shared rail. LT3094 exposed pad is IN (−7 V).
5. Constraints: PCB−A−06, PCB−A−14. Unless noted: resistors 1 % 0603, capacitors X7R.

Title	**Amplifier rails: +−6 V LDOs**
Block	blk.hpa−rails−ldo
Purpose	Two LT3045 and two LT3094 in parallel with trace ballasts (ADR 0015 item 3)
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026−10−25
Sheet	29 / 69

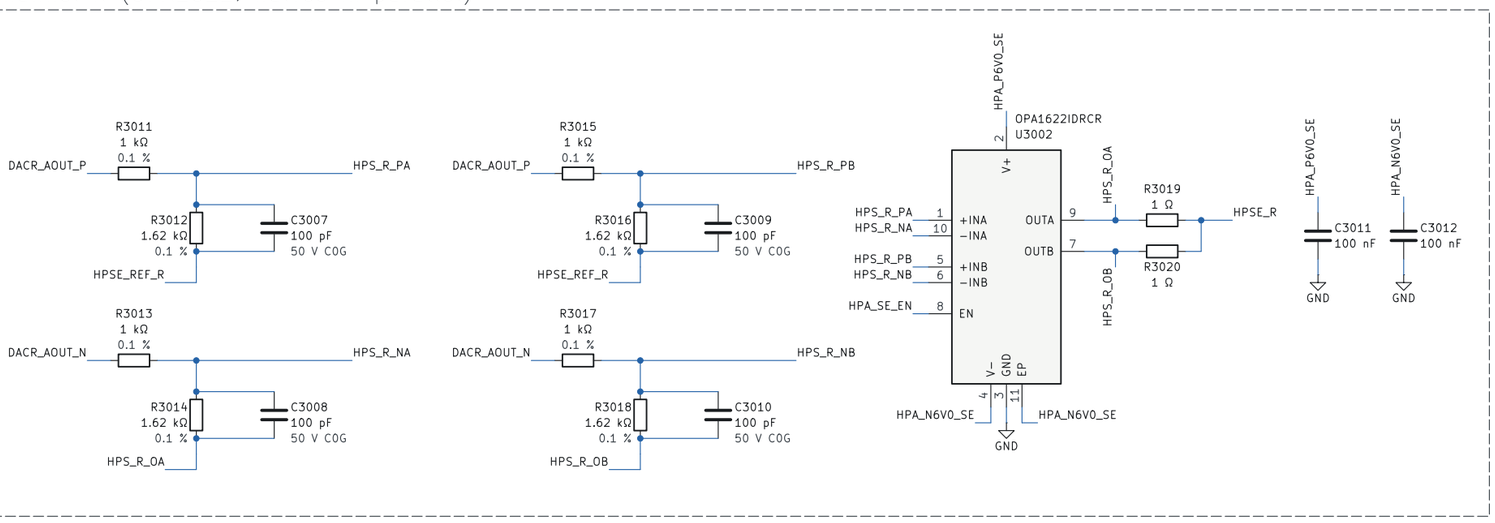
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/Audio/Audio: amplifier rails/Amplifier rails: +−6 V LDOs/

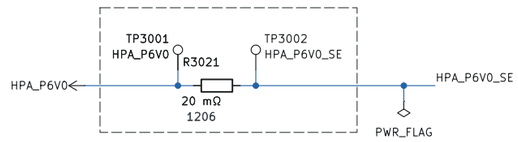
Channel L (OPA1622, halves in parallel)



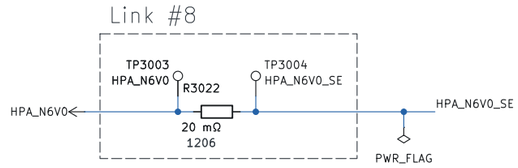
Channel R (OPA1622, halves in parallel)



Link #7



Link #8



{HP_SE}
{DAC_R_OUT}
{DAC_L_OUT}
{AUDIO_CTL}

Interface

Net	Direction	Absent state (block not fitted)
DACx_AOUT_P, DACx_AOUT_N	Input	Both stages load the DAC pins; with this output module left off the other stage is their only load; a stage fitted but unpowered still loads them
HPA_SE_EN	Input	Pulled down on the host side: packages in shutdown (5 µA per channel)
HPSE_L, HPSE_R	Output	To the module's jack only; a disabled OPA1622 output is not high-impedance (SBOS727B §7.4.1), so stage and jack are fitted or left off together (blk.hp-out-se)

Rails

Rail	Capacitors	Requirement and source
HPA_P6V0_SE, HPA_N6V0_SE	100 nF per supply pin, after the links	SBOS727B §10.1; the H20 links sit before them (power tree §4.12)

Notes
1. Function: two OPA1622, one per channel of the 3.5 mm output; each half its own difference amplifier (R1 = R3 1.00 kΩ, R2 = R4 1.62 kΩ, gain 1.62, 100 pF across R2 and R4), feedback at the half's pin, 1.0 Ω ballast to the leg (ADR 0012; BOM v1 lines 19, 37, 107, 108).
2. Absent state: HPA_SE_EN pulled down on the host side: shutdown. A disabled output is not high-impedance; never fitted but unpowered while the other stage plays (feature blocks #33–34); left off only with its jack, as one output module (blk.hp-out-se, -bal: feature blocks #51–52).
3. R4 returns to HPSE_REF_x, the same 3.5 mm sleeve sense as the DAC's REFA and REFB (PCB-A-16).
4. Thermal pad (EP) to V− (PCB-A-08). Halves of a package symmetric, legs symmetric (PCB-A-09, -10).
5. Links H20 #7, #8 on each rail branch, before the local capacitors (power tree §4.12).
6. Unless noted: network resistors 0.1 % 0603 thin film, network capacitors C0G 50 V 0603; ballast 1 % 0603; decoupling X7R 0402. Mute shunt footprints (BOM v1 line 100): open item.

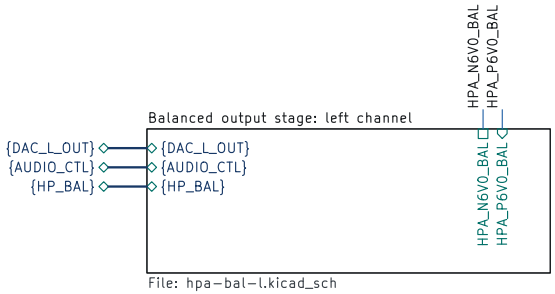
Title
Audio: single-ended output stage

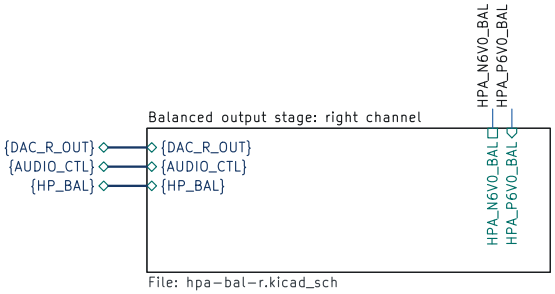
Block
blk.hpa-se

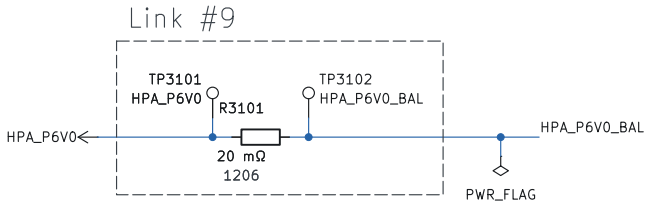
Purpose
Two OPA1622 with paralleled halves, one per channel, difference amplifiers (ADR 0012)

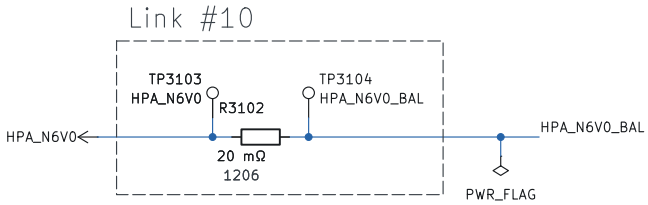
Project, board	Variant	Rev	Date	Sheet
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/Audio/Audio: single-ended output stage/









Interface

Net	Direction	Absent state (block not fitted)
DACx_AOUT_P, DACx_AOUT_N	Input	Both stages load the DAC pins; with this output module left off the other stage is their only load; a stage fitted but unpowered still loads them
HPA_BAL_EN	Input	Pulled down on the host side: packages in shutdown (5 µA per channel)
HPBAL_x_P / _N	Output	To the module's jack only; a disabled OPA1622 output is not high-impedance (SBOS727B §7.4.1), so stage and jack are fitted or left off together (blk.hp-out-bal)

Rails

Rail	Capacitors	Requirement and source
HPA_P6V0_BAL, HPA_N6V0_BAL	100 nF per supply pin, after the links	SBOS727B §10.1; the H20 links sit before them (power tree §4.12)

Notes
1. Function: four OPA1622, one per leg (L+, L−, R+, R−) of the 4.4 mm output, drawn on two sub-sheets (left and right channel); this sheet carries the H20 links #9 and #10 of their rails, before the local capacitors (power tree §4.12; ADR 0012).
2. Absent state: HPA_BAL_EN pulled down on the host side: shutdown. A disabled output is not high-impedance: never fitted but unpowered while the other stage plays (feature blocks #34). Left off with blk.jack-bal as the 4.4 mm output module blk.hp-out-bal (feature blocks #52).
3. Sub-sheet IDs (blk.hpa-bal-*) are sheets of blk.hpa-bal, as the SoC's sub-sheets.

Title	Block
Audio: balanced output stage	blk.hpa-bal

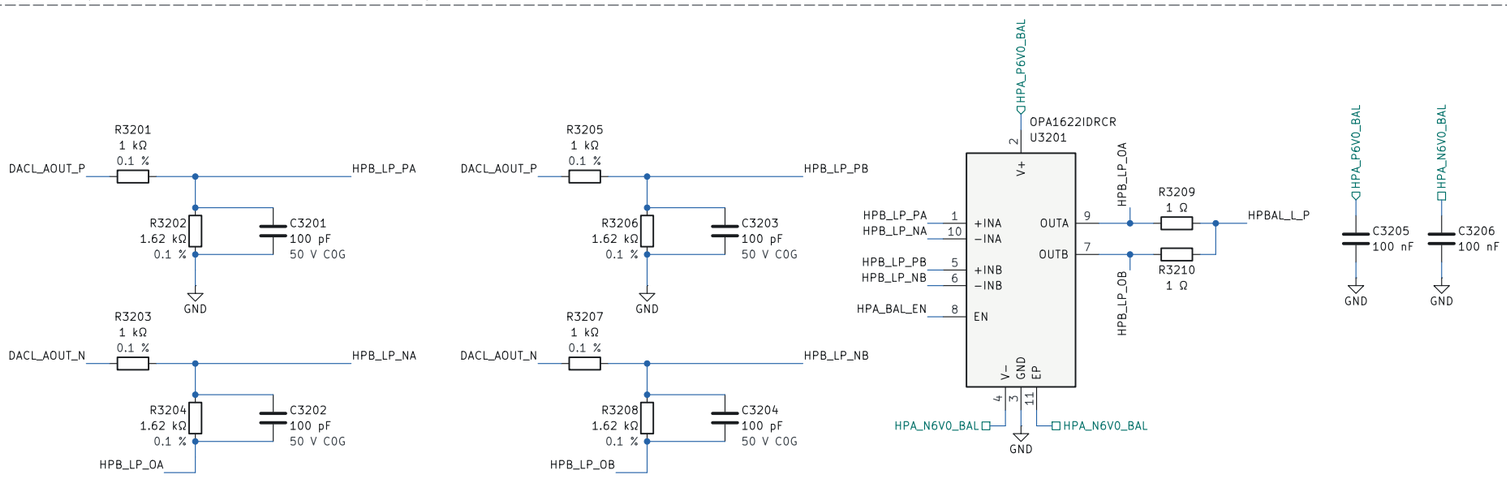
Purpose: Four OPA1622 with paralleled halves, one per leg, and the links of their rails (ADR 0012)

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	31 / 69

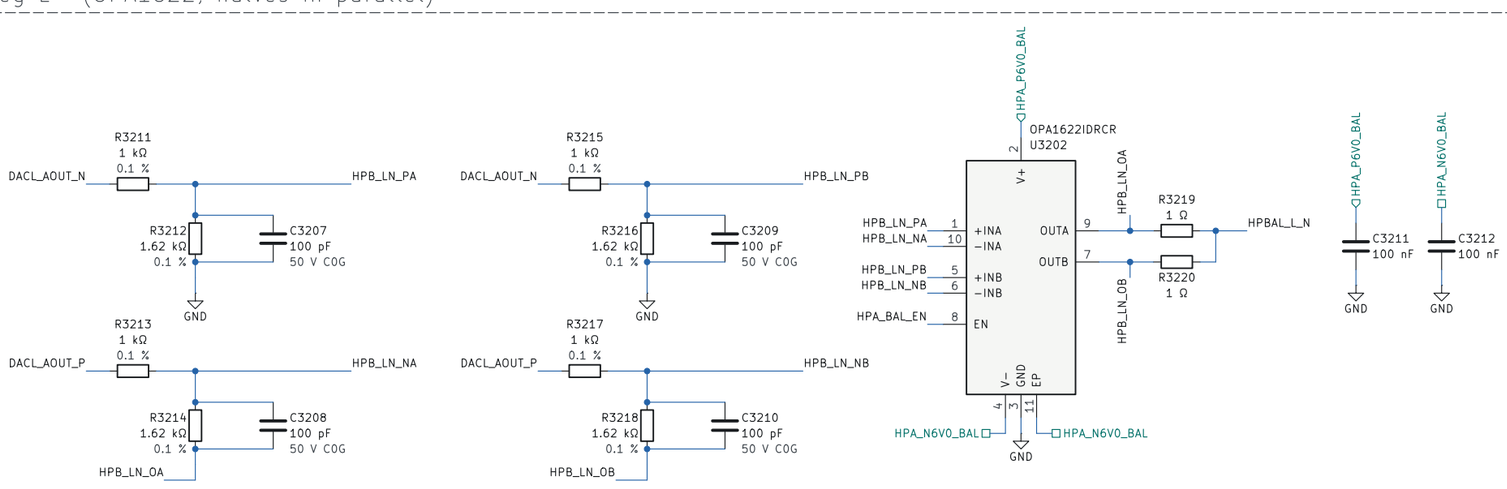
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/Audio/Audio: balanced output stage/

Leg L+ (OPA1622, halves in parallel)



Leg L− (OPA1622, halves in parallel)



{HP_BAL} ◇—
{DAC_L_OUT} ◇—
{AUDIO_CTL} ◇—

Interface

Net	Direction	Absent state (block not fitted)
DACx_AOUT_P, DACx_AOUT_N	Input	Both stages load the DAC pins; with this output module left off the other stage is their only load; a stage fitted but unpowered still loads them
HPA_BAL_EN	Input	Pulled down on the host side: packages in shutdown (5 µA per channel)
HPBAL_x_P / _N	Output	To the module's jack only; a disabled OPA1622 output is not high-impedance (SBOS727B §7.4.1), so stage and jack are fitted or left off together (blk.hp-out-bal)

Rails

Rail	Capacitors	Requirement and source
HPA_P6V0_BAL, HPA_N6V0_BAL	100 nF per supply pin	SBOS727B §10.1; links #9, #10 on blk.hpa-bal

Notes
1. Function: two OPA1622, one per leg of the right channel of the 4.4 mm output; each half its own difference amplifier (R1 = R3 1.00 kΩ, R2 = R4 1.62 kΩ, gain 1.62, 100 pF across R2 and R4). feedback at the half's pin, 1.0 Ω ballast to the leg (ADR 0012; BOM v1 lines 19, 37, 107, 108).
2. Absent state: HPA_BAL_EN pulled down on the host side: shutdown. A disabled output is not high-impedance: never fitted but unpowered while the other stage plays (feature blocks #33–34); left off only with its jack, as one output module (blk.hp-out-se, -bal: feature blocks #51–52).
3. R4 returns to ground; the jack's ground pin joins it at the jack.
4. Thermal pad (EP) to V− (PCB-A-08). Halves of a package symmetric, legs symmetric (PCB-A-09, -10).
5. Rails from blk.hpa-bal, after links H20 #9 and #10 (power tree §4.12).
6. Unless noted: network resistors 0.1 % 0603 thin film, network capacitors C0G 50 V 0603; ballast 1 % 0603; decoupling X7R 0402. Mute shunt footprints (BOM v1 line 100): open item.

Title
Balanced output stage: left channel

Block
blk.hpa-bal-l

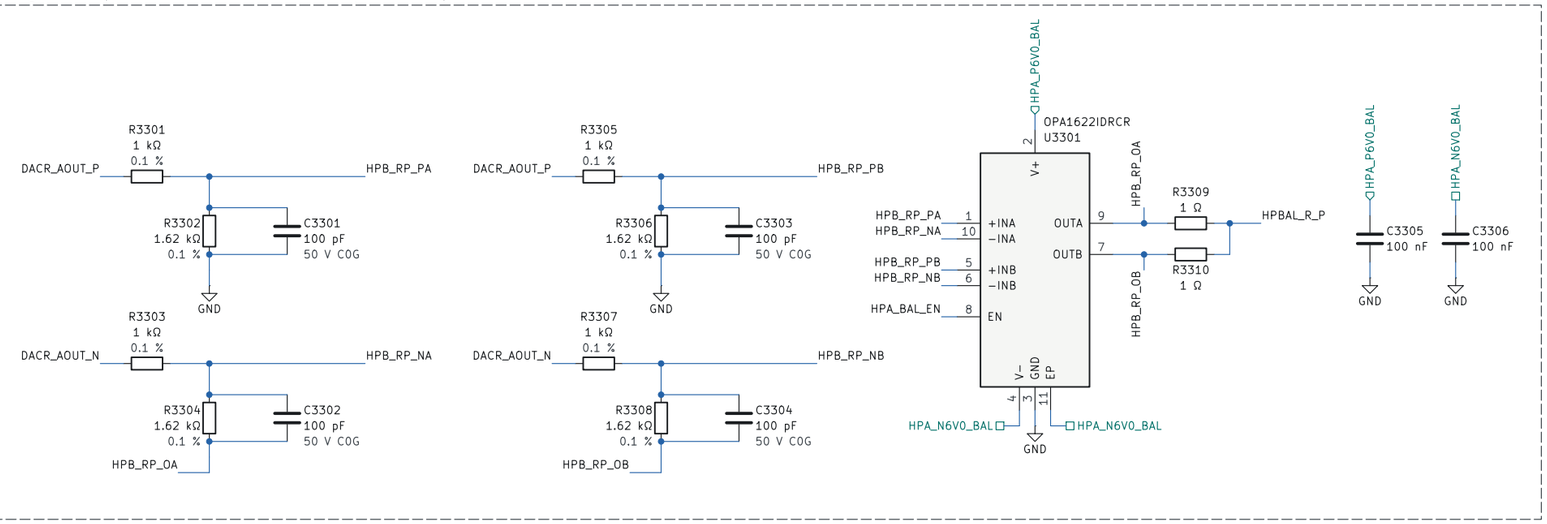
Purpose
Two OPA1622 with paralleled halves for the L+ and L− legs

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	32 / 69

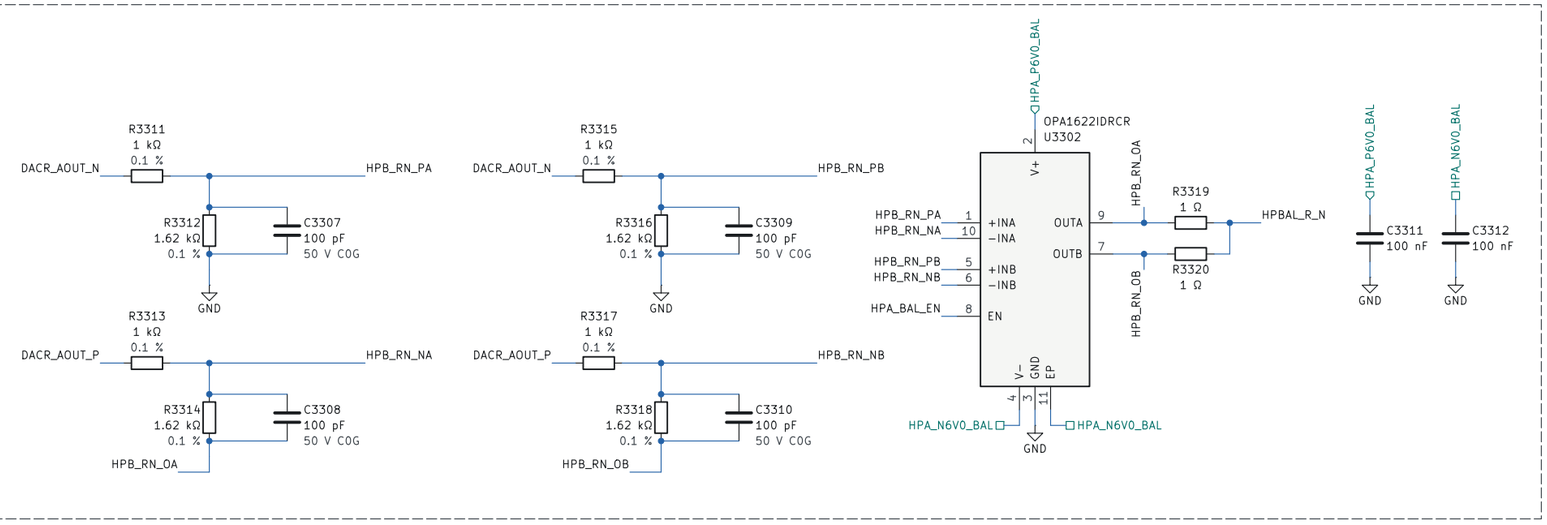
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/Audio/Audio: balanced output stage/Balanced output stage: left channel/

Leg R+ (OPA1622, halves in parallel)



Leg R− (OPA1622, halves in parallel)



{HP_BAL} ◇—
{DAC_R_OUT} ◇—
{AUDIO_CTL} ◇—

Interface

Net	Direction	Absent state (block not fitted)
DACx_AOUT_P, DACx_AOUT_N	Input	Both stages load the DAC pins; with this output module left off the other stage is their only load; a stage fitted but unpowered still loads them
HPA_BAL_EN	Input	Pulled down on the host side: packages in shutdown (5 µA per channel)
HPBAL_x_P / _N	Output	To the module's jack only; a disabled OPA1622 output is not high-impedance (SBOS727B §7.4.1), so stage and jack are fitted or left off together (blk.hp-out-bal)

Rails

Rail	Capacitors	Requirement and source
HPA_P6V0_BAL, HPA_N6V0_BAL	100 nF per supply pin	SBOS727B §10.1; links #9, #10 on blk.hpa-bal

Notes
1. Function: two OPA1622, one per leg of the right channel of the 4.4 mm output; each half its own difference amplifier (R1 = R3 1.00 kΩ, R2 = R4 1.62 kΩ, gain 1.62, 100 pF across R2 and R4). feedback at the half's pin, 1.0 Ω ballast to the leg (ADR 0012; BOM v1 lines 19, 37, 107, 108).
2. Absent state: HPA_BAL_EN pulled down on the host side: shutdown. A disabled output is not high-impedance: never fitted but unpowered while the other stage plays (feature blocks #33–34); left off only with its jack, as one output module (blk.hp-out-se, -bal: feature blocks #51–52).
3. R4 returns to ground; the jack's ground pin joins it at the jack.
4. Thermal pad (EP) to V− (PCB-A-08). Halves of a package symmetric, legs symmetric (PCB-A-09, -10).
5. Rails from blk.hpa-bal, after links H20 #9 and #10 (power tree §4.12).
6. Unless noted: network resistors 0.1 % 0603 thin film, network capacitors C0G 50 V 0603; ballast 1 % 0603; decoupling X7R 0402. Mute shunt footprints (BOM v1 line 100): open item.

Title
Balanced output stage: right channel

Block
blk.hpa-bal-r

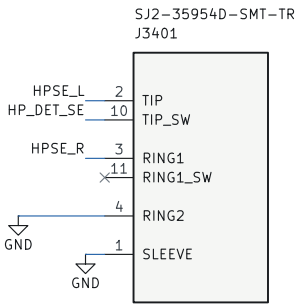
Purpose
Two OPA1622 with paralleled halves for the R+ and R− legs

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	33 / 69

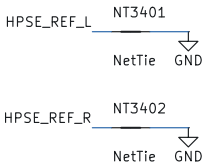
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/Audio/Audio: balanced output stage/Balanced output stage: right channel/

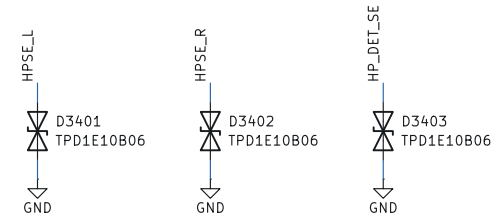
3.5 mm jack (SJ2-35954D)



Reference net ties at the sleeve (PCB-A-03, -16)



ESD and detect (BOM v1 line 36)



{HP_SE}

{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
HPSE_L, HPSE_R	Input	Left off with blk.hpa-se as the 3.5 mm output module (blk.hp-out-se): the nets go with it
HPSE_REF_L, HPSE_REF_R	Passive	Tied to ground at the sleeve land by the net-tie copper, on the board in every variant: the DACs' REFA and REFB stay referenced (DS1156F2 §8.3)
HP_DET_SE	Output	Pull-up on the host side; reads the tip switch

Notes
1. Function: 3.5 mm TRRS jack used as TRS (tip L, ring 1 R, ring 2 and sleeve ground; BOM v1 line 30); TPD1E10B06 on each output and the detect line (line 36).
2. Absent state: passive; left off only with blk.hpa-se (blk.hp-out-se). Plug detect: the tip switch (pin 10, normally closed to the tip, UNVERIFIED by text) on HP_DET_SE, read by expander P10 with its pull-up on the host side, so the line reads "absent" with the jack left off.
3. HPSE_REF_L and _R (DAC REFA, REFB and the SE references) each a separate trace joining ground at the sleeve pin through a net tie (PCB-A-03, PCB-A-16, PCB-A-17); the ties are copper and stay with the jack left off, so REFA and REFB keep their ground (DS1156F2 §8.3).
4. Open: RF filter footprints near the jack (PCB-A-19, Q2); the detect polarity (sample).

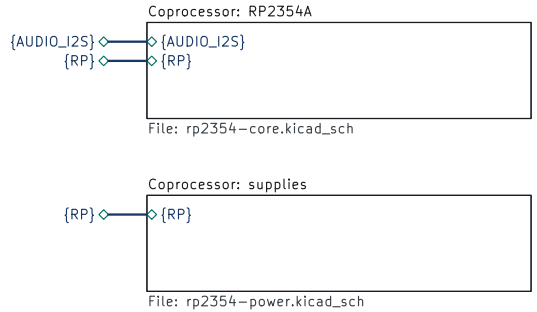
Title	Block
3.5 mm headphone jack	blk.jack-se

Purpose: Single-ended output jack with ESD, plug detect and the reference net ties

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	34 / 69

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/3.5 mm headphone jack/



Notes
1. Function: RP2354A orchestration coprocessor (BRD−014, should); flashed by Blackbird OS over SWD or USB BOOTSEL (BRD−015); never in a data path (BRD−005); the SoC boots without it (BRD−019).
2. Absent state: unfitted or RP_PWR_EN low: rails off; RP_IRQ low (no wake); UART, SWD and RUN parked low by the SoC; the I2S taps reach FT pads only (3.63 V at IOVDD 0 V, G4).
3. Drawn on two sub−sheets (blk.rp2354−core, blk.rp2354−power) holding the units of one part.

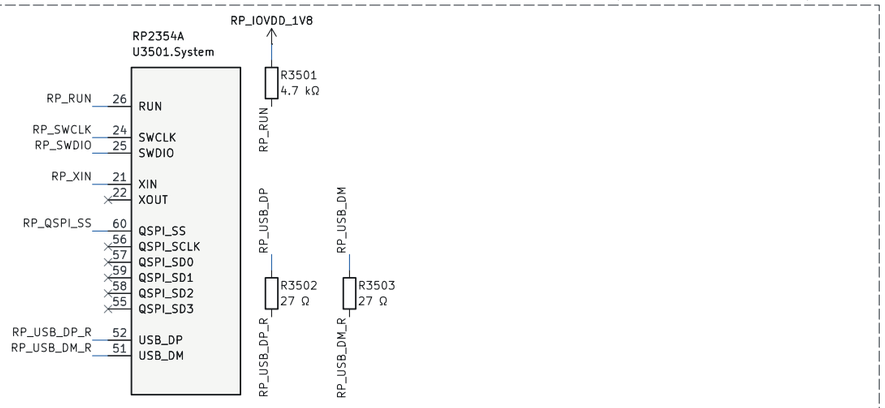
Title	Block
Coprocessor	blk.rp2354

Purpose				
RP2354A orchestration coprocessor, drawn on two sub−sheets				
Project, board	**Variant**	**Rev**	**Date**	**Sheet**
OpenDAP T1, main board	full	A	2026−10−25	35 / 69

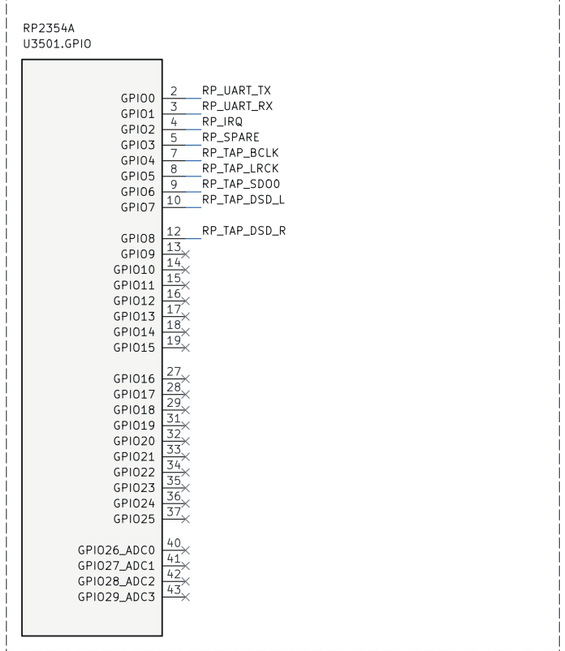
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/Coprocessor/

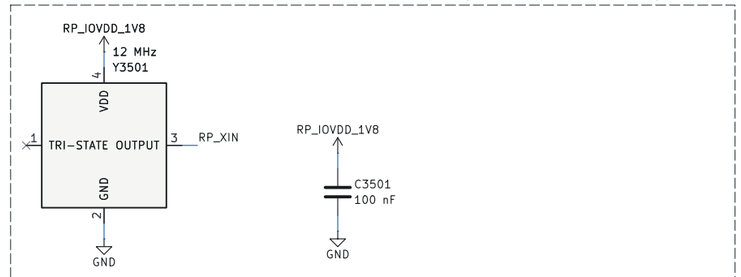
RP2354A system: RUN, SWD, clock, BOOTSEL, USB (BOM v1 line 12)



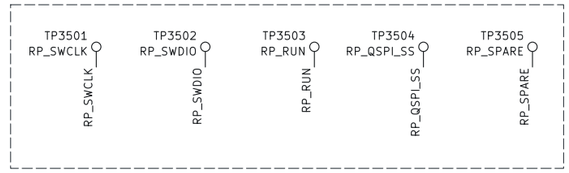
RP2354A GPIO (pinout v1 §3 allocation)



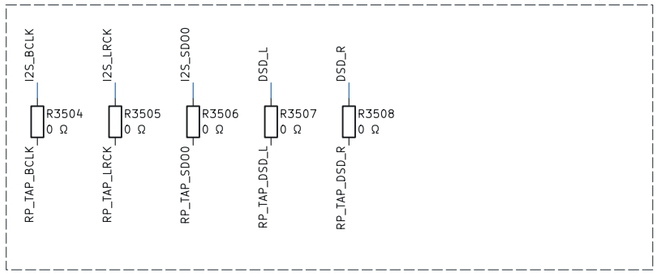
12 MHz oscillator on RP_IOVDD_1V8 (BOM v1 line 14)



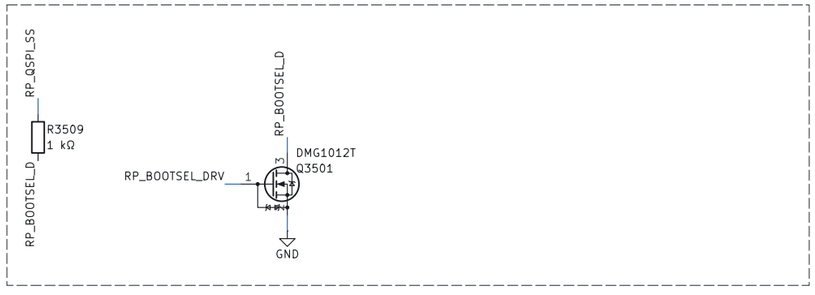
Test pads (BOM v1 line 89; PCB-T-09)



Tap 0 Ω positions (BOM v1 line 134; PCB-A-12)



BOOTSEL: N-FET on QSPI_SS (BOM v1 line 35; pinout v1 C6)



{RP}

{AUDIO_I2S}

Interface

Net	Direction	Absent state (block not fitted)
RP_UART_TX, RP_UART_RX	Bidirectional	UART0 to the SoC's UART0 (PMUIO2); SoC pull-downs park both
RP_IRQ	Output	GPIO2 to 1V20 (wake); 100 kΩ pull-down on the host side (blk.soc-vccio-b), no wake when off or unfitted
RP_RUN, RP_SWCLK, RP_SWDIO	Input	SoC open drain / SWD (VCCIO6); parked low while the rails are off (G4)
RP_BOOTSEL_DRV	Input	N-FET gate, 100 kΩ pull-down on the host side (blk.soc-vccio-b): normal boot
RP_USB_DP, RP_USB_DM	Bidirectional	USB_HOST2 (EHCI/OHCI) of the SoC
I2S_BCLK, I2S_LRCK, I2S_SDO0, DSD_L, DSD_R	Input	Taps on FT GPIO4-8 through 0 Ω (G4); no load on the DAC path

Rails

Rail	Capacitors	Requirement and source
RP_IOVDD_1V8	100 nF at the oscillator	ECS-2016MV (value: capture choice)

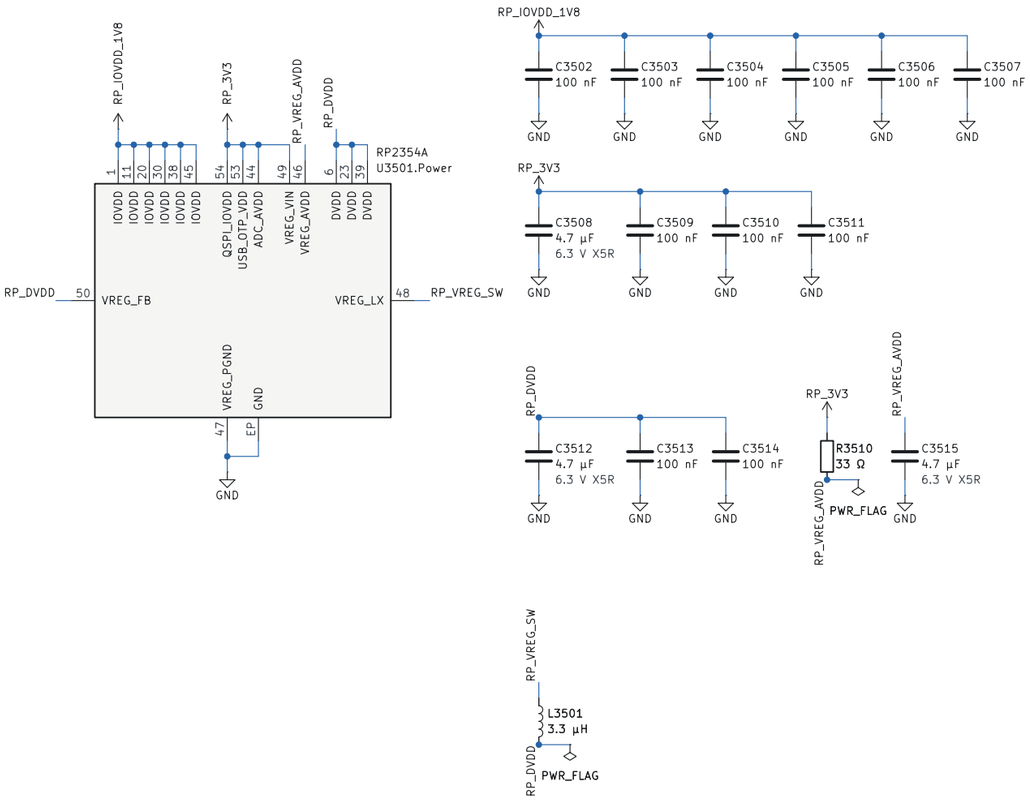
Notes
1. Function: RP2354A system and GPIO units (BRD-014, should: BRD-015, BRD-041): UART0 and RP_IRQ to the SoC, SWD and RUN from it, BOOTSEL, USB to USB_HOST2, passive I2S/DSD taps.
2. Absent state: unfitted or unpowered: RP_IRQ and RP_BOOTSEL_DRV held low on the host side (blk.soc-vccio-b); UART, SWD, RUN parked low by the SoC; taps on FT pads (G4), 0 Ω positions keep a series value possible at bring-up.
3. IOVDD at 1.8 V: firmware sets VOLTAGE_SELECT = 1 for bank 0 at boot (DS §6.1.1).
4. QSPI pads go to the in-package flash only: open here (HDG §3.1); BOOTSEL pulls QSPI_SS low through 1 kΩ and the N-FET. USB (full speed): 27 Ω at the RP2354A (DS Table 1678), 2.2 Ω at the SoC.
5. GPIO allocation (pinout v1 §3, 2026-10-26): UART0 GPIO0/1, RP_IRQ GPIO2, spare GPIO3 (pad), taps GPIO4-8 for one PIO IN (DS §11.5.6); GPIO9-29 open. The I2C2 option is not wired (G16).

Title: **Coprocessor: RP2354A**				Block: blk.rp2354-core
Purpose: System and GPIO units, 12 MHz oscillator, BOOTSEL, USB series, I2S taps, test pads				
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 36 / 69

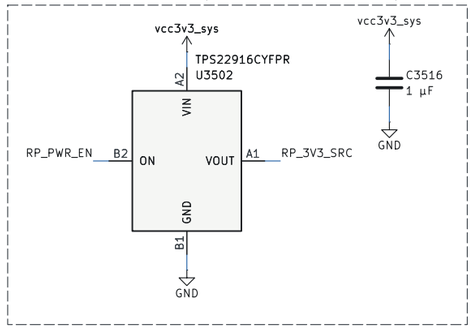
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/Coprocessor/Coprocessor: RP2354A/

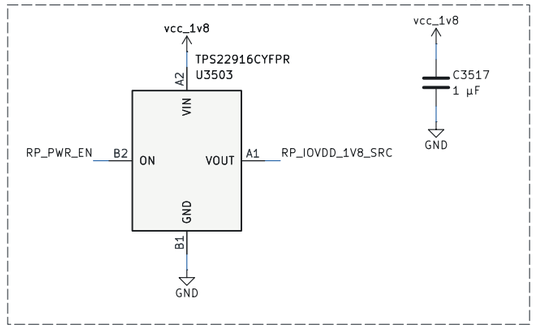
RP2354A supplies, core regulator (DS §6.1, §6.3.8; HDG §2.2.1)



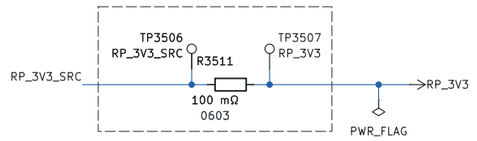
RP_3V3 switch (BOM v1 line 50)



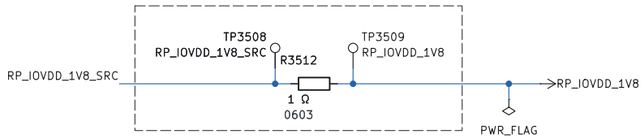
RP_IOVDD_1V8 switch (BOM v1 line 50)



Link #21



Link #30



{RP}

Interface

Net	Direction	Absent state (block not fitted)
RP_PWR_EN	Input	Both switches; SoC pull-down and the TPS22916C ON pull-down: off (BRD-019)
RP_IOVDD_1V8, RP_3V3	Supply out	To blk.rp2354-core (pull-ups, oscillator); off with RP_PWR_EN low

Rails

Rail	Capacitors	Requirement and source
RP_IOVDD_1V8	6 x 100 nF	DS §6.1: 100 nF per power pin
RP_3V3	4.7 µF (VREG_VIN) + 3 x 100 nF	DS §6.1, §6.3.8 (CIN 4.7 µF); QSPI_IOVDD, USB_OTP_VDD, ADC_AVDD
RP_DVDD	4.7 µF at pin 23 + 2 x 100 nF	HDG §2.2.1: 4.7 µF on the furthest DVDD pin
RP_VREG_AVDD	33 Ω + 4.7 µF	HDG p. 7 (RC filter)
vcc3v3_sys, vcc_1v8	1 µF at each switch input	Capture choice (SLVSEN5B)

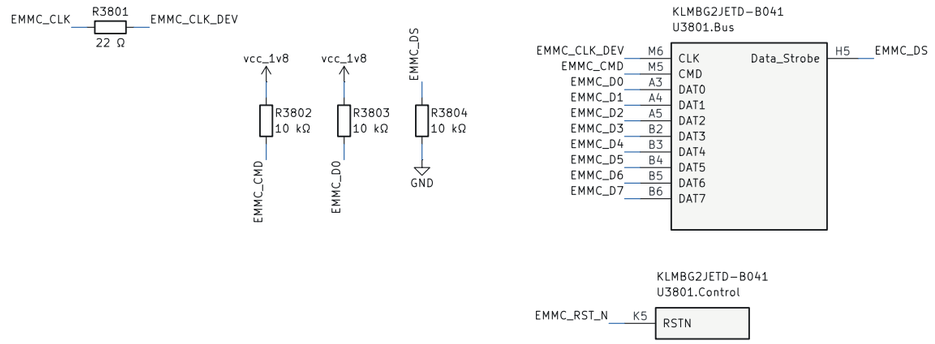
Notes
1. Function: the RP2354A supply unit with its core regulator (3.3 µH, DS §6.3.8) and the two switched rails RP_3V3 (VREG, QSPI, USB, ADC) and RP_IOVDD_1V8 (IOVDD at 1.8 V, BOM v1 line 12).
2. Absent state: RP_PWR_EN low: both switches off (QOD discharges), the part unpowered.
3. VREG_VIN and VREG_AVDD ramp together from RP_3V3 (DS §6.1.8); VREG_AVDD through 33 Ω + 4.7 µF.
4. Links #21 (RP_3V3, 0.1 Ω) and #30 (RP_IOVDD_1V8, 1 Ω), power tree §4.12. The inductor's polarity dot faces VREG_LX as in DS Figure 28: UNVERIFIED until checked on the Pico 2 layout.
5. Supply pins as DS Figure 152: IOVDD 1, 11, 20, 30, 38, 45; QSPI_IOVDD 54 (Table 1679 also lists 54 under IOVDD and leaves out pin 1, which no other function takes: a table erratum).

Title	**Coprocessor: supplies**
Block	blk.rp2354-power
Purpose	RP2354A supply unit, core regulator, RP_3V3 and RP_IOVDD_1V8 switches and links
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026-10-25
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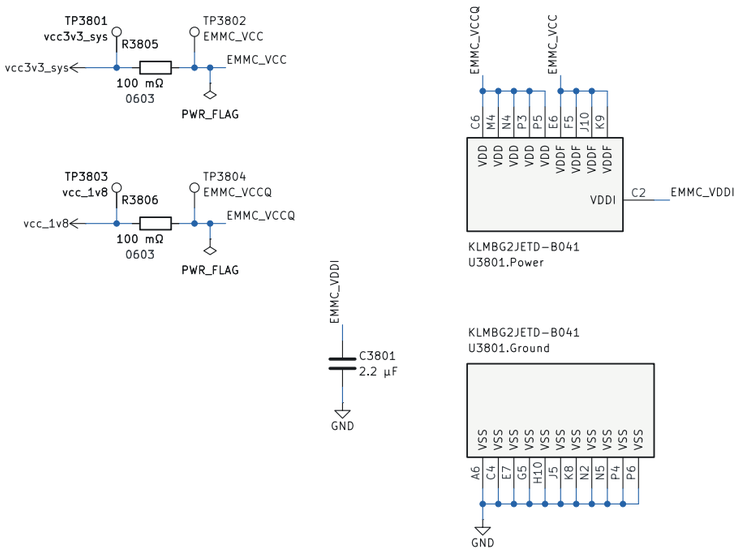
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/Coprocessor/Coprocessor: supplies/

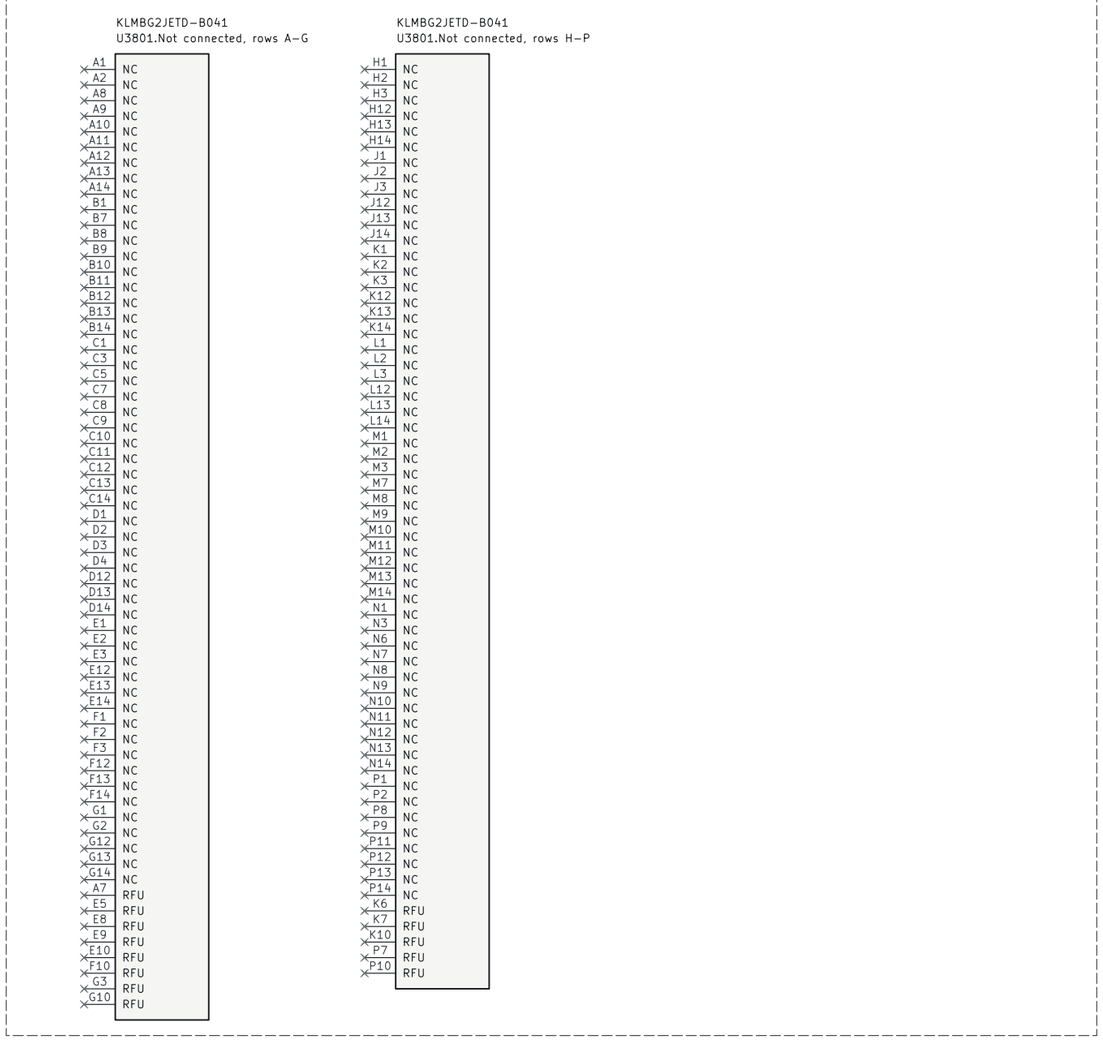
Bus (HDG 1.1 p. 26)



Supplies (power tree §4.12 #14, #15)



Unused balls (NC, RFU: no connection)



{EMMC}

Interface

Net	Direction	Absent state (block not fitted)
EMMC_CLK	Input	Not applicable: blk.emmc is never omitted for boot (BRD-043)
EMMC_CMD, EMMC_D0–D7	Bidirectional	Pull-ups on CMD and D0 here (HDG p. 26)
EMMC_DS	Output	Pull-down here (HDG p. 26)
EMMC_RST_N	Input	SoC ball pull-down at reset (pinout v1)

Rails

Rail	Capacitors	Requirement and source
EMMC_VCC	none drawn	Decoupling not in the data sheet (UNVERIFIED)
EMMC_VCCQ	none drawn	As EMMC_VCC
EMMC_VDDI	2.2 µF	HDG 1.1 p. 26

Notes
1. Function: internal storage. Samsung KLMBG2JETD-B041 eMMC 5.1, 32 GB, 8-bit HS200 (BOM v1 line 10; BRD-037, BRD-043). Variant: production open between 16 and 32 GB; the 16 GB part (KLMAG1JETD-B041, line 138) is a field override of this symbol, written from the variant files.
2. Absent state: not applicable (never omitted). With an empty eMMC the BootROM falls through to microSD or MaskROM (BRD-043).
3. EMMC_CLK: 22 Ω in series at the SoC; CMD and D0 10 kΩ pull-ups; a DS pull-down (HDG 1.1 p. 26). The DS pull-down value (10 kΩ) is a capture choice: the HDG gives none.
4. VDDI: 2.2 µF (HDG 1.1 p. 26; CReg value not in the Samsung data sheet, UNVERIFIED).
5. Links M (0.1 Ω full, 0 Ω production) on VCC and VCCQ (power tree §4.12 #14, #15).
6. Open: VCC and VCCQ decoupling at the eMMC (not in the data sheet, UNVERIFIED). RFU balls: 'do not use for any usage' (Samsung DS p. 5): no connection, as NC.
7. Constraints: PCB-I-06 (EMMC_SD_50SE). The MaskROM pull on D0 belongs to blk.maskrom.

Title: **eMMC storage**

Block: blk.emmc

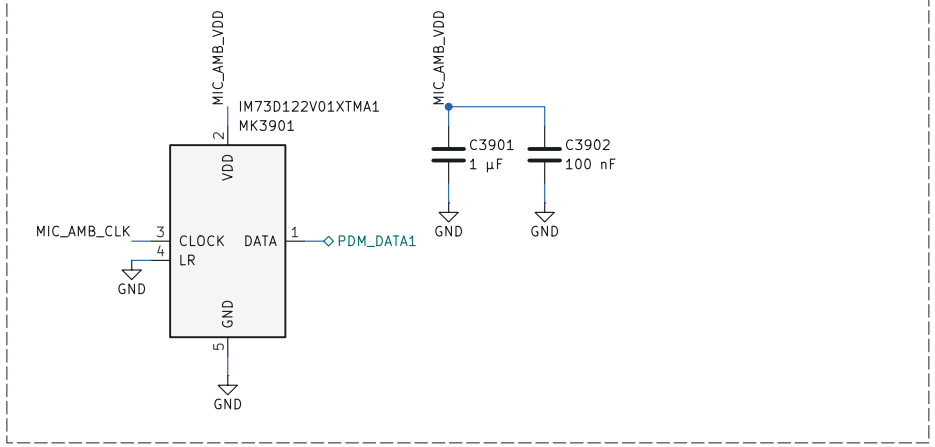
Purpose: 32 GB eMMC 5.1 on the RK3566 eMMC interface, HS200, with its supply links

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	38 / 69

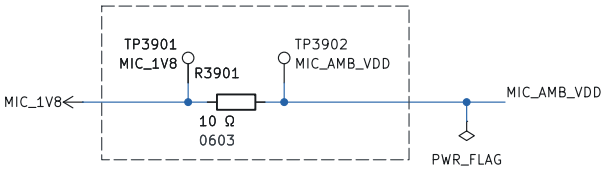
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/eMMC storage/

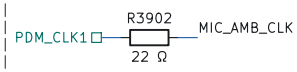
IM73D122 bottom-port PDM microphone (BOM v1 line 34)



Link #35



PDM clock series (pinout v1 AK2)



Interface

Net	Direction	Absent state (block not fitted)
PDM_CLK1	Input	PDM_CLK1_M1 (GPIO4_A0), parked low while MIC_1V8 is off (G10)
PDM_DATA1	Output	PDM_SDI1_M1 (GPIO4_A1); its SoC pull-down holds a level (pinout v1 1P2)
MIC_1V8	Supply in	From blk.mic-main: off with MIC_EN, so the privacy LED covers this microphone

Rails

Rail	Capacitors	Requirement and source
MIC_AMB_VDD	1 µF + 100 nF	IM73D122 DS v1.00 Table 5 note 4 (1 µF) and §3.5 (100 nF)

Notes
1. Function: ambient-noise microphone on the body (BRD-031, should); the second PDM channel of the SoC (PDM CLK1, SDI1), described as a sensor. BRD-027: it shares MIC_1V8 with the faceplate microphone, so it never runs without the privacy LED's supply.
2. Absent state: unfitted: SDI1 held by the SoC pull-down, CLK1 parked low.
3. LR to GND: data driven on the falling clock edge (DATA1, DS v1.00 Table 8), the only microphone on SDI1. 22 Ω series on the clock at the SoC (pinout v1 AK2).
4. VDD from MIC_1V8 through link #35, 10 Ω (power tree §4.12: 1 mA or less, est.).
5. PCB: 0.8 mm acoustic port under the microphone, ground ring as the land (DS v1.00 Fig. 13).

Title	Block
Ambient microphone	blk.mic-ambient

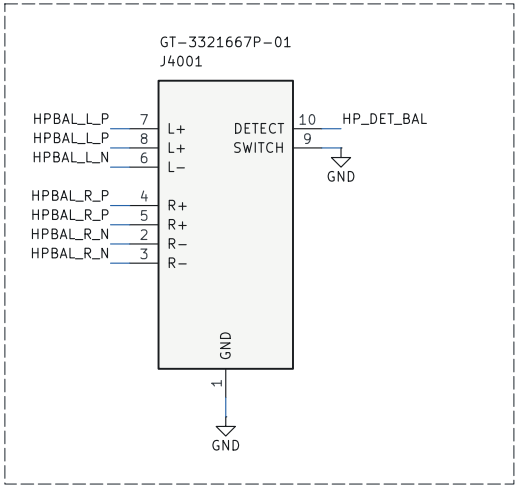
Purpose: IM73D122 PDM microphone on PDM CLK1 and SDI1, on its own MIC_1V8 branch (link #35)

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	39 / 69

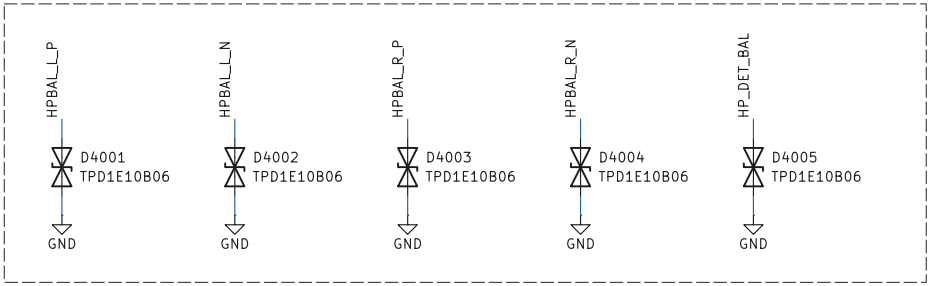
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/Ambient microphone/

4.4 mm jack (GT-3321667P-01)



ESD (BOM v1 line 36)



{HP_BAL}

{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
HPBAL_L_P, _L_N, _R_P, _R_N	Input	Left off with blk.hpa−bal as the 4.4 mm output module (blk.hp−out−bal): the nets go with it
HP_DET_BAL	Output	Pull−up on the host side; reads the detect switch

Notes
1. Function: 4.4 mm 5−pole jack (L+, L−, R+, R−, GND; JEITA RC−8141C, BOM v1 line 31); TPD1E10B06 on each leg and the detect line (line 36).
2. Absent state: passive; left off only with blk.hpa−bal (blk.hp−out−bal). Detect: pin 10 to HP_DET_BAL, pin 9 to ground (switch state UNVERIFIED on the drawing), read by expander P11 with its pull−up on the host side, so the line reads "absent" with the jack left off.
3. Pin−to−pad assignment UNVERIFIED (libs kicad/ note): check against a sample before boards.
4. Open: RF filter footprints near the jack (PCB−A−19, Q2); the Pentaconn alternate land.

Title	Block
4.4 mm headphone jack	blk.jack−bal

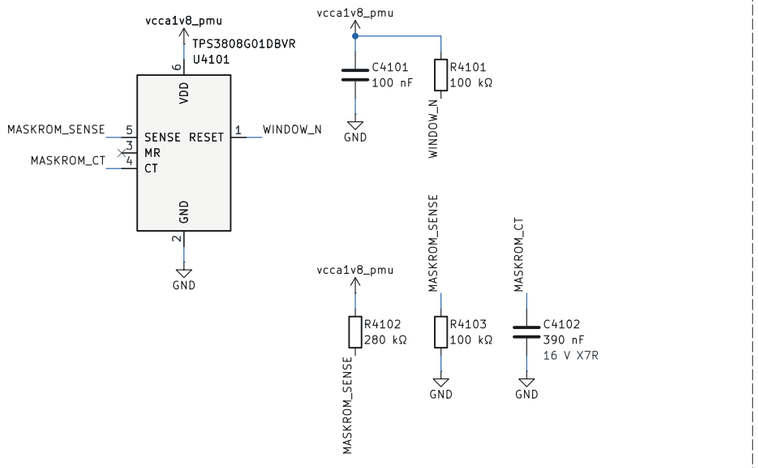
Purpose: Balanced output jack (JEITA RC−8141C) with ESD and plug detect

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026−10−25	40 / 69

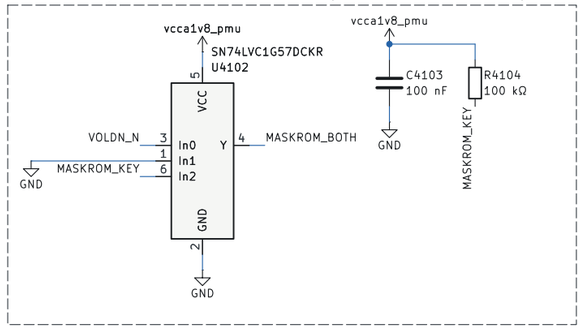
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/4.4 mm headphone jack/

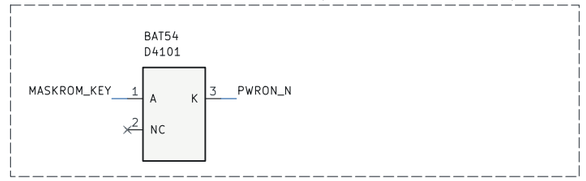
Boot window: TPS3808G01, 1.21–3.43 s (BOM v1 line 97)



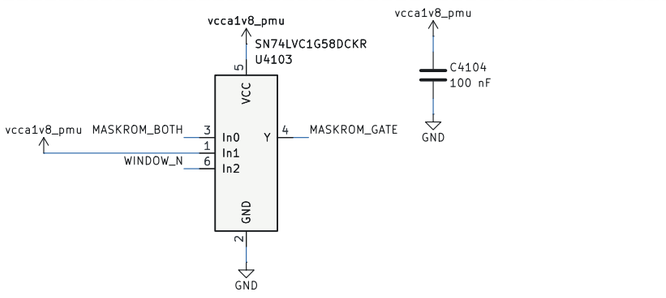
Both keys: SN74LVC1G57 (BOM v1 line 98)



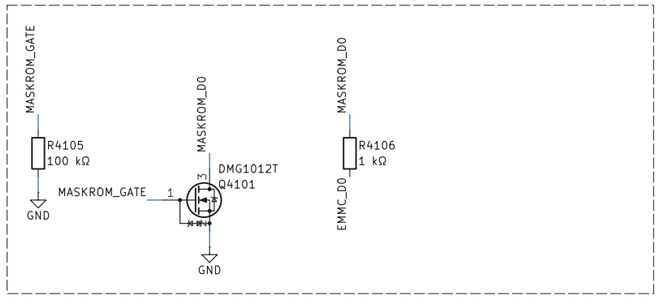
PWRON isolation: BAT54 (BOM v1 line 78)



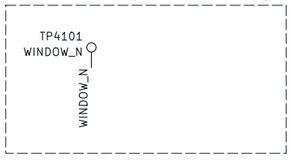
Inside the window: SN74LVC1G58 (BOM v1 line 98)



eMMC D0 pull: DMG1012T, 1 kΩ (BOM v1 line 35)



Test pad (PCB-T-09)



{KEYS}

{EMMC}

Interface

Net	Direction	Absent state (block not fitted)
PWRON_N	Input	RK809 PWRON, through the BAT54 (cathode on PWRON): no load on the power key
VOLDN_N	Input	Vol− with its 100 kΩ pull-up on blk.buttons
EMMC_D0	Bidirectional	Never omitted; the FET off (gate pull-down) leaves D0 free, even with vcca1v8_pmu down

Rails

Rail	Capacitors	Requirement and source
vcca1v8_pmu	100 nF at the TPS3808 and at each gate	SBVS050N (0.1 µF at VDD); gates: capture choice
MASKROM_CT	390 nF X7R ±10 %	BOM v1 line 97: td = CT / 175 nF + 0.5 ms

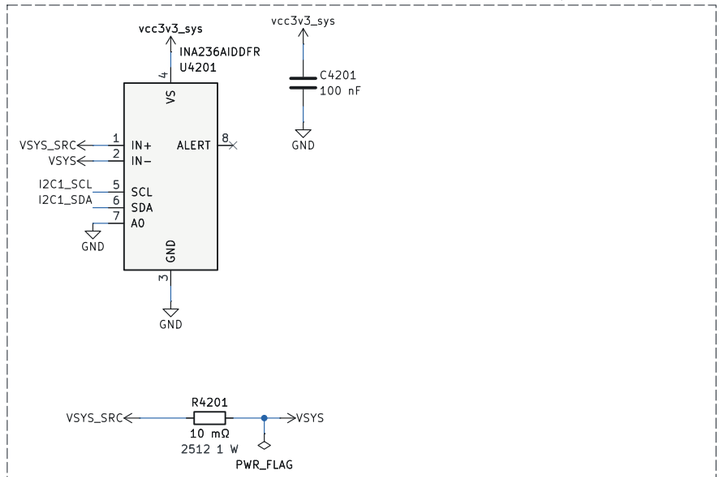
Notes
1. Function: MaskROM entry from the buttons (BRD-035, BRD-043 must; board ADR 0009): Power and Vol- held at a cold power-on pull eMMC_D0 low, so the BootROM falls through to USB download.
2. Absent state: never omitted (always on, vcca1v8_pmu); the gate pull-down holds the FET off and D0 free while vcca1v8_pmu is down (feature-block record 48).
3. Window: TPS3808G01 on vcca1v8_pmu, SENSE 280 kΩ / 100 kΩ (trip 1.54 V), CT 390 nF: td 2.23 s typical, 1.21–3.43 s (the ±40 % of the 130 pF and 180 nF rows, td linear in CT; SBVS050N §7.3.2, 220 nA to 1.23 V; CT ±10 %); WINDOW_N is RESET, open drain, 100 kΩ to vcca1v8_pmu; MR open.
4. Logic on vcca1v8_pmu (Schmitt inputs): 1G57 In2 = PWRON through the BAT54 with 100 kΩ up, In1 = GND, In0 = Vol-: high when both are pressed; 1G58 In2 = WINDOW_N, In1 = VCC, In0 = that: high only inside the window, into the DMG1012T gate (100 kΩ down); its drain pulls D0 through 1 kΩ.
5. Simulated (sim/maskrom): D0 held 1.20–3.42 s after RESETB, never at resume, normal boot untouched; the 1 kΩ stub costs 0.04 V of HS200 margin. A warm restart keeps vcca1v8_pmu up and does not reopen the window: MaskROM through the buttons needs a full power-off.
6. PCB: the 1 kΩ sits on the D0 trace with a stub of 3 mm or less (BOM v1 line 35).

Title: **MaskROM entry**				Block: blk.maskrom
Purpose: Power + Vol− at power-on pull eMMC D0 low inside a boot window: TPS3808, two gates, BAT54, FET				
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 41 / 69

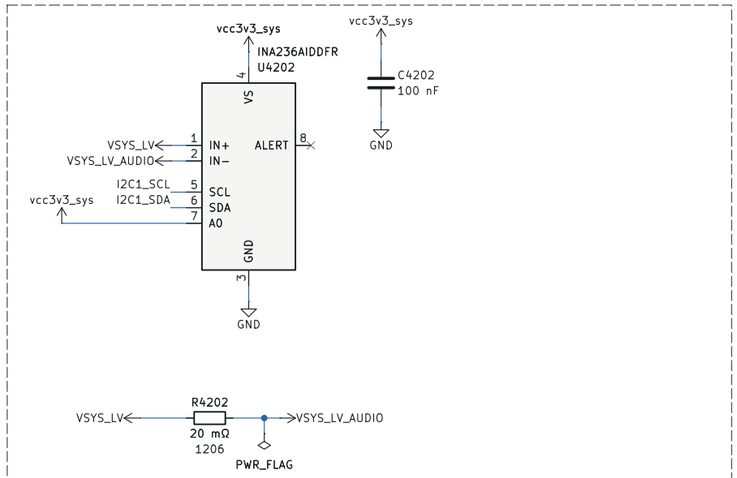
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/MaskROM entry/

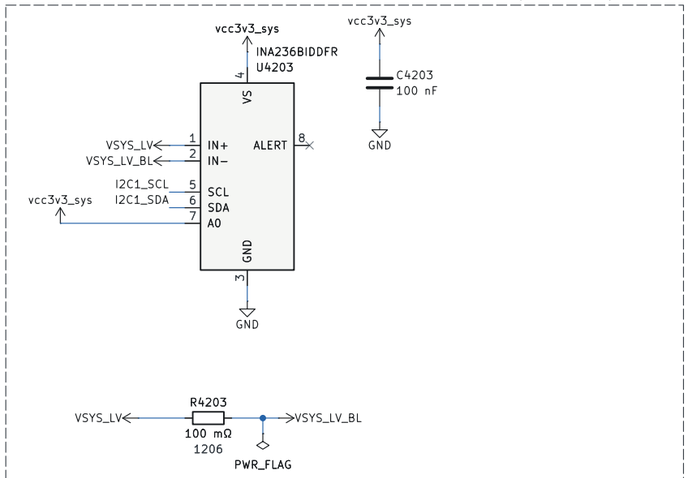
VSYS at 0x40: BQ25798 SYS output, 10 mΩ (line 54)



Audio, VSYS_LV side, at 0x41: 20 mΩ (line 54, G11)



Display at 0x49: TPS61160 VIN, 100 mΩ (line 54)



HAT at 0x43: 5.1 V buck input, 20 mΩ (line 54)

Audio, VSYS side, at 0x48: LMR33630C VIN, 20 mΩ (line 136)

{I2C1}

Interface

Net	Direction	Absent state (block not fitted)
I2C1_SCL, I2C1_SDA	Bidirectional	0x40, 0x41, 0x43 (INA236A), 0x48, 0x49 (INA236B); pull-ups on blk.soc-pmuio
VSYS_SRC, VSYS, VSYS_LV	Supply in	The shunts stay in circuit (power tree §5); VSYS_SRC from the BQ25798 SYS
VSYS_LV_AUDIO, VSYS_AUDIO	Supply out	Through their shunts: blk.audio-supply, blk.mclk, blk.hpa-rails-pre
VSYS_LV_BL, VSYS_P5V1	Supply out	Through their shunts: TPS61160 VIN (blk.fp-backlight), TPS62903 VIN (blk.pwr-5v1)

Rails

Rail	Capacitors	Requirement and source
vcc3v3_sys	100 nF per INA236	SBOSA81D §8.3: 0.1 µF at VS

Notes
1. Function: power instrumentation (T1-026, must): five INA236 on I2C1 at 1.8 V (VIH 0.9 V at any VS, SBOSA81D §6.5), VS on vcc3v3_sys; positions and shunts per power tree §4.12 (G11).
2. Absent state: an INA236 unfitted leaves its shunt in circuit and its address free (power tree §5). The parts power up converting (MODE 111b, 0.3 mA each): Blackbird OS writes power-down.
3. Shunts (Vishay WSL, doc 30100): 10 mΩ 2512 1 W (VSYS: 58 mW at 2.4 A, 0.36 W at the 6 A BATFET rating); 20 mΩ 2010 0.5 W (HAT, line 54 asks for 0.5 W or more); 20 mΩ 1206 (audio); 100 mΩ 1206 (display: 11 mW at 0.33 A). Full scale 81.92 mV / R: 8.2, 4.1, 0.82 A.
4. A0 (SBOSA81D Table 7-1): GND, VS, SCL for 0x40, 0x41, 0x43; INA236B GND, VS for 0x48, 0x49. No A0 = SDA: it needs 100 ns more hold (p. 16), and U-Boot leaves the RK3566 hold at 1 clock.
5. ALERT open (may float, SBOSA81D Table 5-1). Kelvin traces from IN+ and IN- to the shunt pads (SBOSA81D §8.4.1); never issue a general-call reset on I2C1 (it resets all five, p. 18).

Title	Block
Current monitors	blk.current-monitors

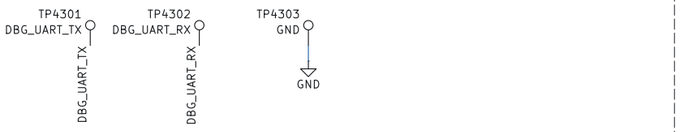
Purpose: Five INA236 on I2C1 and their shunts: VSYS, audio (VSYS_LV and VSYS sides), display, HAT

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	42 / 69

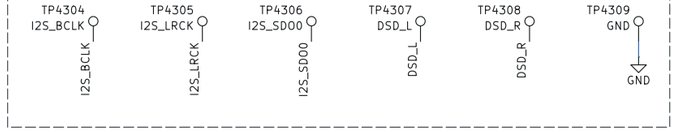
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/Current monitors/

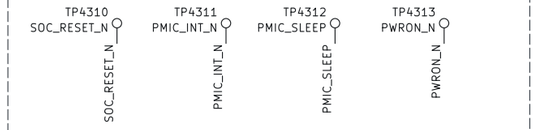
Debug UART2 M0, 1.8 V (BOM v1 line 87; PCB-T-01)



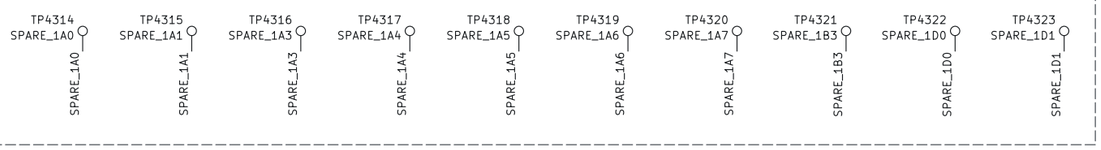
I2S and DSD (BOM v1 line 88; BRD-040)



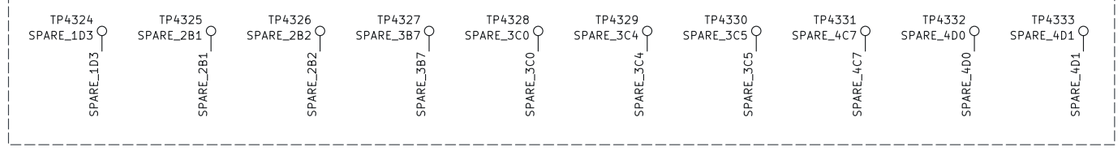
Reset, PMIC and power key (PCB-T-09)



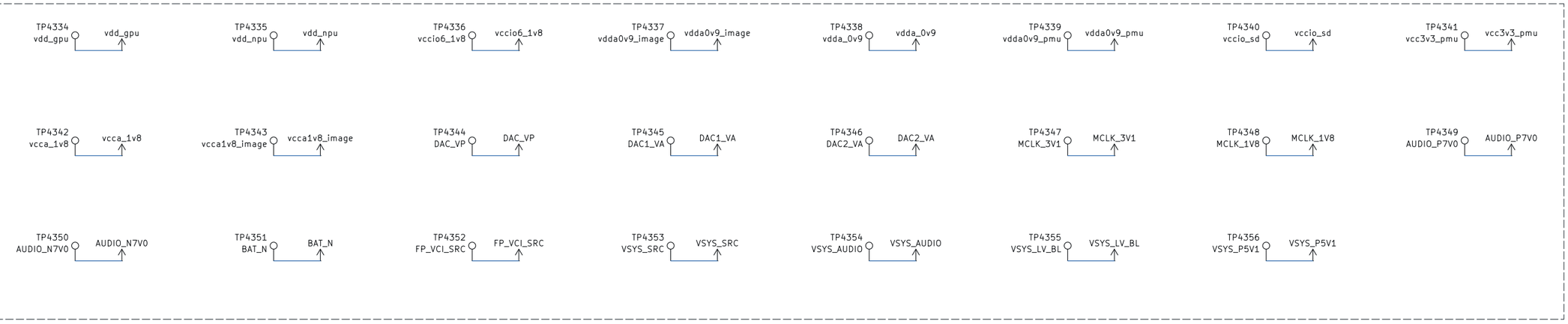
Spare SoC GPIOs (pinout v1; PCB-T-09)



Spare SoC GPIOs, continued



Rail test pads (PCB-T-02, BOM v1 line 86): the rails no other sheet probes



{PMIC_CTL}

{KEYS}

{DEBUG}

{AUDIO_I2S}

Interface

Net	Direction	Absent state (block not fitted)
DBG_UART_TX, DBG_UART_RX	Bidirectional	Pads only (1.8 V, PMUIO2); the SoC pull-ups hold the idle level
I2S_BCLK, I2S_LRCK, I2S_SDO0, DSD_L, DSD_R	Input	Pads only on the DAC path (BRD-040); no load
SOC_RESET_N, PMIC_INT_N, PMIC_SLEEP, PWRON_N	Input	Pads only
SPARE_*	Bidirectional	Pads only; each ball keeps its reset pull (pinout v1)

Rails

Rail	Capacitors	Requirement and source
(none)	–	Pads only: no decoupling on this sheet

Notes
1. Function: test and debug access (BRD-040, BRD-047 must; T1-026): the BootROM, TPL and U-Boot console on UART2 M0 at 1.8 V (a 1.8 V USB-UART adapter), the I2S/DSD lines at the DAC path, reset and PMIC pads, the spare SoC GPIOs, and a pad on every rail no other sheet probes.
2. Absent state: pads only, nothing fitted. The block's measurement links sit on their blocks' sheets (power tree §4.12); the RP2354A SWD pads on blk.rp2354-core; one MCLK pad on each DAC sheet.
3. Spares pinout v1 leaves without a pad (the audio-bank balls next to I2S1, FSPI_D3) carry no-connect flags on the SoC sheets; GPIO0_A6 is the cover detect input (blk.cover-detect).
4. I2S/DSD pads: BCLK is also the DSD clock (DSDCLK from I2S_BCLK, blk.dac-l/-r); stubs from the DAC path at most 1 mm, as the coprocessor taps (PCB-A-12).

Title	Block
Test and debug access	blk.debug

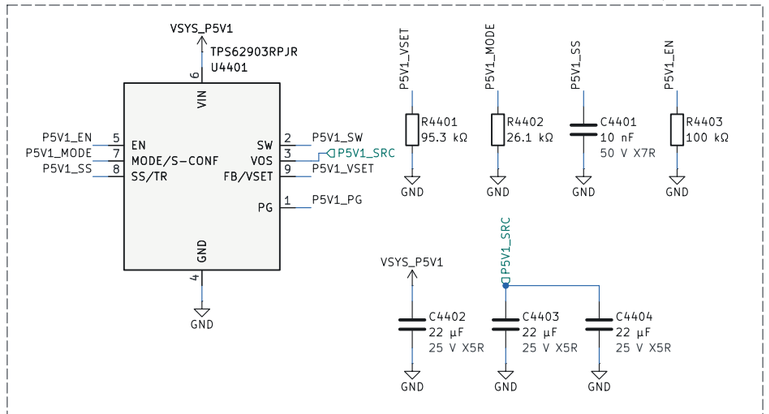
Purpose: Debug UART, I2S/DSD and bring-up pads, spare GPIO pads and rail test pads

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	43 / 69

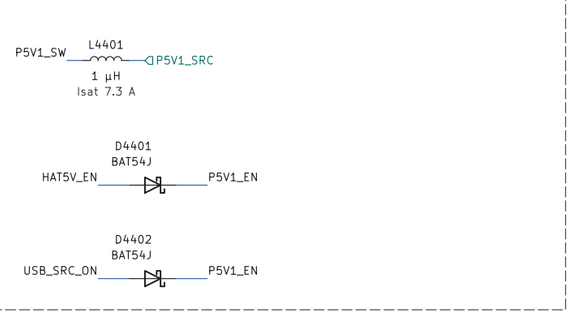
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/Test and debug access/

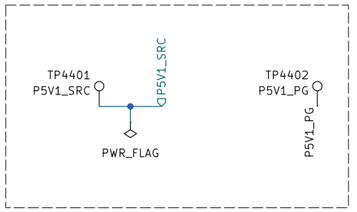
TPS62903, 5.1 V, 2.5 MHz (BOM v1 lines 91, 92, 121)



Inductor and enable OR (BOM v1 line 91)



Test pads



{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
VSYS_P5V1	Supply in	VSYS through the HAT INA236 shunt (0x43, blk.current-monitors)
HAT5V_EN, USB_SRC_ON	Input	Expander P02, P03, pulled down: the buck off
P5V1_SRC	Supply out	Off, discharged (output discharge on, MODE 26.1 kΩ)

Rails

Rail	Capacitors	Requirement and source
VSYS_P5V1	22 µF 25 V	SLVSES3A Table 8-1: 3 µF effective minimum (§6.3): 3.9 µF at 8.8 V (BOM v1.18 item 1)
P5V1_SRC	2 x 22 µF 25 V 0805	BOM v1 line 121: ≥ 12 µF effective at 5.1 V: 13.5 µF (BOM v1.18 item 5)

Notes
1. Function: TPS62903 VSYS (VSYS_P5V1) to 5.1 V, 1.6 A design (53 % of 3 A), for the HAT VBUS and the USB-C source through eFuse #D (ADR 0014: power tree §4.6).
2. Absent state: EN pulled down (100 kΩ): off, output discharged.
3. EN = HAT5V_EN OR USB_SRC_ON in hardware: one BAT54J from each (BOM v1 line 91).
4. VSET 95.3 kΩ (5.1 V), MODE/S-CONF 26.1 kΩ (2.5 MHz, discharge on, auto PFM/PWM, Table 7-1), SS 10 nF: capture choice (ISS 2.5 µA). Phase margin at VSYS 6.0 V: bench item (§7 item 27).

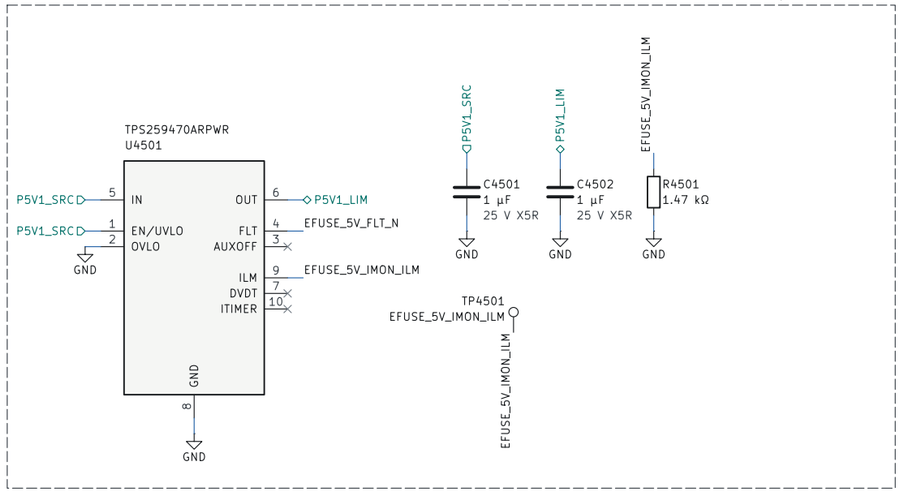
Title	Block
5.1 V source	blk.pwr-5v1

Purpose
TPS62903 buck from VSYS for the HAT and USB-C sources, enabled by either port

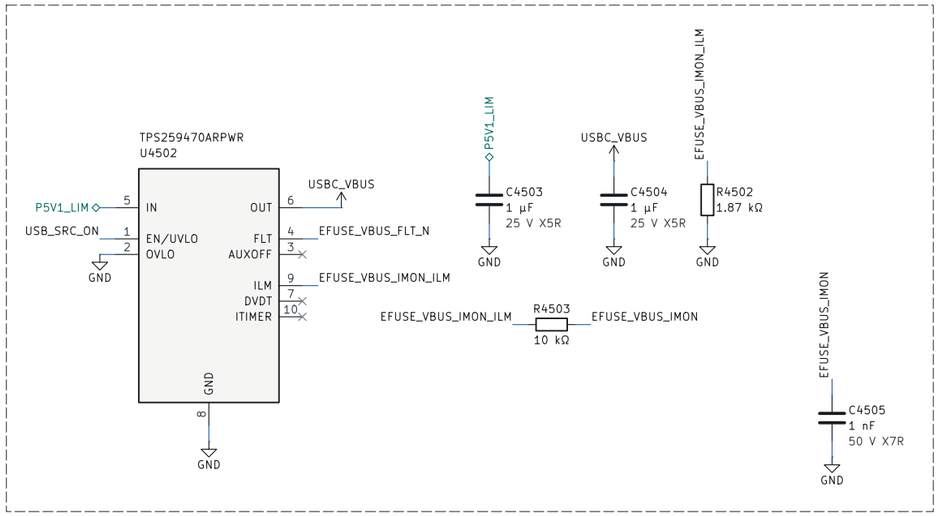
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	44 / 69

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/5.1 V source/

eFuse #D, shared limiter, RILM 1.47 kΩ (BOM v1 line 44, v1.14)



eFuse #C, USB-C source, RILM 1.87 kΩ (line 44)



{IMON}

{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
P5V1_SRC	Supply in	From blk.pwr-5v1
P5V1_LIM	Supply out	To eFuse #A (blk.hat-5v)
USB_SRC_ON	Input	Expander P03, pulled down: #C off
EFUSE_5V_FLT_N, EFUSE_VBUS_FLT_N	Output	Open drain to expander P16, P17 (pull-ups there)
EFUSE_VBUS_IMON	Output	SARADC_VIN3, 1 nF at the ball

Rails

Rail	Capacitors	Requirement and source
P5V1_SRC, P5V1_LIM, USBC_VBUS	1 µF 25 V at each IN and OUT	SLVSFC9C §8.3

Notes
1. Function: eFuse #D limits the 5.1 V source for both ports together (2.021 / 2.268 / 2.520 A); eFuse #C sources the USB-C VBUS (1.589 / 1.783 / 1.981 A) under USB_SRC_ON (RFC 0015).
2. Absent state: USB_SRC_ON pulled down: #C off, TPS2121 OV1 released.
3. #D EN/UVLO on its own input: on with the buck, OVLO to GND: no overvoltage lockout.
4. ITIMER and DVDT open (fastest response, BOM v1 line 44). IMON through 10 kΩ, keeping the 1 nF of the SARADC off the ILM node (< 50 pF, SLVSFC9C p. 35); #D IMON on a test pad.

Title	Block
USB-C source path	blk.usb-vbus-out

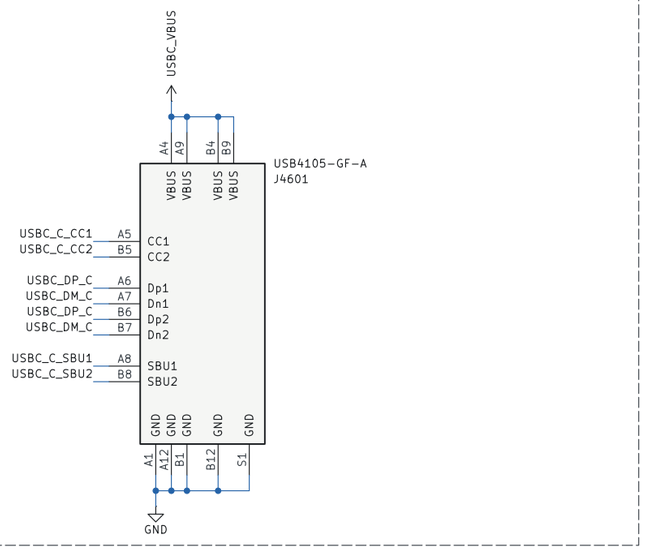
Purpose: eFuse #D, the shared 5.1 V limiter, and eFuse #C to the USB-C VBUS

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	45 / 69

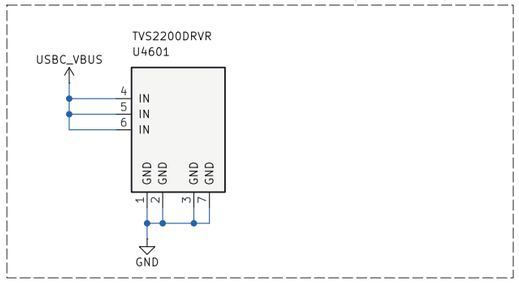
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/USB-C source path/

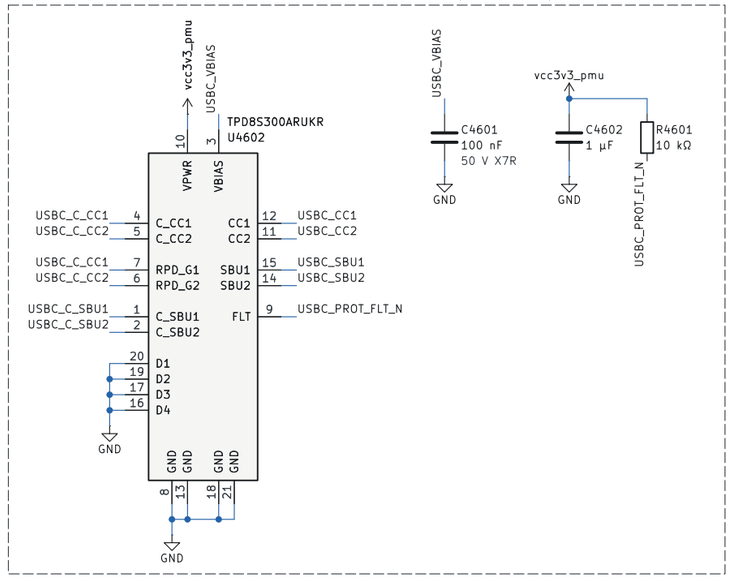
Receptacle: GCT USB4105-GF-A (BOM v1 line 61)



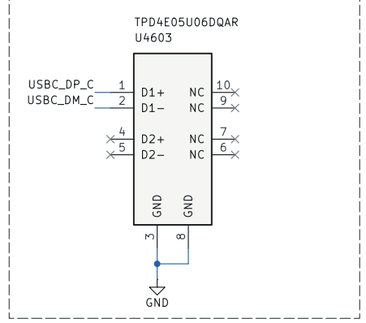
VBUS TVS, 22 V flat clamp (line 64)



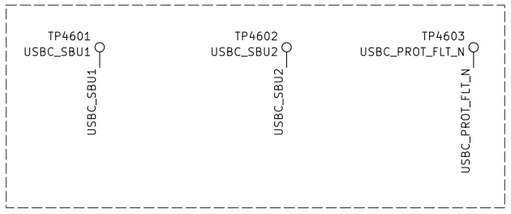
CC and SBU protection: TPD8S300A (line 63)



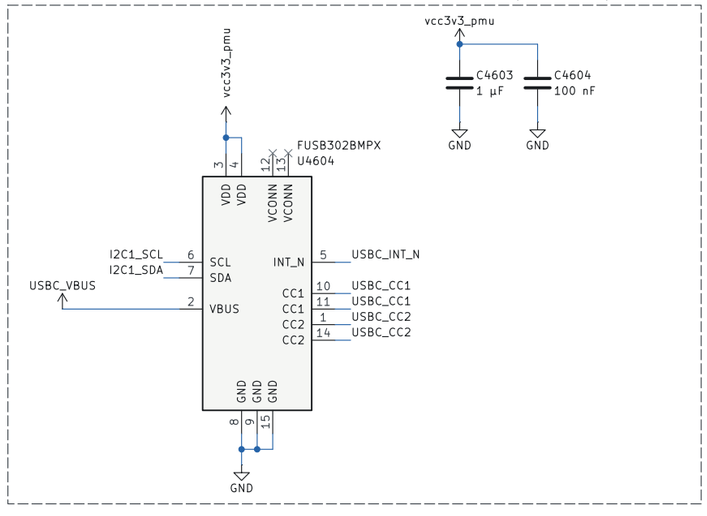
D± ESD, 0.5 pF (line 62)



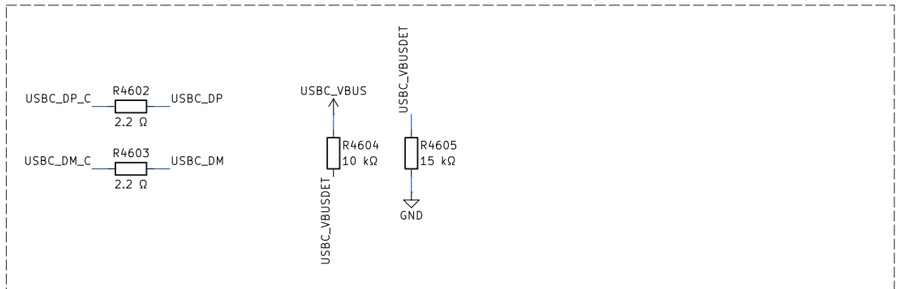
SBU and fault test pads



Port controller: FUSB302B at I2C1 0x22 (line 41)



D± series 2.2 Ω (HDG p. 56); VBUSDET divider (HDG Figure 2-43)



{USB} ◇—

{IO_IRQ} ◇—

{I2C1} ◇—

Interface

Net	Direction	Absent state (block not fitted)
USBC_VBUS	Supply in / out	Sink to the TPS2121 IN1 (blk.charger); source from eFuse #C
USBC_DP, USBC_DM	Bidirectional	OTG0, 2.2 Ω here; MaskROM download port
USBC_VBUSDET	Output	OTG0 VBUSDET, 10 kΩ / 15 kΩ from VBUS
USBC_INT_N	Output	FUSB302B INT_N, open drain; pull-up on blk.soc-pmuio
I2C1_SCL, I2C1_SDA	Bidirectional	FUSB302B at 0x22

Rails

Rail	Capacitors	Requirement and source
vcc3v3_pmu	1 µF + 100 nF (FUSB302B); 1 µF (TPD8S300A VPWR)	Datasheets: TPD8S300A §10
USBC_VBIAS	100 nF 50 V	TPD8S300A §9.2.2.1: ≥ 35 V rated

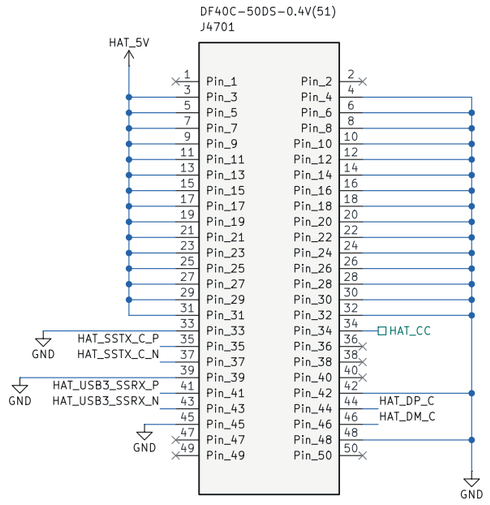
Notes
1. Function: the main USB-C receptacle: USB 2.0 OTG0 and USB PD sink (5–12 V into the TPS2121) or source (5 V from eFuse #C); FUSB302B with the mainline fusb302 + tcpm drivers.
2. Absent state: never omitted. Dead battery: TPD8S300A RPD_Gx tied to C_CCx so a source sees Rd with the protection unpowered (§9.2.2.2); its CC FETs pass once vcc3v3_pmu is up.
3. D±: TPD4E05U06 (0.5 pF) at the receptacle; the TPD8S300A D1–D4 clamps are grounded as unused (§9.2.2.6). Common-mode choke footprint (HDG Figure 2-44): not fitted, open item.
4. VBUSDET: HDG divider 10 kΩ / 15 kΩ (3.0 V at 5 V); the pin's rating at 9–12 V VBUS is UNVERIFIED (BOM §7 item 41): bench item before a PD contract above 5 V is enabled.
5. VCONN not sourced (no e-marked cable supply): VCONN pins open, UNVERIFIED termination.
6. SBU to test pads (debug UART option, feature blocks #9); the TPD8S300A FLT on a test pad.

Title	**Main USB-C port**
Block	blk.usb-c
Purpose	USB4105 receptacle, VBUS TVS, CC/SBU protection, D± ESD and the FUSB302B port controller
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026-10-25
Sheet	46 / 69

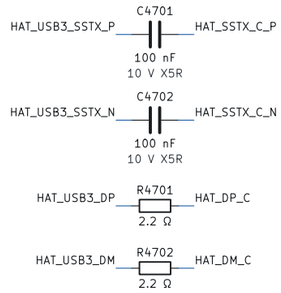
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/Main USB-C port/

Flat-cable receptacle: DF40C-50DS (BOM v1 line 66; interface v2 §9.2)



USB 3 TX AC coupling, USB 2 series (lines 70; HDG p. 59)



{USB}

Interface

Net	Direction	Absent state (block not fitted)
HAT_USB3_SSTX_P/N, _SSRX_P/N	Bidirectional	USB3 HOST1 SuperSpeed (90 Ω)
HAT_USB3_DP, HAT_USB3_DM	Bidirectional	USB3 HOST1 USB 2 half, 2.2 Ω here
HAT_CC	Bidirectional	From the FUSB302B01 (blk.hat-5v)
HAT_5V	Supply in / out	15 contacts, 0.3 A each (interface v2 §9.2)

Rails

Rail	Capacitors	Requirement and source
None	No decoupling: the 100 nF are the TX AC-coupling capacitors	RK3566 HDG 1.1 p. 59

Notes
1. Function: the HAT USB port on the board end of the flat cable (RFC 0015): USB 3 Gen 1, USB 2, CC and VBUS to the device-side piece and its pogo field (hats standard/ interface v2).
2. Absent state: HAT or cable absent: no attach; the AC caps keep a dangling cable off the TX pair.
3. Contacts per interface v2 §9.2: VBUS on 3-31 odd, GND on 4-32 even, 33, 39, 42, 45, 48; CC 34; SSTX 35/37; SSRX 41/43; D± 44/46; 36, 38, 40, 47 spare; 1, 2, 49, 50 metal fittings. Contact numbering against the Hirose drawing is UNVERIFIED (interface v2 §11 item 5).
4. ESD and the VBUS TVS are on the device-side piece, none repeated here (BOM lines 65, 68, 69).

Title	Block
HAT USB port	blk.hat-signals

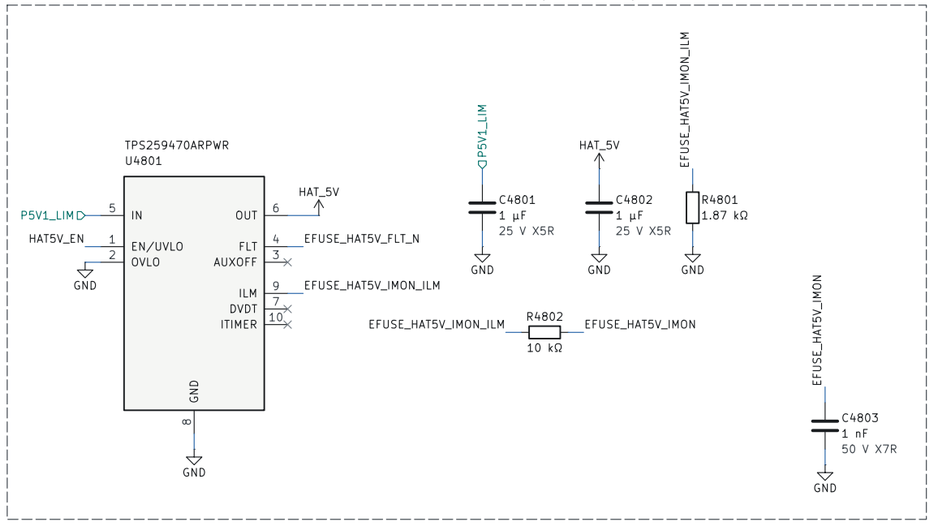
Purpose: DF40C-50 flat-cable receptacle: USB 3 with AC coupling, USB 2, CC, VBUS and GND

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	47 / 69

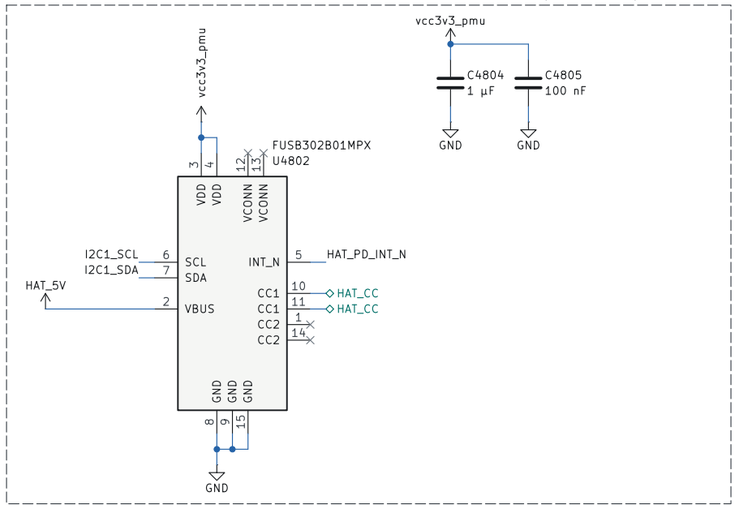
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/HAT USB port/

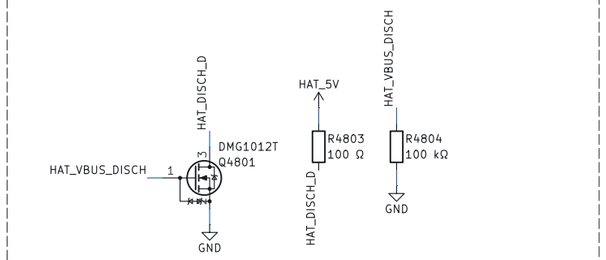
eFuse #A, HAT VBUS source, RILM 1.87 kΩ (line 44)



Port controller: FUSB302B01 at I2C1 0x23 (line 41)



VBUS discharge (BOM v1.13 changes item 3)



{IO_IRQ}
{IMON}
{I2C1}
{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
P5V1_LIM	Supply in	From eFuse #D (blk.usb-vbus-out)
HAT_5V	Supply in / out	HAT VBUS: source through #A, sink into the TPS2121 IN2 (blk.charger)
HAT5V_EN, HAT_VBUS_DISCH	Input	Expander P02, P01, pulled down: source off, no discharge
HAT_CC	Bidirectional	To the HAT connector (blk.hat-signals); Rd present unpowered
HAT_PD_INT_N	Output	Open drain; 4.7 kΩ pull-up to vcc3v3_pmu on blk.soc-pmuio
EFUSE_HAT5V_IMON	Output	SARADC_VIN2, 1 nF at the ball

Rails

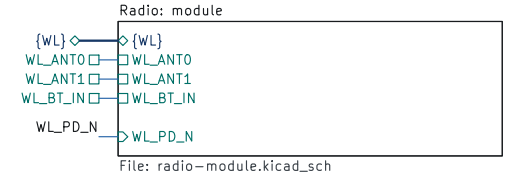
Rail	Capacitors	Requirement and source
vcc3v3_pmu	1 µF + 100 nF	FUSB302B/D
P5V1_LIM, HAT_5V	1 µF 25 V at IN and OUT	SLVSFC9C §8.3

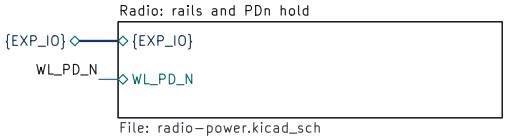
Notes
1. Function: the HAT port's VBUS under USB PD (hats standard/ interface v2, RFC 0014): eFuse #A sources 5 V (TCPM vbus-supply on HAT5V_EN), the TPS2121 IN2 sinks 5 V from a source HAT.
2. Absent state: HAT5V_EN low, VBUS discharged; the FUSB302B01 presents Rd on CC, also unpowered (dead-battery support), so a source HAT can feed the T1.
3. One CC (the pogo field is keyed): CC2 open; VCONN not sourced, open (UNVERIFIED termination).
4. Discharge: DMG1012T with 100 Ω (50 mA at 5 V), gate on HAT_VBUS_DISCH after the source is off; the value against the PD discharge time and the port capacitance is UNVERIFIED (§7 item 47).
5. The VBUS TVS and the ESD arrays sit on the device-side piece at the pogo field (BOM lines 65, 68, 69).

Title: **HAT port VBUS and USB PD**	Block: blk.hat-5v
Purpose: eFuse #A to the HAT VBUS, the FUSB302B01 port controller and the VBUS discharge	
Project, board: OpenDAP T1, main board	Variant: full / Rev: A / Date: 2026-10-25 / Sheet: 48 / 69

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/HAT port VBUS and USB PD/





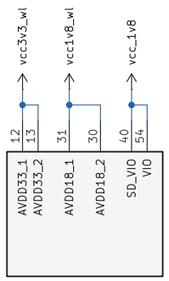
Notes
1. Function: Murata Type 2EL (NXP IW612): Wi−Fi 6 on SDIO 3.0, Bluetooth 5.4 on UART1 with flow control (feature blocks #39), drawn on two sub−sheets: the module, and its rails with the PDn hold translator.
2. Absent state: WL_PWR_EN low: both module rails off, PDn low, wake lines pulled to no wake.
3. Reserved for a PCIe + USB 2x2 module variant (a block revision): the RK3566 PCIe sideband balls and USB HOST3, unconnected (pinout v1, 2026−10−25).

Title	Block
Wi−Fi and Bluetooth module	blk.radio
Purpose: Murata Type 2EL (IW612): SDIO, UART, wake lines, its 3.3 V and 1.8 V rails	

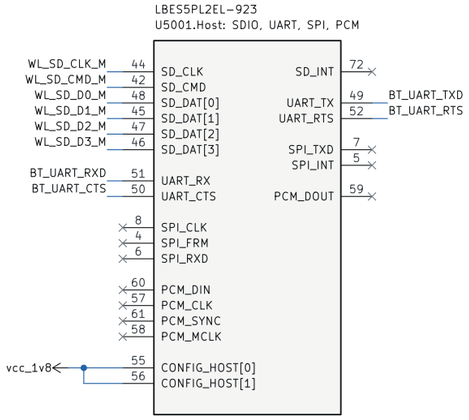
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026−10−25	49 / 69

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/Wi−Fi and Bluetooth module/

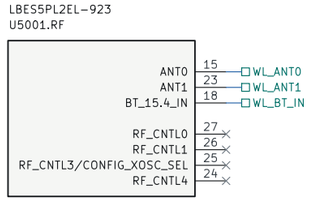
Murata Type 2EL, LBES5PL2EL-923 (BOM v1 line 55)



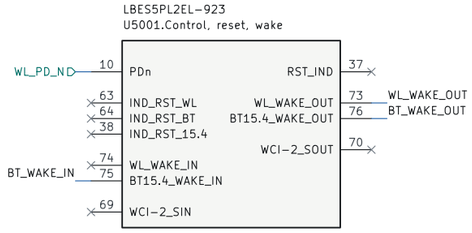
LBES5PL2EL-923
U5001.Power



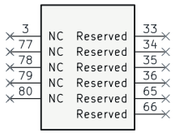
LBES5PL2EL-923
U5001.Host: SDIO, UART, SPI, PCM



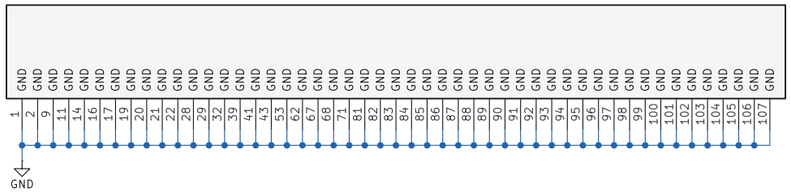
LBES5PL2EL-923
U5001.RF



LBES5PL2EL-923
U5001.Control, reset, wake

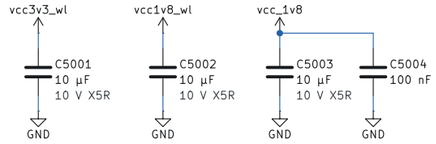


LBES5PL2EL-923
U5001.NC and reserved

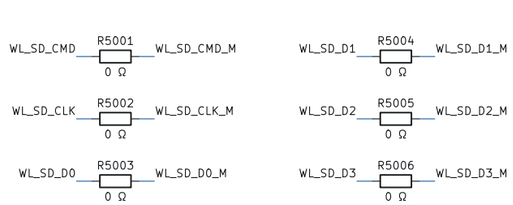


LBES5PL2EL-923
U5001.Ground

Module decoupling (E2B-09-1413 Figure 5)



SDIO series positions (E2B-09-1413 Figure 5 note 3)



{WL}

Interface

Net	Direction	Absent state (block not fitted)
WL_SD_CLK, _CMD, _D0–D3	Bidirectional	SDMMC1 at 1.8 V (VCCIO4); SoC pulls, parked low
BT_UART_TXD, _RXD, _CTS, _RTS	Bidirectional	UART1 MO with flow control (module-side names)
WL_WAKE_OUT, BT_WAKE_OUT	Output	PMUIO2 wake balls, SoC pull-downs: no wake while off
BT_WAKE_IN	Input	SoC pull-down
WL_PD_N	Input	From blk.radio-power
WL_ANT0, WL_ANT1, WL_BT_IN	Passive	To blk.antennas

Rails

Rail	Capacitors	Requirement and source
vcc3v3_wl	10 µF at AVDD33	E2B-09-1413 Figure 5
vcc1v8_wl	10 µF at AVDD18	E2B-09-1413 Figure 5
vcc_1v8	10 µF at SD_VIO, 100 nF at VIO	E2B-09-1413 Figure 5 (VIO not drawn there: capture choice)

Notes
1. Function: Murata Type 2EL (NXP IW612): Wi-Fi 6 on SDIO 3.0, Bluetooth 5.4 on UART1 with flow control; nxpwifi and btnxpuart (DKMS until Linux 7.3). 802.15.4 SPI not wired (T1-077).
2. Absent state: rails off (blk.radio-power), wake lines pulled to no wake by the SoC.
3. CONFIG_HOST[1:0] = 11 (SDIO, UART, SPI hosts; Figure 5 note 2). RF_CNTL, GPIO, PCM, WCI-2, reset indicators and reserved pins open (E2B-09-1413 §3.3); module land from Murata's DXF, mask and stencil UNVERIFIED (module datasheet not obtained).
4. SDIO series positions fitted with 0 Ω; damping values at bring-up. UART direct.

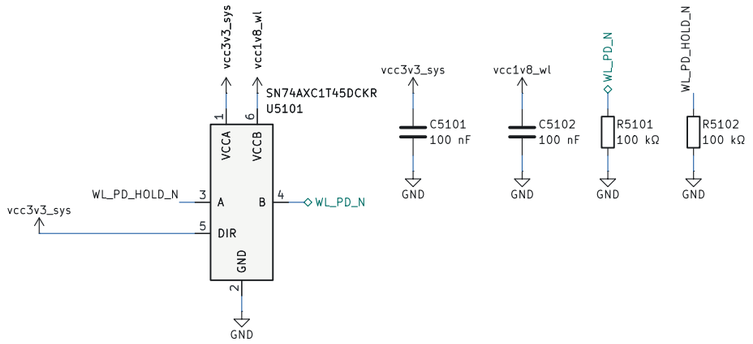
Title	Block
Radio: module	blk.radio-module
Purpose	Murata Type 2EL by unit, its decoupling and the SDIO series positions

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	50 / 69

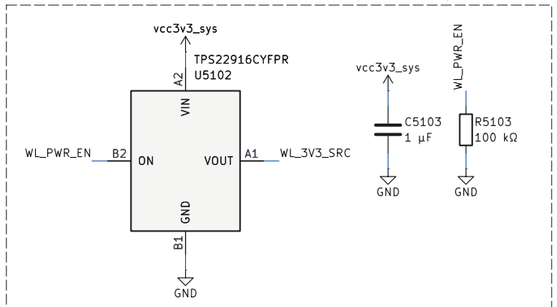
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/Wi-Fi and Bluetooth module/Radio: module/

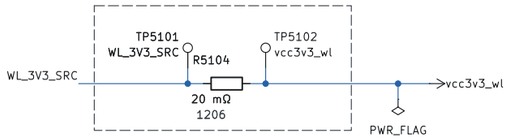
PDn hold translator: SN74AXC1T45 (line 148)



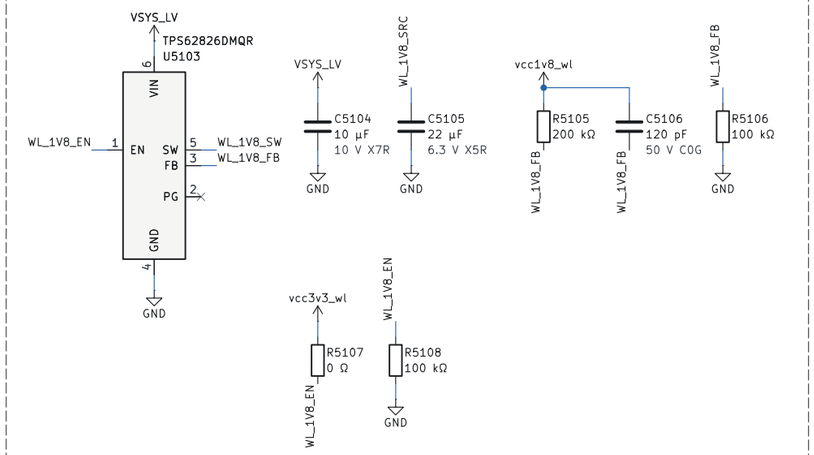
AVDD33 switch: TPS22916C (line 50)



Link #11



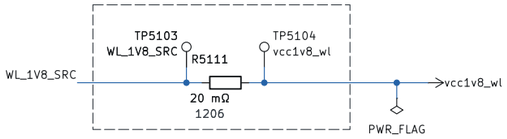
AVDD18 buck: TPS62826, 1.8 V (lines 129–131)



Inductor, EN option and discharge positions (lines 130, 131)



Link #12



{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
WL_PWR_EN, WL_PD_HOLD_N	Input	Expander P06, P07, 100 kΩ pull-downs here: rails off, PDn low
vcc3v3_wl, vcc1v8_wl	Supply out	Module AVDD33 and AVDD18 (blk.radio-module)
WL_PD_N	Output	Module PDn, held low while the radio is off

Rails

Rail	Capacitors	Requirement and source
vcc3v3_sys	1 µF at the switch, 100 nF at VCCA	SLVSEN5B; SCES882E
vcc1v8_wl	22 µF at the buck; 100 nF at VCCB	BOM v1 line 131
VSYS_LV	10 µF 10 V X7R	BOM v1 line 131

Notes
1. Function: the module rails: vcc3v3_wl through a TPS22916C on WL_PWR_EN, vcc1v8_wl from a TPS62826 enabled by vcc3v3_wl, links #11 and #12 (power tree §4.12), and the PDn hold.
2. Absent state: WL_PWR_EN low: vcc3v3_wl off and the AVDD18 buck with it (EN from vcc3v3_wl); PDn low through the SN74AXC1T45 (its B port unpowered, Ioff).
3. AVDD18: 200 kΩ / 100 kΩ (1.80 V), 120 pF; link #12 inside the feedback (power tree §4.12). The DNP 0 Ω from WL_1V8_EN_GPIO and the DNP 1 kΩ discharge are bring-up options (lines 129, 131).
4. Sequencing between AVDD33, AVDD18 and VIO: UNVERIFIED (module datasheet not obtained).

Title	Block
Radio: rails and PDn hold	blk.radio-power

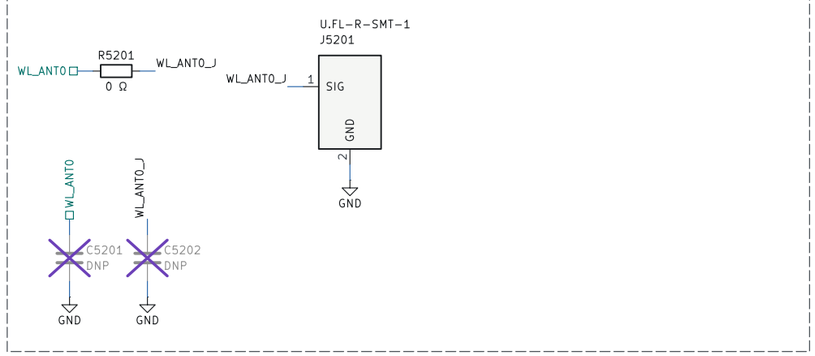
Purpose: AVDD33 switch, AVDD18 buck, links #11 and #12, the PDn hold translator

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	51 / 69

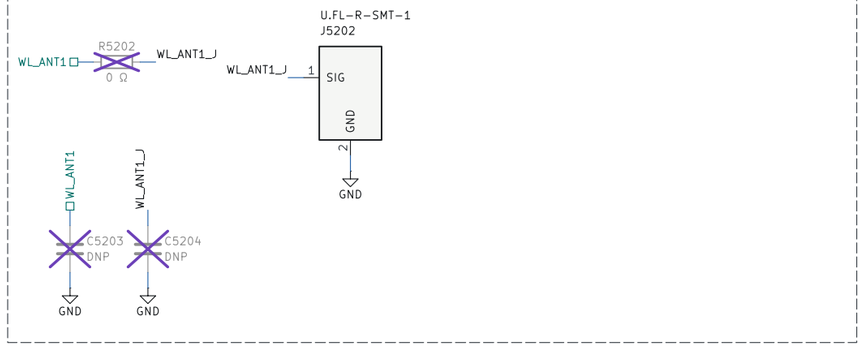
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/Wi-Fi and Bluetooth module/Radio: rails and PDn hold/

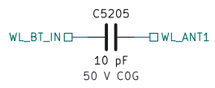
ANT0: Wi-Fi 2.4 / 5 GHz and Bluetooth (shared, ADR 0018)



ANT1: dedicated configuration, not fitted (ADR 0018 item 4)



Shared link: pin 18 to pin 23, 10 pF (E2B-54-1953 §4.3.3)



Interface

Net	Direction	Absent state (block not fitted)
WL_ANT0, WL_ANT1, WL_BT_IN	Passive	Module RF pins (blk.radio-module)

Rails

Rail	Capacitors	Requirement and source
None	No supply: RF matching positions and the 10 pF link only	ADR 0018

Notes
1. Function: both RF paths as Murata's copy (0603 positions, LQP03 / GRM03 class): ANT0 to the U.FL of the Molex 146153-0050 (fitted), ANT1 to a second U.FL (not fitted) (ADR 0018).
2. Absent state: passive. Shared configuration: series 0 Ω on ANT0 fitted, shunts DNP; the 10 pF link from BT_15.4_IN to ANT1 fitted; the ANT1 path not fitted.
3. Dedicated configuration: fit the ANT1 path and terminate BT_15.4_IN per E2B-09-1413 Figure 10 (values UNVERIFIED as drawn) instead of the 10 pF link; calibration file per ADR 0018 item 6.
4. RF microstrip 50 Ω, 7-25 mm, ground 230-240 µm below (Murata): ≥ 0.8 mm keep-out under each U.FL.

Title	Block
Antenna feeds	blk.antennas

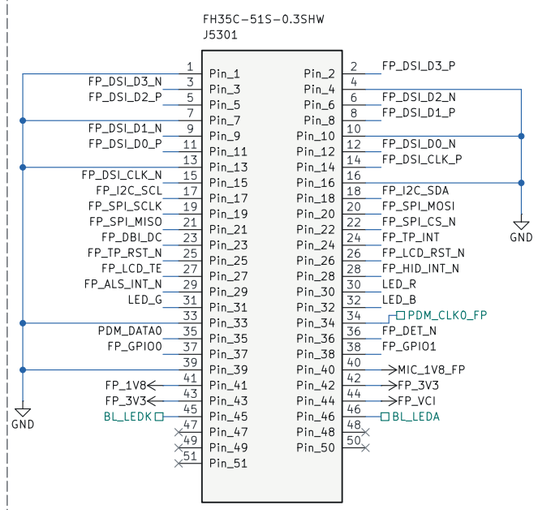
Purpose: ANT0 and ANT1 paths as Murata's copy: matching positions, U.FL, the 10 pF shared link

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	52 / 69

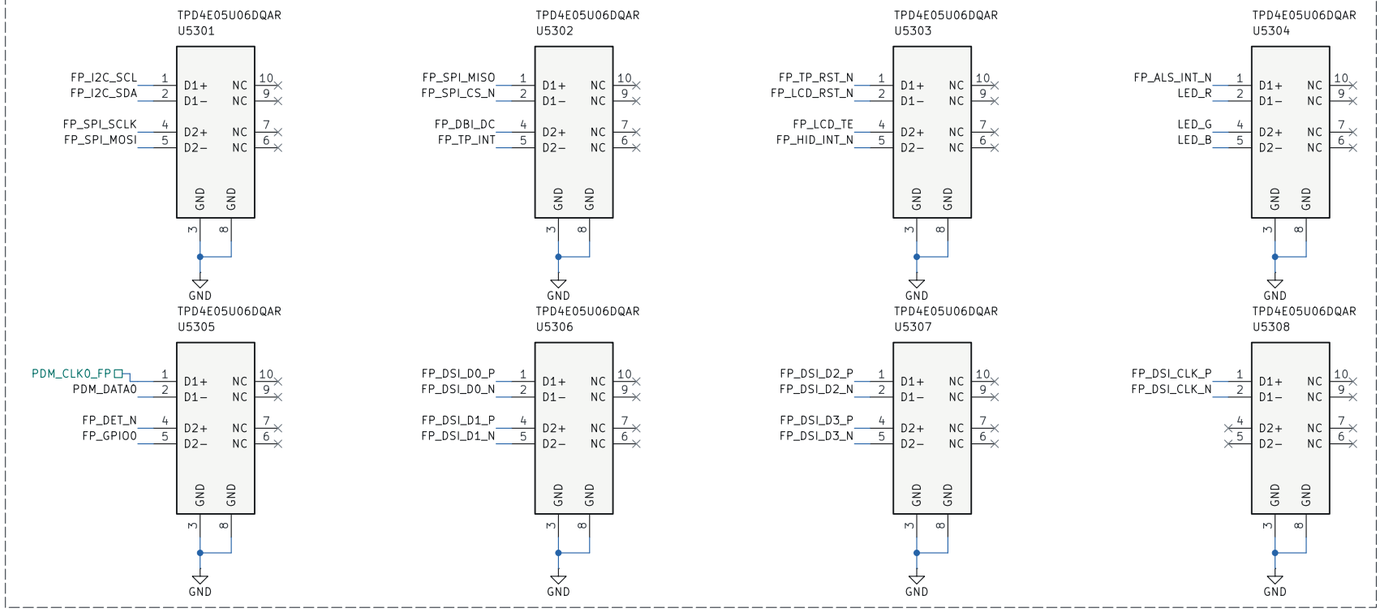
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/Antenna feeds/

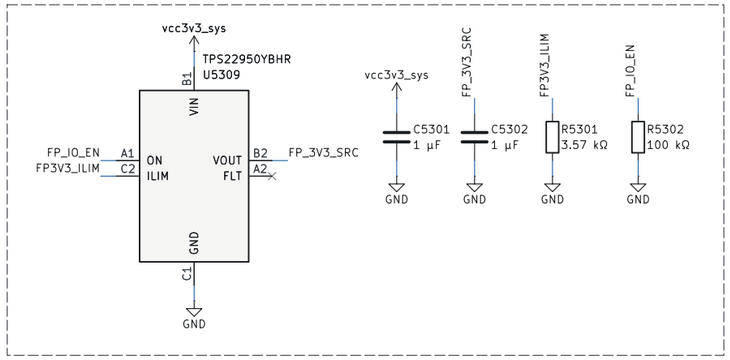
FH35C-51S-0.3SHW, contacts per pinout v1 §8.2 (BOM v1 line 71)



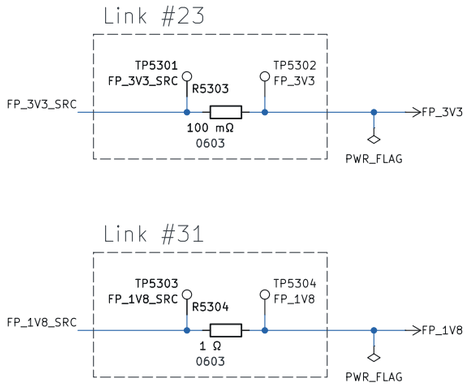
ESD: 5 x TPD4E05U06 low-speed, 20 lines (BOM v1 line 75); 3 x TPD4E05U06 DSI, 10 lines (line 72)



FP_3V3: TPS22950, 0.22–0.38 A (line 145)

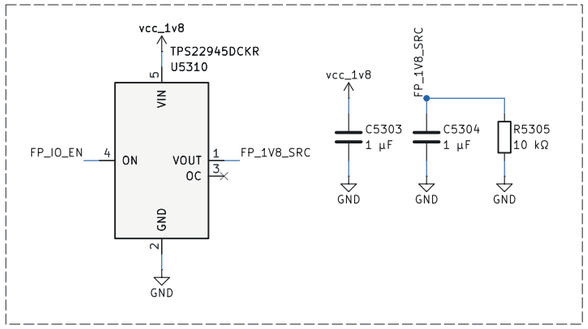


Link #23

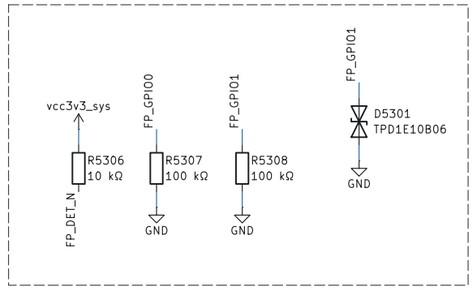


Link #31

FP_1V8: TPS22945, 100–200 mA (line 146)



Detect, spares, FP_GPIO1 ESD



{EXP_IO} {FP} {LED}

Interface

Net	Direction	Absent state (block not fitted)
FP_DSI_*	Output	From blk.fp-display (DSI0 D-PHY, 100 Ω); ESD here (BOM v1 line 72)
FP_I2C_*, FP_SPI_*, FP_TP_*, FP_LCD_*, FP_*_INT_N	Bidirectional	From blk.fp-controls
LED_R, LED_G, LED_B	Output	From blk.status-led (LP5521 current sources): open with the LP5521 unfitted
PDM_CLK0_FP, PDM_DATA0	Bidirectional	blk.mic-main: clock through 22 Ω; data to the SoC
FP_IO_EN	Input	Expander P00, 100 kΩ pull-down here: both switches off
FP_DET_N, FP_GPIO0, FP_GPIO1	Bidirectional	Expander P12–P14; detect pulled up: absent
FP_VCI, MIC_1V8_FP, BL_LEDA, BL_LEDK	Supply	From blk.fp-display, blk.mic-main, blk.fp-backlight

Rails

Rail	Capacitors	Requirement and source
vcc3v3_sys, FP_3V3_SRC	1 µF at VIN and VOUT	SLVSFJ2B (value: capture choice)
vcc_1v8, FP_1V8_SRC	1 µF at IN and OUT	SLVS832D (value: capture choice)

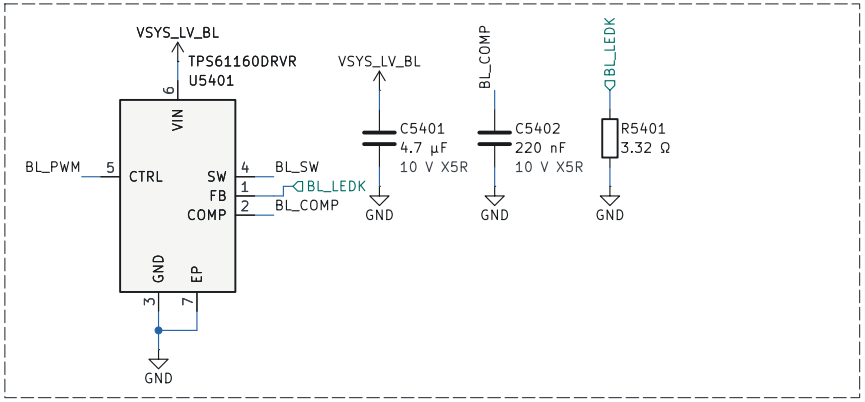
Notes
1. Function: the FH35C-51S faceplate connector (t1 faceplate interface v1): DSI, I2C3, SPI/DBI, touch, panel, interrupts, status LED, PDM, detect, spares, power; 5 contacts reserved.
2. Absent state: faceplate detached: FP_DET_N reads high (absent): FP_IO_EN low keeps FP_3V3 (TPS22950 QOD) and FP_1V8 (10 kΩ discharge, the TPS22945 has none) off.
3. Fault ceilings (t1 ADR 0005): FP_3V3 0.4 A (ILIM 0.22–0.38 A, UNVERIFIED at 3.57 kΩ), FP_1V8 0.2 A (100–200 mA at 3.3 V; limit at 1.8 V UNVERIFIED, SLVS832D §7.5).
4. FP_DET_N 10 kΩ pull-up and the spares' 100 kΩ pull-downs: capture choices. The TPS22950 FLT output is left open (auto-retry; the fault shows as FP_3V3 low): open item.
5. Contacts 47–51 reserved (47, next to BL_LEDA, never a logic line). Contact numbering against the Hirose drawing: pinout v1 §8.1.
6. Links (power tree §4.12): #23 FP_3V3, 0.1 Ω (80 mA, 8 mV); #31 FP_1V8, 1 Ω (5 mA, 5 mV).
7. ESD at the connector: DSI CLK and D0–D3 on 3 x TPD4E05U06 (line 72, 0.42–0.5 pF, one pair per two channels); 20 low-speed lines on 5 x TPD4E05U06 and FP_GPIO1 on a TPD1E10B06 (line 75).

Title	**Faceplate connector**
Block	blk.fp-connector
Purpose	FH35C-51S contacts, low-speed ESD, and the current-limited FP_3V3 and FP_1V8 switches
Project, board	OpenDAP T1, main board
Variant	full
Rev	A
Date	2026-10-25
Sheet	53 / 69

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/Faceplate connector/

TPS61160, RSET 3.32 Ω for 60 mA (BOM v1 line 51)



Inductor, rectifier, output capacitor (lines 52, 53)



{FP}

Interface

Net	Direction	Absent state (block not fitted)
VSYS_LV_BL	Supply in	VSYS_LV 3.8 V through the display INA236 shunt (0x49, blk.current-monitors)
BL_PWM	Input	PWM8_M0, SoC pull-down: off
BL_LEDA, BL_LEDK	Output	Contacts 46, 45: the LED string between them

Rails

Rail	Capacitors	Requirement and source
VSYS_LV_BL	4.7 µF	BOM v1 §8 (CIN 4.7 µF); SLVS791E §8.2.1.2.5: 1–4.7 µF
BL_LEDA	1 µF 50 V	BOM v1 §8; SLVS791E: 0.47–10 µF

Notes
1. Function: TPS61160 boost LED driver from VSYS_LV_BL, 60.2 mA (RSET 3.32 Ω, VFB 200 mV); strings up to 22 V; open-LED threshold 25–27 V at SW keeps BL_LEDA below 30 V (SLVS791E §6.5).
2. Absent state: CTRL low (SoC pull-down): off; an open string is bounded by the threshold.
3. COMP 220 nF as the datasheet's typical application (SLVS791E Figure 15); inductor 22 µH XFL3012-223MEC (Coilcraft doc 747, start lead on BL_SW); PMEG6010CEH rectifier.
4. Software: pwm-backlight on CTRL (PWM 5–100 kHz, SLVS791E §6.3).

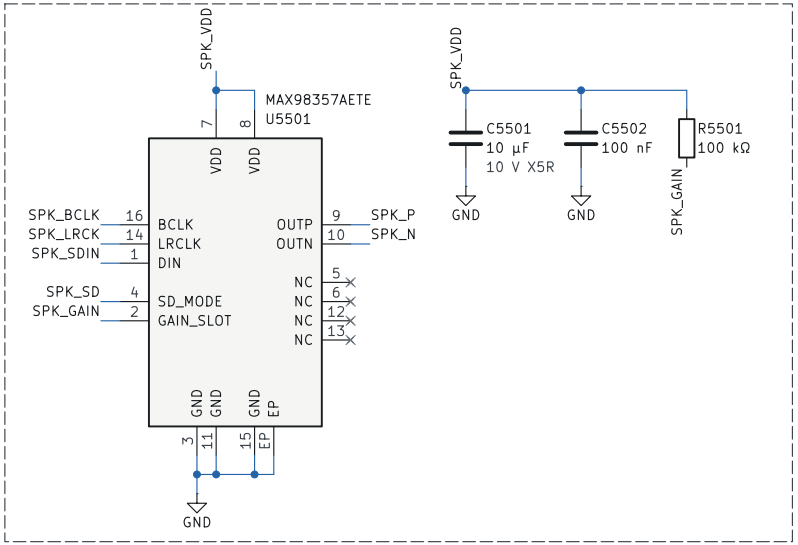
Title	**Faceplate backlight**
Block	blk.fp-backlight
Purpose	TPS61160 boost LED driver, 60 mA, open-LED threshold below 30 V

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	54 / 69

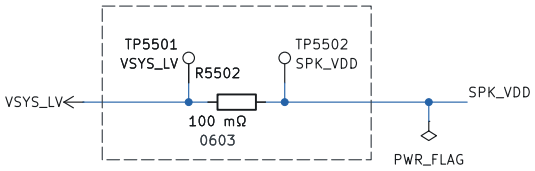
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/Faceplate backlight/

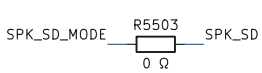
MAX98357A, I2S3 (BOM v1 line 32)



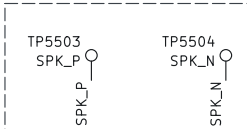
Link #17



SD_MODE select position (DS Figure 5)



Speaker spring-contact pads (BOM v1 line 33)



{SPK_I2S}

Interface

Net	Direction	Absent state (block not fitted)
SPK_BCLK, SPK_LRCK, SPK_SDIN	Input	I2S3_M0 at 3.3 V; 22 Ω at the SoC for the clocks
SPK_SD_MODE	Input	GPIO3_A6, SoC pull-down and the 100 kΩ internal pull-down: shutdown

Rails

Rail	Capacitors	Requirement and source
SPK_VDD	10 µF + 100 nF	MAX98357A Rev 11 p. 33

Notes
1. Function: MAX98357A mono class-D amplifier for the internal speaker (0.7 W, BOM line 33).
2. Absent state: SD_MODE low: shutdown (0.6 µA).
3. GAIN_SLOT 100 kΩ to VDD: 3 dB, the lowest gain (Rev 11 Table 8), with the Blackbird OS limiter keeping the speaker inside its rating; the gain is a capture choice.
4. SD_MODE driven at 3.3 V selects the left channel; the 0 Ω position takes the select resistor of Figure 5 if (L+R)/2 is wanted (pinout v1 B8).
5. VDD from VSYS_LV through link #17, 0.1 Ω (power tree §4.12: 0.3 A peak, 30 mV).

Title	Block
Speaker amplifier	blk.speaker

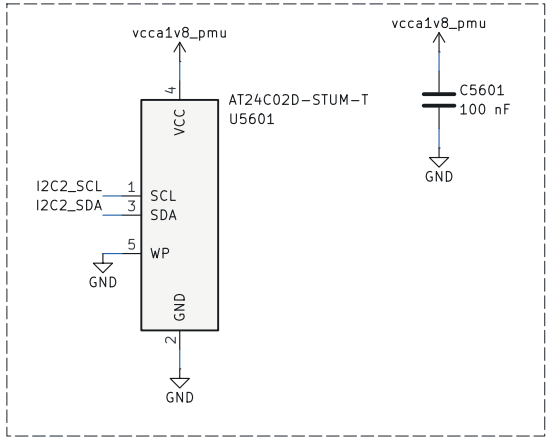
Purpose: MAX98357A on I2S3, SD_MODE shutdown, speaker spring-contact pads

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	55 / 69

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/Speaker amplifier/

AT24C02D at I2C2 0x50



{I2C2}

Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	0x50; pull-ups on blk.soc-pmuio

Rails

Rail	Capacitors	Requirement and source
vcca1v8_pmu	100 nF	8871F (value: capture choice)

Notes
1. Function: the board identity EEPROM (board revision, variant, serial), read by Blackbird OS.
2. Absent state: never omitted. Supply vcca1v8_pmu, as the rest of I2C2 (G16 proposal).
3. WP tied low: writable; write protection by software. The supply rail is the G16 proposal, not stated in the BOM.

Title
Board EEPROM

Block
blk.board-id

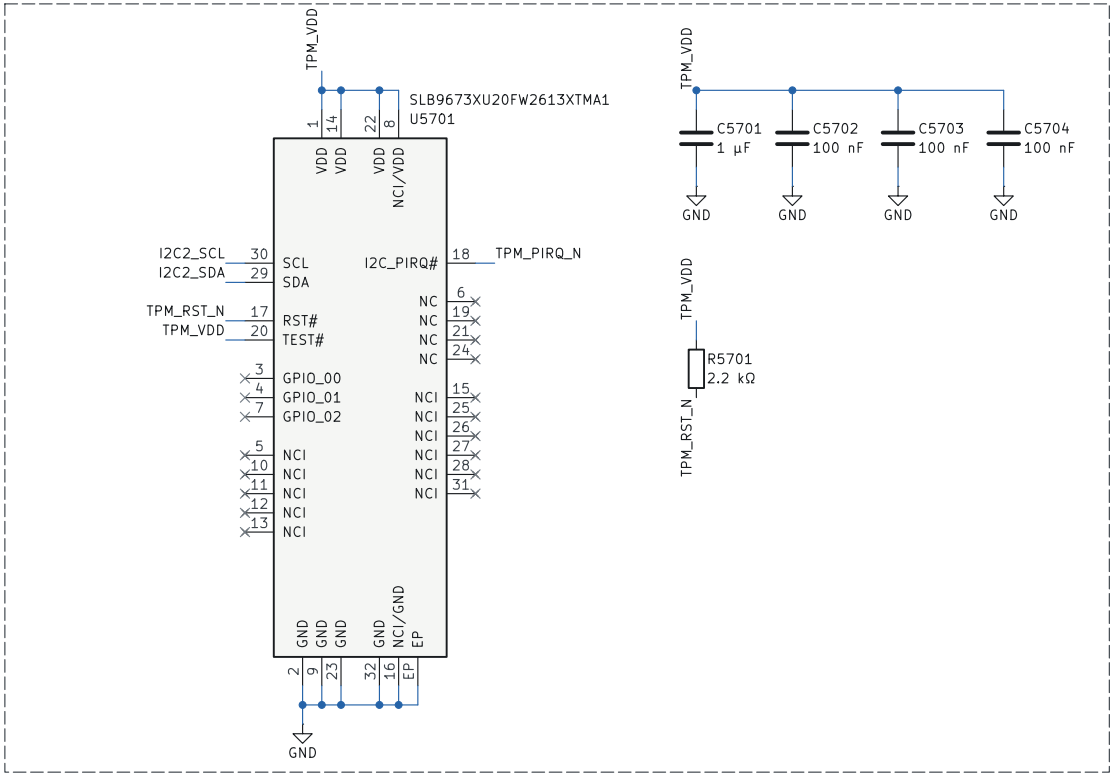
Purpose
AT24C02D board identity EEPROM on I2C2 at 0x50

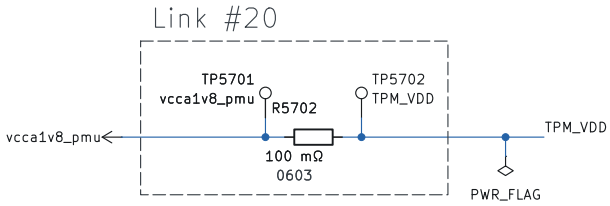
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	56 / 69

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/Board EEPROM/

SLB9673 TPM 2.0 at I2C2 0x2E (BOM v1 line 122)





{I2C2} {TPM}

Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	0x2E; pull-ups on blk.soc-pmuio
TPM_RST_N	Input	GPIO0_D3; 2.2 kΩ pull-up here wins over the SoC pull-down
TPM_PIRQ_N	Output	Open drain; pull-up on vcc_1v8 on blk.soc-vccio-a

Rails

Rail	Capacitors	Requirement and source
TPM_VDD	1 µF + 3 x 100 nF	SLB9673 DS Figure 2

Notes
1. Function: SLB9673 TPM 2.0, the keystore's hardware key holder (BRD-059, should; OS-032); mainline tpm_tis_i2c (DKMS on Debian).
2. Absent state: unfitted: 0x2E free; reset pulled to its inactive level; PIRQ pulled up.
3. Always on (vcca1v8_pmu, G5). TEST# held high; GPIO_00-02, NC and NCI open (DS Table 4).
4. VDD from vcca1v8_pmu through link #20, 0.1 Ω (power tree §4.12: 30 mA est., 3 mV).

Title	Block
Hardware key holder	blk.tpm

Purpose: SLB9673 TPM 2.0 on I2C2 at 0x2E, always on

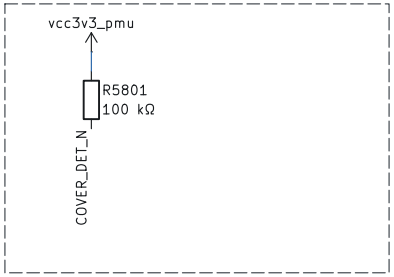
Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	57 / 69

© The OpenDAP Project · CERN-OHL-S-2.0 · github.com/opendap-project/board /Hardware key holder/

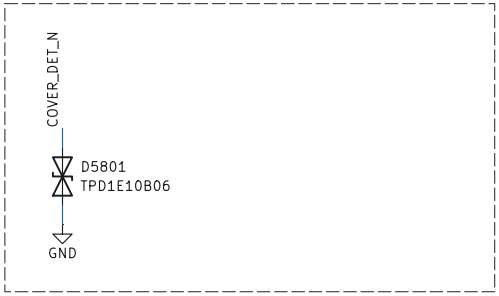
Contact pads at the board edge (BOM v1 line 143; PCB-L-10)



Pull-up to vcc3v3_pmu



ESD at the pad (TPD1E10B06)



{KEYS}

Interface

Net	Direction	Absent state (block not fitted)
COVER_DET_N	Output	The SoC ball's own reset pull-down (GPIO0_A6) reads cover closed; the description omits the input (dnp)

Rails

Rail	Capacitors	Requirement and source
(none)	–	No supply on this sheet: the pull-up hangs from vcc3v3_pmu

Notes
1. Function: back-cover detect (ADR 0020 item 7, should): low while the cover is closed; on a rise Blackbird OS syncs storage and powers off in order (ADR 0017 item 3), before the pack can be reached. It protects data, never safety (T1-086 is met without it).
2. Absent state: dnp: the ball's reset pull-down holds the cover-closed level and the device description omits the input; no other block depends on it.
3. GPIO0_A6 (PMUIO1, 3.3 V, wake; pinout v1 §7, G22). 100 kΩ to vcc3v3_pmu: 33 µA with the cover closed; open, 3.3 V − (10 µA ball + 0.1 µA TPD1E10B06) x 101 kΩ = 2.28 V > VIH 2.0 V. Pinctrl turns the ball's 16-43 kΩ pull-down off. Software: gpio-keys (a switch event).
4. Open (case input): the actuator. Two pads, the input and ground, take a conductive bridge on the cover or a two-terminal switch on the same lands; pad size UNVERIFIED.

Title	Block
Cover detect	blk.cover-detect

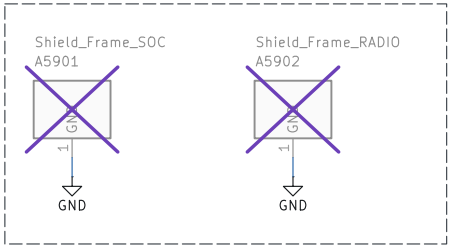
Purpose: Back-cover detect input: board-edge contact pads, pull-up, ESD

Project, board	Variant	Rev	Date	Sheet
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/Cover detect/

Frames



Interface

Net	Direction	Absent state (block not fitted)
GND	Ground	Each frame land is one ground pad ring (pad 1)

Rails

Rail	Capacitors	Requirement and source
(none)	–	No supply on this sheet

Notes
1. Function: shield-can frames on the back (PCB-T-06, PCB-EMC-03), one per frame of shields.toml: SoC and LPDDR4X; Radio module.
2. Absent state: dnp in both variants: fitted only if EMC or audio measurements require (BOM v1 line 90); the lands and keep-outs stay on the board either way.
3. Two-piece, etched to the board's outline (amended T1-038): frame soldered, lid clipped. Height with lid 1.5 mm (PCB-Z-01, the back pocket): lid 0.1 mm and 0.1 mm clearance leave 1.3 mm for the parts inside. None on the front (PCB-Z-03).
4. Geometry: hardware/t1-main-board/shields.toml: land 0.4 mm, keep-out 0.5 mm outside (PCB-T-06) and 0.3 mm inside; lid, clearance, land and the inner keep-out UNVERIFIED until the maker's drawing.

Title	Block
Shield-can frames	blk.shields

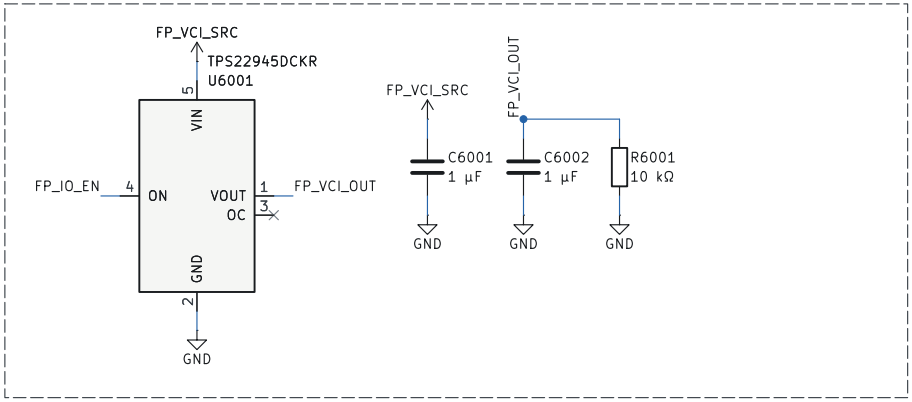
Purpose: Etched two-piece shield-can frames on the back: SoC and LPDDR4X, radio module (not fitted)

Project, board	Variant	Rev	Date	Sheet
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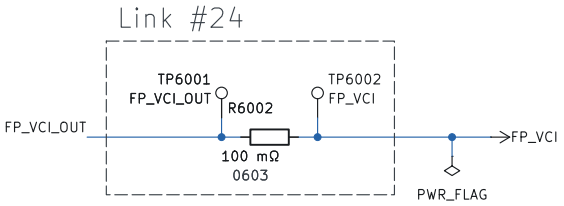
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/Shield-can frames/

FP_VCI: TPS22945 after RK809 LDO4 (BOM v1 line 146)



Link #24



{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
FP_VCI_SRC	Supply in	RK809 LDO4, 2.8 V (DT-adjustable)
FP_IO_EN	Input	Expander P00 (pull-down on blk.fp-connector)
FP_VCI	Supply out	Contact 44: off with FP_IO_EN low, whatever the LDO4 OTP state
FP_DSI_CLK_P/N, FP_DSI_D0-D3_P/N	Output	DSI0 from blk.soc-phy straight to the connector

Rails

Rail	Capacitors	Requirement and source
FP_VCI_SRC, FP_VCI_OUT	1 µF at IN and OUT	SLVS832D (value: capture choice)

Notes
1. Function: the panel supply FP_VCI through a 100-200 mA switch, since the RK809 states no LDO4 trip and its OTP start-up state is unknown (power tree §2, v0.5); DSI0 lanes to contacts 2-15.
2. Absent state: FP_IO_EN low: FP_VCI off, discharged by 10 kΩ; panel RST low and TE pulled down by the SoC pulls (pinout v1).
3. Blackbird OS opens the switch before LDO4 goes down (no reverse blocking, SLVS832D §9.1.3).
4. Open: the DSI common-mode filter positions (PCB-EMC-01) wait for the layout study.
5. Link #24, 0.1 Ω, after the switch (power tree §4.12: 0.1 A design, 10 mV); its Kelvin pads are the rail test pads (PCB-T-02).

Title	Block
Faceplate display supply	blk.fp-display

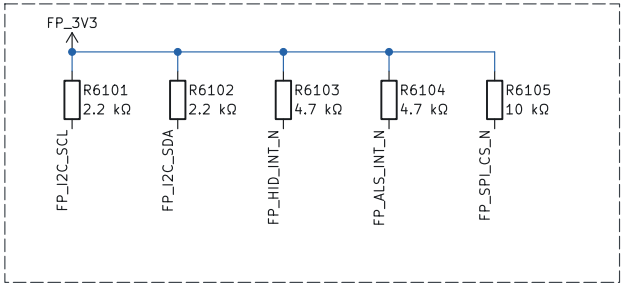
Purpose: FP_VCI current-limited switch after RK809 LDO4; DSI lanes pass to the connector

Project, board	Variant	Rev	Date	Sheet
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/Faceplate display supply/

Pull-ups on FP_3V3 (G8; BOM v1.11)



Touch address strap footprint (G9)



{FP}

Interface

Net	Direction	Absent state (block not fitted)
FP_I2C_SCL, FP_I2C_SDA	Bidirectional	I2C3_M1 at 3.3 V; pulled to FP_3V3, so nothing back-powers
FP_HID_INT_N, FP_ALS_INT_N	Output	Open drain on the faceplate; read low while FP_3V3 is off
FP_SPI_*, FP_DBI_DC, FP_TP_*, FP_LCD_*	Bidirectional	SPI1_M1 and GPIO on VCCIO5; SoC pulls

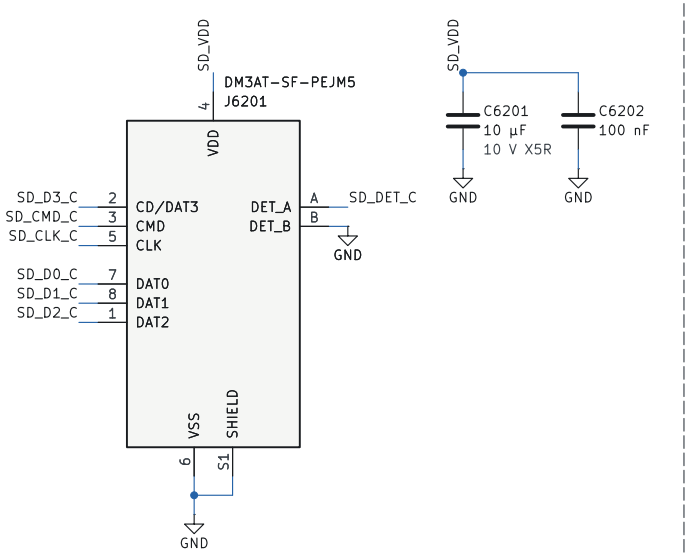
Notes
1. Function: the faceplate control bus and its interrupts (record 24): I2C3 at 400 kHz with 2.2 kΩ (≤ 100 pF from the faceplate), HID_INT (wake) and ALS_INT 4.7 kΩ, all on FP_3V3.
2. Absent state: FP_3V3 off: the pull-ups are unpowered, no faceplate part is back-powered; the interrupts read low and are masked by software.
3. SPI CS 10 kΩ pull-up (inactive at faceplate power-up, pinout v1): value a capture choice.
4. GT911 address strap: a DNP pull-down footprint on FP_TP_INT (Alternate gt911-strap); the SoC pull-down sets the default address.

Title: **Faceplate controls**				Block: blk.fp-controls
Purpose: I2C3, SPI/DBI and interrupt pulls on FP_3V3, the touch address strap				
Project, board: OpenDAP T1, main board	Variant: full	Rev: A	Date: 2026-10-25	Sheet: 61 / 69

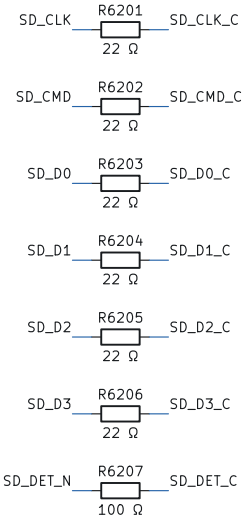
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/Faceplate controls/

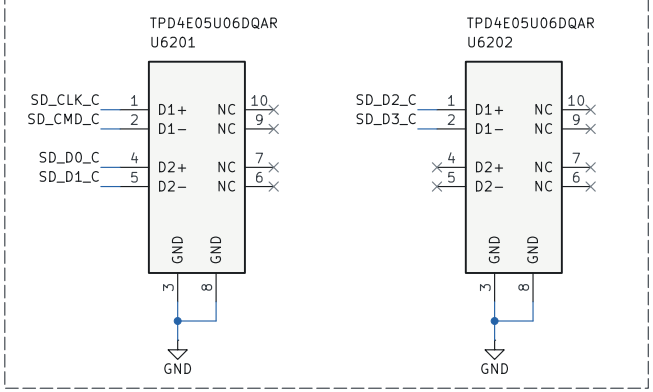
DM3AT-SF-PEJM5 socket (BOM v1 line 11)



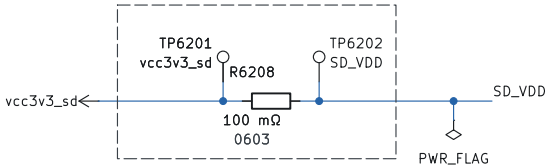
Series resistors (HDG p. 52)



ESD: 2 x TPD4E05U06 (BOM v1 line 73)



Link #16



{SDMMC0}

Interface

Net	Direction	Absent state (block not fitted)
SD_CLK, SD_CMD, SD_D0–D3	Bidirectional	SDMMC0 on VCCIO3 (vccio_sd, 1.8 / 3.3 V)
SD_DET_N	Output	SDMMC0_DET boot strap: SoC pull-up = no card; the socket switch pulls low
vcc3v3_sd	Supply in	RK809 SWOUT2

Rails

Rail	Capacitors	Requirement and source
SD_VDD	10 µF + 100 nF	Capture choice (socket datasheet silent)

Notes
1. Function: microSD socket on SDMMC0 (UHS-I SDR104 at 1.8 V signalling).
2. Absent state: card absent is normal: SD_DET_N high through the SoC pull-up; while it is high SDMMC0 carries UART2/JTAG M1 (HDG p. 20).
3. 22 Ω series on every line, 100 Ω on detect (HDG p. 52): TPD4E05U06 (0.5 pF) at the socket.
4. Detect switch wiring (DET_A to the line, DET_B to GND) against the Hirose drawing: UNVERIFIED.
5. VDD from vcc3v3_sd through link #16, 0.1 Ω (power tree §4.12: 0.2 A, 20 mV).

Title	Block
microSD socket	blk.microsd

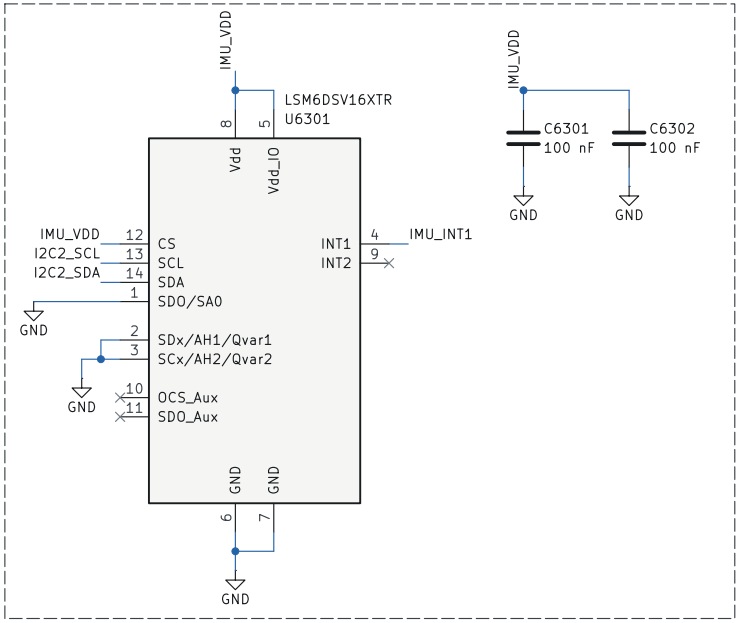
Purpose: DM3AT socket, SDMMC0 series resistors, card detect and ESD

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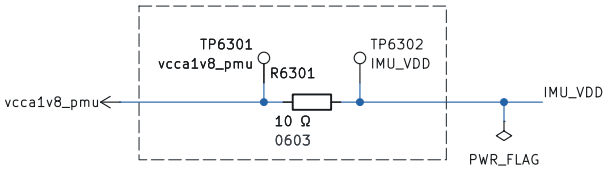
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/microSD socket/

LSM6DSV16X at I2C2 0x6A (BOM v1 line 59)



Link #32



{IO_IRQ}

{I2C2}

Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	0x6A (SA0 low); pull-ups on blk.soc-pmuio
IMU_INT1	Output	GPIO0_C5 (PMUIO2), wake on motion; SoC pull-down: no wake unfitted

Rails

Rail	Capacitors	Requirement and source
IMU_VDD	100 nF at Vdd, 100 nF at Vdd_IO	DS13510 §6 (C1, C2)

Notes
1. Function: LSM6DSV16X 6-axis IMU, wake on motion through INT1 (BRD-032, should).
2. Absent state: unfitted: INT1 pulled down by the SoC, 0x6A free. Always on (vcca1v8_pmu, G1).
3. CS to Vdd_IO selects I2C; SA0 low; SDx, SCx to GND (analog hub and Qvar off); OCS_Aux and SDO_Aux open (internal pull-ups, DS13510 Table 23); INT2 open.
4. Vdd and Vdd_IO from vcca1v8_pmu through link #32, 10 Ω (BOM v1 lines 59, 137; power tree §4.12: 0.65 mA, 6.5 mV).

Title	Block
Motion sensor	blk.imu

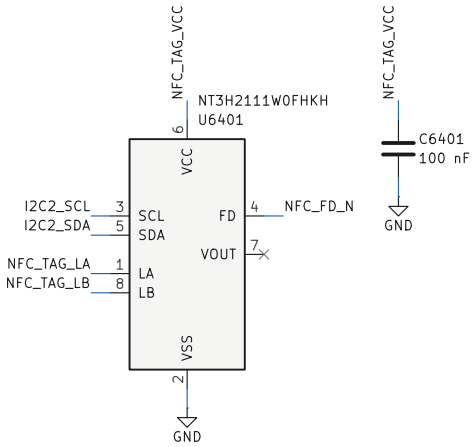
Purpose: LSM6DSV16X on I2C2 at 0x6A, INT1 wake on motion, always on

Project, board	Variant	Rev	Date	Sheet
OpenDAP T1, main board	full	A	2026-10-25	63 / 69

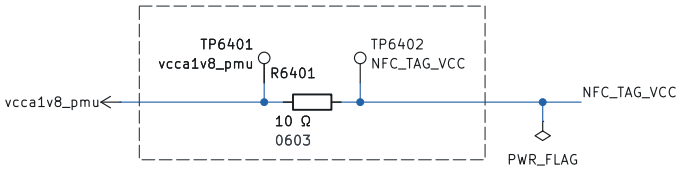
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/Motion sensor/

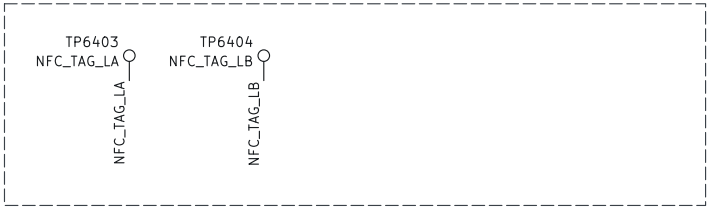
NT3H2111 NTAG I2C plus at I2C2 0x55 (BOM v1 line 125)



Link #33



Loop antenna contacts (BOM v1 line 126)



{NFC}

{I2C2}

Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	0x55; pull-ups on blk.soc-pmuio
NFC_FD_N	Output	Open drain, field detect; 4.7 kΩ pull-up on blk.soc-pmuio; wake on tap

Rails

Rail	Capacitors	Requirement and source
NFC_TAG_VCC	100 nF	Capture choice: the DS gives no value (UNVERIFIED)

Notes
1. Function: NTAG I2C plus dynamic tag for tap-to-pair (BRD-057, should): phones read it with the T1 off; the T1 writes it over I2C2.
2. Absent state: unfitted: 0x55 free; FD pulled up. Always on (vcca1v8_pmu, G6).
3. VOUT (energy harvesting) not used: open. The loop antenna (about 20 x 30 mm, line 126) and its tuning against the 50 pF input are open items; the tag with VCC at 0 V and a live I2C2 is UNVERIFIED (G6).
4. VCC from vcca1v8_pmu through link #33, 10 Ω (power tree §4.12: 1 mA est., UNVERIFIED).

Title	Block
NFC tag	blk.nfc-tag

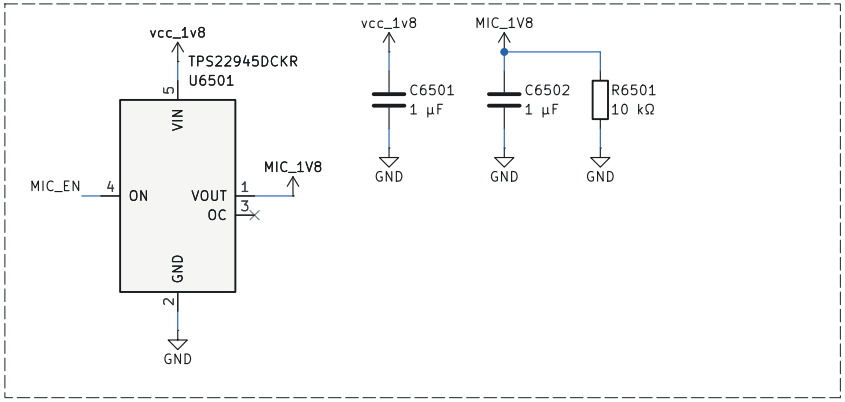
Purpose: NTAG I2C plus on I2C2 at 0x55, field detect and the loop antenna contacts

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/NFC tag/

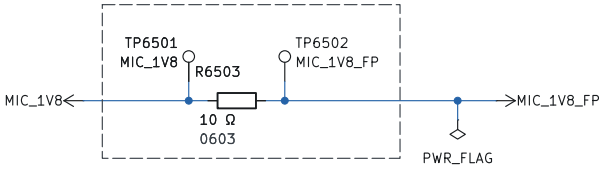
MIC_1V8: TPS22945, 100–200 mA (BOM v1 line 146)



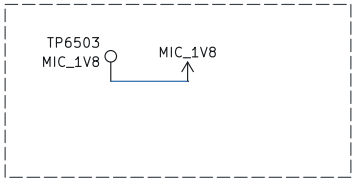
PDM clock series (HDG p. 72)



Link #34



Test pad (PCB–T–02)



{FP}

Interface

Net	Direction	Absent state (block not fitted)
MIC_EN	Input	GPIO4_C0, SoC pull–down: off
MIC_1V8, MIC_1V8_FP	Supply out	Off with MIC_EN low; MIC_1V8_FP to contact 40, MIC_1V8 also to blk.mic–ambient
PDM_CLK0, PDM_CLK0_FP	Output	PDM_CLK0_M1, parked low while MIC_1V8 is off (G10)

Rails

Rail	Capacitors	Requirement and source
vcc_1v8, MIC_1V8	1 µF at IN and OUT	SLVS832D (value: capture choice)

Notes
1. Function: MIC_1V8 for the faceplate microphone through a 100–200 mA switch (≤ 0.2 A ceiling).
2. Absent state: MIC_EN low: MIC_1V8 off, discharged by 10 kΩ; PDM clock parked low.
3. Link #34 (L10, 10 Ω) on the faceplate branch, MIC_1V8_FP; the ambient branch takes link #35 on blk.mic–ambient (power tree §4.12, G10).
4. Open: the privacy–LED gate (BOM v1 line 35: DMG1012T, gate = MIC_1V8, LED anode from FP_3V3) has no faceplate contact in pinout v1 §8.2; it is drawn when the LED's place is set.

Title	Block
Microphone supply	blk.mic–main

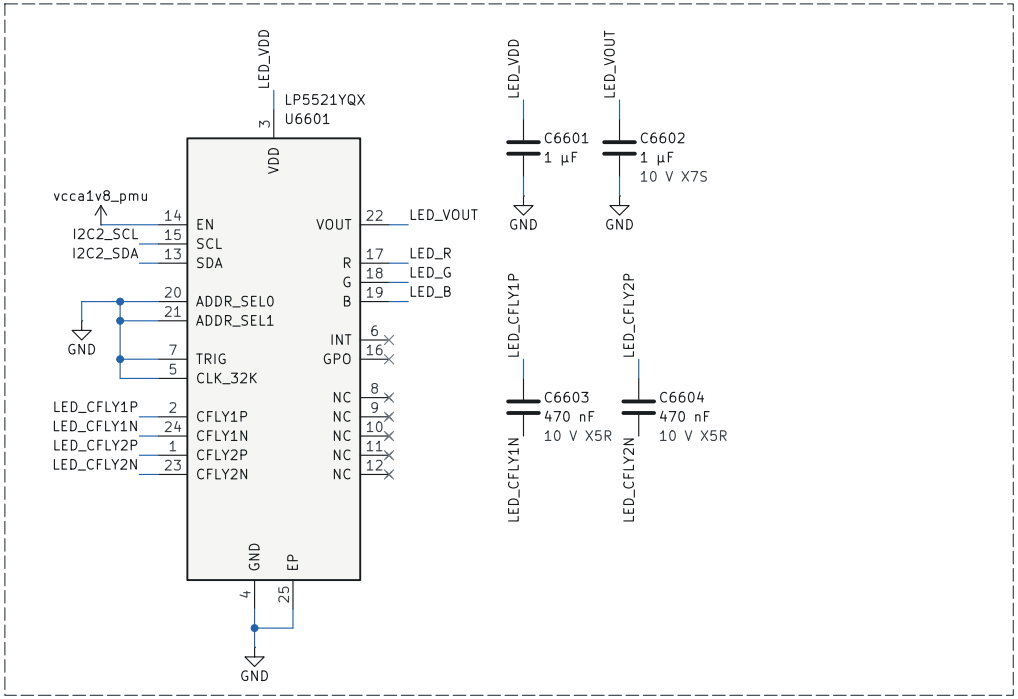
Purpose: MIC_1V8 current–limited switch and the PDM clock series resistor

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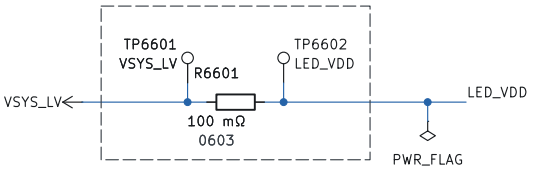
/Microphone supply/

LP5521 at I2C2 0x32 (BOM v1 line 105; SNVS441I Figure 26)



{I2C2} ◇—— {LED} ◇——

Link #22



Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	0x32 (ADDR_SEL0, ADDR_SEL1 low); pull-ups on blk.soc-pmuio. Unfitted: 0x32 free
LED_R, LED_G, LED_B	Output	Open: nothing drives or back-powers them; in use, current sources up to 25.5 mA to contacts 30-32

Rails

Rail	Capacitors	Requirement and source
LED_VDD	1 µF	SNVS441I p. 37: CIN 1 µF
LED_VOUT	1 µF 10 V	SNVS441I p. 37: COUT 1 µF, 10 V for DC bias
LED_CFLY1, LED_CFLY2	2 x 470 nF	SNVS441I p. 37: CFLY1, CFLY2 470 nF

Notes
1. Function: LP5521 RGB driver for the faceplate status LED (FPT-003, FPT-016 must; FPT-025 should): three current sources with PWM and program engines; mainline leds-lp5521 (DKMS).
2. Absent state: faceplate detached: the outputs are open, nothing back-powers. Unfitted: 0x32 free, LED lines open. Standby 1 µA with EN high and CHIP_EN = 0 (SNVS441I p. 6).
3. EN on vcca1v8_pmu: always enabled, and the I2C levels follow EN (1.8 V, I2C2). VDD from VSYS_LV 3.8 V through link #22, 0.1 Ω (power tree §4.12: 0.1 A est., 10 mV).
4. Internal clock: CLK_32K to GND; TRIG to GND (SNVS441I §7.3.5.3, §7.3.5.5): INT and GPO open (no SoC line in pinout v1). The R channel runs from VDD (R_TO_BATT, BOM line 105).
5. Each channel is capped at 20 mA in the devicetree (80 % of the LED's 25 mA IF, line 79).

Title	Block
Status LED driver	blk.status-led

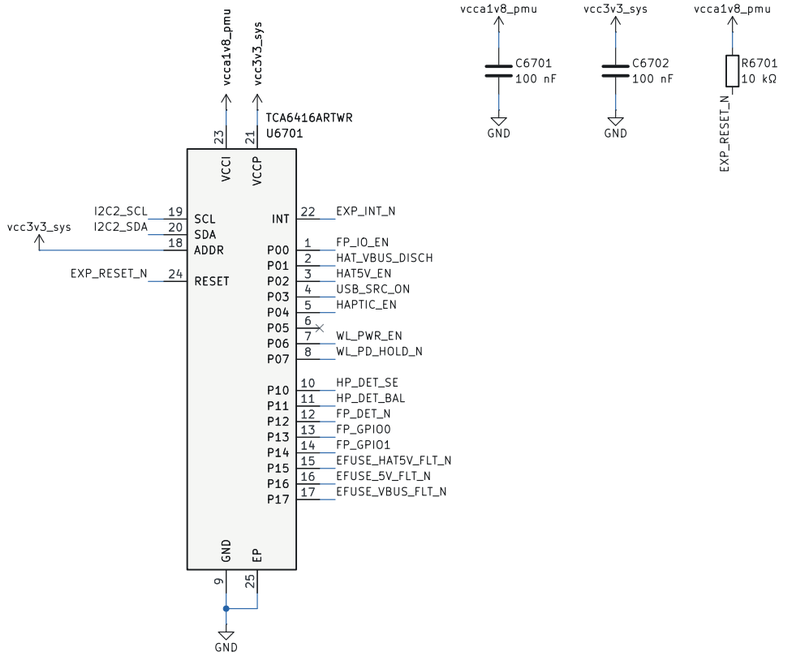
Purpose
LP5521 RGB LED driver on I2C2 at 0x32: charge pump, LED_R/G/B to the faceplate, link #22

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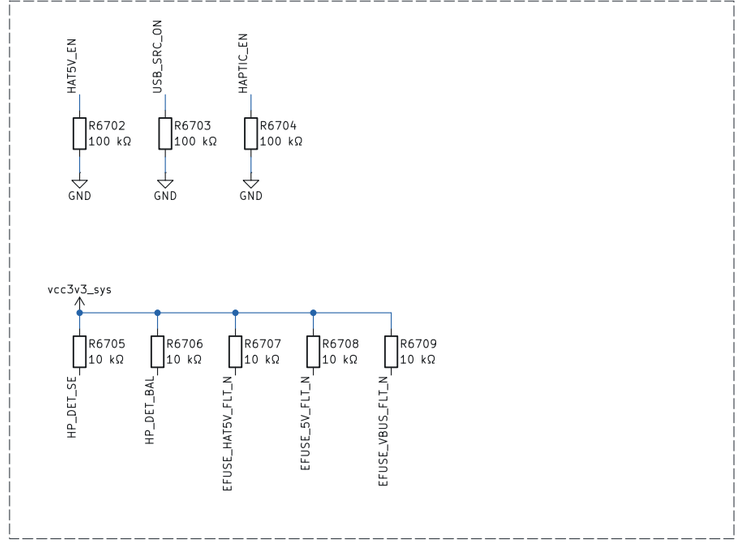
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/Status LED driver/

TCA6416A at I2C2 0x21 (BOM v1 line 93)



Off-state pull-downs and input pull-ups (PCB-P-18)



{IO_IRQ}
{I2C2}
{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	Pull-ups on blk.soc-pmuio (vcca1v8_pmu)
EXP_INT_N	Output	Open drain; 4.7 kΩ pull-up on blk.soc-pmuio
P00-P07 outputs	Output	Ports power up as inputs: each output's board pull-down holds it off
P10-P17 inputs	Input	Jack detects, faceplate detect and spares, eFuse faults

Rails

Rail	Capacitors	Requirement and source
vcca1v8_pmu, vcc3v3_sys	100 nF at VCCI and VCCP	SCPS194F (value: capture choice)

Notes
1. Function: TCA6416A, the slow enables and detects (pinout v1 expander table); VCCI on vcca1v8_pmu, VCCP on vcc3v3_sys; outputs hold through suspend.
2. Absent state: never omitted. At power-up every port is an input: P02-P04 pulled down here; P00 on blk.fp-connector, P01 on blk.hat-5v, P06, P07 on blk.radio.
3. ADDR to VCCP: address 0x21 (SCPS194F). RESET 10 kΩ to VCCI (not driven by the SoC).
4. Input pull-ups to VCCP: jack detects (HP_DET_*, the jack switch pulls low) and the eFuse faults (open drain), 10 kΩ: capture choices.
5. P05 spare, no-connect (was NFC_PWR_EN: the NFC reader removed, maintainer decision 2026-10-27); the driver sets it as an output, low.

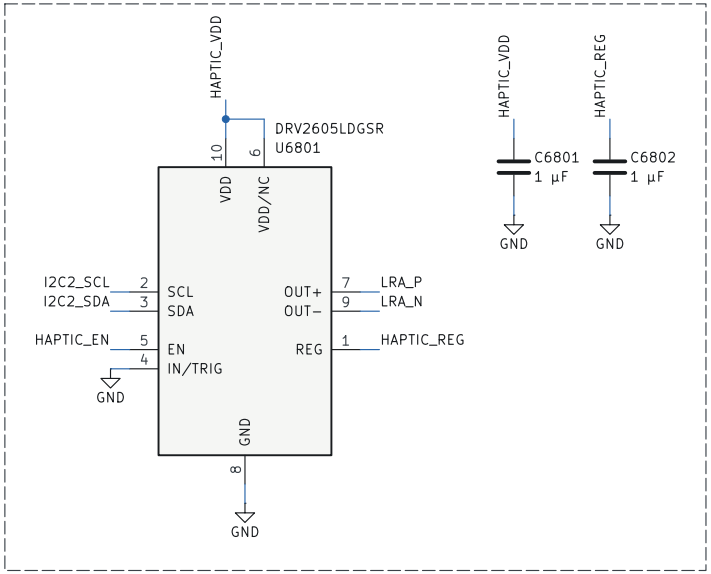
Title	Block
GPIO expander	blk.gpio-expander
Purpose	TCA6416A on I2C2: slow enables with their off-state pulls, detects and fault inputs

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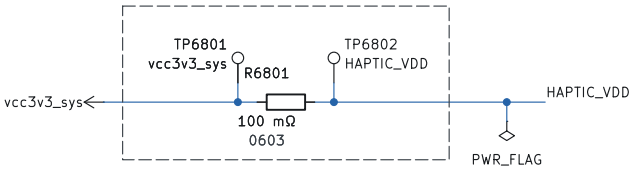
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/GPIO expander/

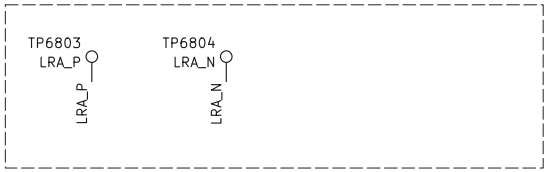
DRV2605L at I2C2 0x5A (BOM v1 line 123)



Link #19



LRA contacts (BOM v1 line 124)



{I2C2}

{EXP_IO}

Interface

Net	Direction	Absent state (block not fitted)
I2C2_SCL, I2C2_SDA	Bidirectional	0x5A; pull-ups on blk.soc-pmuio
HAPTIC_EN	Input	Expander P04, pulled down on blk.gpio-expander: standby

Rails

Rail	Capacitors	Requirement and source
HAPTIC_VDD	1 µF	SLOS854D Table 32
HAPTIC_REG	1 µF	SLOS854D Table 32

Notes
1. Function: DRV2605L closed-loop LRA driver with auto-resonance (BRD-058, should) for the Vybronics VG1040003D coin LRA on the case; mainline drv260x (DKMS on Debian).
2. Absent state: EN low: standby; unfitted: 0x5A free, LRA contacts open.
3. IN/TRIG to GND (effects over I2C, SLOS854D p. 5); VDD/NC tied to VDD.
4. VDD from vcc3v3_sys through link #19, 0.1 Ω (power tree §4.12: 0.1 A est., 10 mV).

Title	Block
Haptic driver	blk.haptics

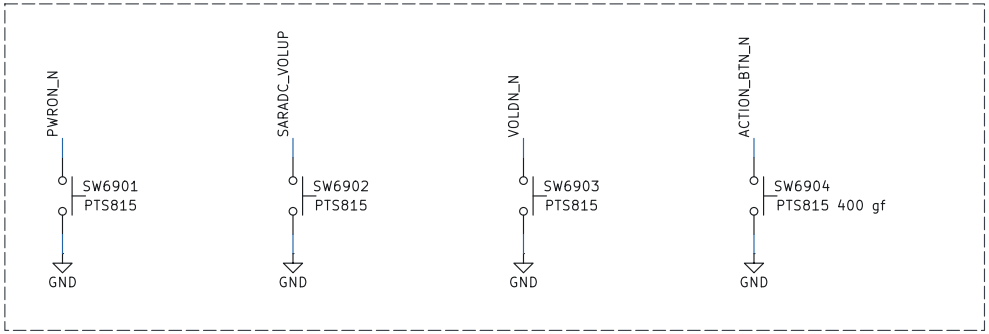
Purpose
DRV2605L LRA driver on I2C2 at 0x5A and the actuator contacts

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/Haptic driver/

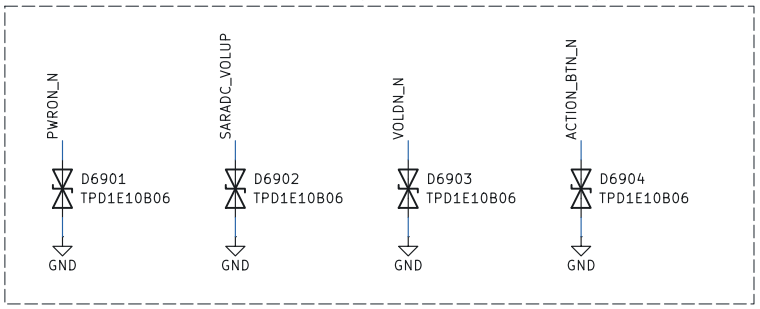
Buttons (BOM v1 lines 76, 77)



Vol− pull−up 100 kΩ (pinout v1 §4)



ESD: 4 x TPD1E10B06 (BOM v1 line 74)



{KEYS}

Interface

Net	Direction	Absent state (block not fitted)
PWRON_N	Output	RK809 PWRON (blk.pmic, 100 nF there); also blk.maskrom through the BAT54
SARADC_VOLUP	Output	SARADC_VIN0: 10 kΩ to vcca_1v8 and 1 nF on blk.soc−pmuio; reads released
VOLDN_N	Output	SARADC_VIN1 and the MaskROM gate: 100 kΩ to vcca1v8_pmu here, 1 nF at the SoC
ACTION_BTN_N	Output	GPIO0_C2 wake: 4.7 kΩ to vcca1v8_pmu on blk.soc−pmuio; reads released

Notes
1. Function: the four body buttons (T1−014, T1−015; BRD−016, −017, −035, −036): Power to the RK809 PWRON, Vol+ and Vol− on their own SARADC inputs, Action on a PMUIO2 wake ball.
2. Absent state: never omitted; released lines read high through their pulls.
3. Vol− is pulled to the always−on vcca1v8_pmu, which also feeds the MaskROM gate: a pull to vcca_1v8 would read pressed while that rail is still off (pinout v1 §4, Vol− row).
4. Action: PTS815 G force code (400 gf, UNVERIFIED) on the side buttons' land (BOM v1 line 76); its deeper travel (T1−022) is the case cap's free travel. PTS815 land per p. B−44: terminals 1−2 are pad 1, 3−4 pad 2 (the circuit diagram on the same page).

Title: **Side buttons**				Block: blk.buttons
Purpose: Power, Action, Vol+ and Vol−, each with its ESD diode				
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/Side buttons/